

Cost, Volume and profit Analysis:

Cost: Sacrifice for the resources

Cost Classification:

- 1) Variable Cost
- 2) Fixed Cost
- 3) Mixed cost

Variable cost: cost that varies with unit. As unit changes total variable cost changes. Variable cost has some proportional relation with unit.

For example: if

1 unit cost = Tk. 5

2 unit cost = Tk. 10

5 Units cost = Tk. 50

10 units cost = 100

Question papers for Students, total number of mark sheets or certificate issued for the students is VC for a particular class.

Fixed Cost: cost that does not vary with unit. As unit changes fixed cost does not change. Fixed cost facilitates a capacity within that capacity FC never be changed. For example: class room decoration cost is a FC. So FC is fixed in total but per unit FC changes as unit changes

Mixed Cost: combination of Variable cost and FC. Mixed cost follows the following equation

$Y = a + bx$Mixed cost

Y= total cost

a= FC

b= variable cost per unit

x= number of units

For example: Salary of any worker, Electricity bill

$TC(\text{Total Cost}) = TVC(\text{Total Variable Cost}) + FC(\text{Fixed Cost})$

$TC = VC \text{ per unit}(V_p) * \text{quantity}(Q) + FC$

$\text{Total Revenue}(TR) / \text{Total sales} = P * Q$

$\text{Profit} = TR - TC$

$\text{Profit} = (P * Q) - (V * Q) - FC$

$0 = Q(P - V) - FC$

$Q = [FC / (P - VC)] \dots \dots \text{Break Even Point(BEP)}$

$P = 10$

$VC_p = 7$

$\text{Contribution Margin (CM)} = 3$

$\text{Contribution Margin Ratio} = (\text{CM} / \text{Sales per unit}) * 100$

$\text{Variable cost ratio} = (\text{VC} / \text{Sales per unit}) * 100$

$\text{CM ratio} = 1 - \text{VC ratio}$

$\text{VC ratio} = 1 - \text{CM ratio}$

$FC = 30,000$

Capacity : 0-15,000 units

$BEP = [30,000 / 10 - 7]$

$Q = 10,000 \text{ Units}$

If I have sold 15,000 units total profit will be $= [5000 * 3] = 15,000 \text{ taka}$

If I want to make profit of Tk. 5000. How many units I need to produce??

$$(30,000 + 5000) / 3 = 11,667 \text{ units}$$

Safety Margin = Current Production - Break even sale

$$= 12,00,000 - 9,60,000$$

$$= 2,40,000$$

Degree of operating leverage (DOL) = $[CM / \text{Operating Profit}]$

= Times

Capital Budgeting and Relevant Costing:

Long term investment decision.

5-100 years

Series of **cash inflows** and **outflows**

Time value of money is considered:

Discount rate /opportunity cost/ cost of capital

0	1	2	3	4	5
(5,00,000)	140,000	230,000	1,10,000	1,90,000	1,40,000
	127272	190082	82644	129772	86928

7% inflation rate

10%

Capital Budgeting tools:

Payback period (PBP): 3.11 years

Discounted Payback Period (DPBP): 3.78 years

Net Present Value (NPV): 1,16,453

Internal Rate of Return (IRR)

Modified Internal Rate of Return (MIRR)

Projects Classification:

Independent Project:

Mutually Exclusive Project :

Relevant Costing:

Cost that does differ between or among alternatives. That has impact on decision making

Opportunity Cost: next best alternative that one may sacrifice to get something

On Capital Budgeting

- a) A firm with a WACC of 10 percent is considering the following mutually exclusive projects:

Year	Project : A	Project: B
0	(Tk. 400)	(Tk.600)
1	55	300
2	55	300
3	55	50
4	225	50
5	225	50
6	225	50

- i) According to the payback criterion, which project should be accepted?
 - ii) According to the discounted payback criterion, which project should be accepted?
 - iii) According to the NPV criterion, which project should be accepted?
 - iv) According to the IRR criterion, which project should be accepted?
- b) Your Company is considering two mutually exclusive projects, X and Y, whose costs and cash flows are shown below:

Year	X	Y
0	(Tk. 1,000)	(Tk. 1,000)
1	100	1,000
2	300	100
3	400	50
4	700	50

The projects are equally risky, and their cost of capital is 12 percent. You must make recommendation, and you must base it on the modified IRR (MIRR). What is the MIRR of the better project?

- c) Consider the projects below with respective cash flows:

Project	<u>Years</u>					
	0	1	2	3	4	5
A	(200,000)	35,000	80,000	90,000	75,000	20,000
B	(200,000)	28,000	40,000	50,000	90,000	10,000

Assume that cost of capital for both project is 10%.

Requirements:

- i. Calculate Pay Back Period (PBP) for both projects and comment.
- ii. Calculate Net Present Value(NPV) of each project and comment
- iii. Calculate Internal Rate of Return (IRR) for both projects. Which method will you used to select the project? Why?

On CVP

- a) The following information is given for the calculation of break-even-point:

Selling price per unit	Tk. 5
Variable cost per unit	3
Contribution margin per unit	2
Fixed costs	24,000

You are required to calculate break-even point and desired sales volume if (i) the desired profit is Tk. 30,000 and (ii) desired profit after income tax is Tk. 20,000 and income tax rate is 40 %.

- b)

Here is the data of XYZ Ltd:		Per-Unit	Percentage
Sales (\$20 per unit)	\$270,000	20	
Variable expenses	(189,000)	?	?
Contribution Margin	81,000	?	?
Fixed expenses	(90,000)		
Net operating loss	<u>\$(9,000)</u>		
i) Fill up missing (?) figure in the above statement.			
ii) Draw a graph identifying BEP and profit and loss region.			
iii) If actual sales are \$ 400000 then find out margin of safety and comment about the company.			

- c) Last month when Holiday Creations Inc. sold 50,000 units for total sales value of Tk.200,000.

Total variable cost and total fixed expense was Tk. 120,000 and Tk. 24,000 respectively

Required: 1) what is the company's contribution margin (CM) ratio?

- 2) Estimate the changes in the company's net income if it were to increase its total sale by Tk. 1,000

- d) Voltar Company manufactures and sells a specialized cordless telephone for high electromagnetic radiation environments. The company's contribution format income statement for the most recent year is given below:

	Total	Per Unit	Percent of
Sales			
Sales (20,000 units)	\$1,200,000	\$60	100%
Variable expenses	900,000	45	? %
Contribution margin	300,000	\$15	? %
Fixed expenses	240,000		
Net operating income	\$ 60,000		

Management is anxious to increase the company's profit and has asked for an analysis of a number of items.

Required:

- I. Compute the company's CM ratio and variable expense ratio.
- II. Compute the company's break-even point in both units and sales dollars.

- III.** Assume that sales increase by \$400,000 next year. If cost behavior patterns remain unchanged, by how much will the company's net operating income increase?
- IV.** Refer to the original data. Assume that next year management wants the company to earn a profit of at least \$90,000. How many units will have to be sold to meet this target profit?
- V.** Refer to the original data. Compute the company's margin of safety in both dollar and percentage form.
- VI.**
- a.** Compute the company's degree of operating leverage at the present level of sales.
 - b.** Assume that through a more intense effort by the sales staff, the company's sales increase by 8% next year. By what percentage would you expect net operating income to increase? Use the degree of operating leverage to obtain your answer.
 - c.** Verify your answer to (b) by preparing a new contribution format income statement showing an 8% increase in sales.
- VII.** In an effort to increase sales and profits, management is considering the use of a higher-quality speaker. The higher-quality speaker would increase variable costs by \$3 per unit, but management could eliminate one quality inspector who is paid a salary of \$30,000 per year. The sales manager estimates that the higher-quality speaker would increase annual sales by at least 20%.
- a)** Assuming that changes are made as described above, prepare a projected contribution format income statement for next year. Show data on a total, per unit, and percentage basis.
 - b)** Compute the company's new break-even point in both units and dollars of sales. Use the formula method.
 - c)** Would you recommend that the changes be made?

You are examining the viability of a capital investment that your firm is interested in. The project will require an initial investment of \$500,000 and the projected revenues are \$400,000 a year for five years. The projected cost-of-goods-sold is 40% of revenues and the tax rate is 40%. The initial investment is primarily in plant and equipment and can be depreciated straight-line over five years. (The salvage value is zero.) The project makes use of other resources that your firm already owns:

Two employees of the firm, each with a salary of \$40,000 a year, who are currently employed by another division, will be transferred to this project. The other division has no alternative use for them, but they are covered by a union contract which will prevent them from being fired for three years (during which they would be paid their current salary.)

The project will use excess capacity in the current packaging plant. While this excess capacity has no alternative use now, it is estimated that the firm will have to invest \$ 250,000 in a new packaging plant in year 4 as a consequence of this project using up excess capacity (instead of year 8 as originally planned).

The project will use a van currently owned by the firm. While the van is not currently used, it can be rented out for \$ 3000 a year for five years. The book value of the van is \$10,000 and it is being depreciated straight line (with five years remaining for depreciation).

The discount rate to be used for this project is 10%.

- i.** What (if any) is the opportunity cost associated with using the two employees from another division (described in para 2)?
- ii.** What, if any, is the opportunity cost associated with the use of excess capacity of the packaging plant?
- iii.** What, if any, is the opportunity cost associated with the use of the van?
- iv.** What is the after-tax operating cash flow each year on this project?
- v.** What is the net present value of this project?

Determining unit cost of products:

Components of products:

- 1) Direct Material
- 2) Direct Labor
- 3) Factory Overhead (Indirect Cost)

- What is the difference between direct and indirect cost?

Cost sheets:

Raw Material purchase	xx
[+] Freight in	xx
[+] Loading and Unloading charge	xx
[+] Insurance cost	<u>xx</u>
Purchase cost	XX
[-] Purchase Discount	(xx)
[-] Purchase return and Allowance	<u>(xx)</u>
Net Purchase	xx
[+] Beginning RM	xx
[-] Closing RM	<u>(xx)</u>
RM Used (DM)	xx
Direct Labor	<u>xx</u>
Prime Cost	xx
Applied FOH	<u>XX</u>
Total Manufacturing Cost/Work cost	xx
[+] Beginning WIP	xx
[-] Closing WIP	<u>(xx)</u>
Cost of goods Manufactured	XX
[+] Beginning FG	xx
[-] Closing FG	<u>(xx)</u>
COGS (Cost of goods Sold)	XX
[+] Administrative and Selling cost	<u>XX</u>
Total sales cost	XXX
[+] Profit	<u>XX</u>
Selling Price	<u><u>XX</u></u>

Unit cost: (Total sales cost/No. of units produced)

Administrative and Selling cost :

Sales commission
Sales person salary
Sales discount
Freight out
Salary to manager

The following data are forming the accounts of Millville Company:

Inventories:

	July 1, 20A	June 30, 20B
Finished Goods	Tk. 20,000	Tk. 28,000
Work in process	60,000	36,000
Materials	40,000	48,000

Sales discounts..... Tk.8, 000
Purchase discounts..... 3,200
Sales.....1,800,000
Purchase returns and allowances..... 20,000
Depreciation-factory machinery..... 160,000
Factory insurance..... 50,000
Freight out..... 8,000
Other factory expenses..... 16,000
Bond interest expenses.....50,000
Sales salaries.....100,000
Freight in.....12,000
Direct factory labor.....800,000
Materials purchases.....400,000
Advertising expenses..... 12,000

Required: Prepare a Cost of Goods Sold statement for the year ended June 30, 20B

Financial Statements Analysis:

Assets classification:

Current Asset:

Asset that can be converted into cash/cash equivalents/any other resources within one year.

Example:

Accounts Receivable, Inventory, Loan paid, Bank deposit

Non-current assets:

Asset that can be converted into cash/cash equivalents/any other resources beyond one year.

Example: **land, property plant and equipment, Furniture, patent, copyright, trademarks**

Mr. x issued a loan in favor of Mr. Y for taka 10,000. R. Y is committed to pay loan for taka 7500 within one year and remaining he will in 18th month.

Current asset: 7500

Non-current asset: 2,500

Liability Classification:

Current Liabilities: obligation that should be paid within one year.

Example: Notes payable, Accounts payable, Salaries payable, utilities payable

Non-current liabilities/Long term liabilities : obligation that should be paid after one year. It is long term liabilities

Example: Debenture, Notes payable

Working capital= Current assets – Current liabilities, WC is used to maintain business operation.

WC loan taken from bank

Different Ratios:

Current Ratio: $(\text{Current Assets} / \text{Current liabilities})$

Standard: 2:1

Quick Ratio: $(\text{Current asset} - \text{Inventories}) / (\text{current liabilities} - \text{Bank overdraft})$

Standard 1:1

Return on Assets: $(\text{Net income} / \text{Average total assets})$

Gross profit Ratio: $(\text{Gross profit} / \text{Net Sales})$

Net Profit Ratio: $(\text{Net profit} / \text{net sales})$

Debt to assets ratio: $(\text{Debt} / \text{Total assets})$

Debt to equity ratio: $(\text{Debt} / \text{Equity})$

EPS (Earning per share): $(\text{Net income attributable to equity holder} / \text{Number of share outstanding})$

Interest coverage ratio: $(\text{operating profit or EBIT} / \text{interest expenses})$

Financial Statement Analysis:

Common Size Analysis: through preparing common size statement

1) Common size analysis for the income statement: Show each item in the income statement as a percentage of gross sale

2) Common Size analysis for the balance sheet: Show each item in the balance sheet as a percentage of total assets

Again the analysis can be made in two ways

1) Horizontal Analysis

2) Vertical Analysis

The accompanying balance sheet and profit and loss account relate to Hypothetical Ltd is as follows.

Balance Sheet as at March 31

(Amount in Lakh of Taka)

Particulars	Previous Year	Percentage	Current Year	Percentage
Liabilities				
Equity share capital(of Tk. 10 each)	240	40%	240	37%
General reserves	96	16%	182	28%
Long-term loans	182	30%	169.5	26%
Creditors	67	12%	52	8%
Outstanding expenses	6	1%	0	0%
Other current liabilities	9	1.5%	6.5	1%
Total Liabilities	<u>600</u>		<u>650</u>	
Assets:				
Plant (net of accumulated depreciation)	402	67%	390	60%
Cash	54	9%	78	12%
Debtors/ AR	60	10%	65	10%
Inventories	84	14%	117	18%
Total Assets	<u>600</u>		<u>650</u>	

Income Statement for the year Ended March 31

(Amount in Lakh of Taka)

Particulars	Previous year	Percentage	Current year	Percentage
Gross sales	370	100%	480	100%
(Less) Returns	(20)	(5%)	(30)	(7%)
Net Sales	350	95%	450	93%
(Less) Cost of goods sold	(190)	51%	(215)	(45%)
Gross profit	160	43%	235	48%
(Less) Selling, general and Administrative cost	(50)	(14%)	(72)	(15%)
Operating profit(EBIT)	110	29%	163	33%
(Less) Interest Expenses	(20)	(5%)	(17)	(4%)
EBT	90	24%	146	29%
(Less) Taxes	(31.5)	(9%)	(51.5)	(11%)
Net Income(EAT)	58.5	15%	94.9	18%

Requirement:

- Convert these into common-size statements(Only show percentage not amounts)
- Calculate Cash Conversion Cycle (CCC) for the current year and make comments on company's operating efficiency and liquidity.
- Calculate Current Ratio, Quick Ratio (Both Year)
- Calculate Debt to Assets Ratio, Debt to Equity Ratio, Financial Leverage Ratio, and Interest Coverage Ratio and make comments on company's solvency (Both Year).

Cash Conversion Cycle (CCC)/ Cash Cycle: IHP+ARCP-APPP

Operating Cycle: IHP+ARCP

Accounts Receivable Turnover: (Net credit sales/average accounts receivable)

Avg A/R= (Beginning AR+Ending AR)/2

ARCP (Accounts Receivable Collection Period): (360/AR turnover)

Accounts Payable Turnover: (net credit purchase/Avg. accounts payable)

Accounts payable Payment period (APPP): (360/ AP turnover)

Inventory Turnover: (Cost of Goods sold/Average inventory)

Inventory Holding Period (IHP): (360 days/ Inventory turnover)

Janu 1	(APPP)	Feb 15	March 20	(ARCP)	April 25
(IHP)					

Purchase= cost of goods sold+ closing inventory –beginning inventory

January 1 , beginning inventory 5 taka

Purchase during the year 100 taka

Sale during year 85 (COGS) taka cost sold Tk.155

Closing inventory 20 tk

Product Costs versus Period Costs

In addition to classifying costs as manufacturing or nonmanufacturing costs, there are other ways to look at costs. For instance, they can also be classified as either *product costs* or *period costs*. To understand the difference between product costs and period costs, we must first discuss the matching principle from financial accounting.

Generally, costs are recognized as expenses on the income statement in the period that benefits from the cost. For example, if a company pays for liability insurance in advance for two years, the entire amount is not considered an expense of the year in which the payment is made. Instead, one-half of the cost would be recognized as an expense each year. The reason is that both years—not just the first year—benefit from the insurance payment. The unexpensed portion of the insurance payment is carried on the balance sheet as an asset called prepaid insurance.

The *matching principle* is based on the *accrual* concept that *costs incurred to generate a particular revenue should be recognized as expenses in the same period that the revenue is recognized*. This means that if a cost is incurred to acquire or make something that will eventually be sold, then the cost should be recognized as an expense only when the sale takes place—that is, when the benefit occurs. Such costs are called *product costs*.

LEARNING OBJECTIVE 2

Distinguish between product costs and period costs and give examples of each.

Product Costs

For financial accounting purposes, **product costs** include all costs involved in acquiring or making a product. In the case of manufactured goods, these costs consist of direct materials, direct labor, and manufacturing overhead. Product costs “attach” to units of product as the goods are purchased or manufactured, and they remain attached as the goods go into inventory awaiting sale. Product costs are initially assigned to an inventory account on the balance sheet. When the goods are sold, the costs are released from inventory as expenses (typically called cost of goods sold) and matched against sales revenue. Because product costs are initially assigned to inventories, they are also known as **inventoriable costs**.

We want to emphasize that product costs are not necessarily treated as expenses in the period in which they are incurred. Rather, as explained above, they are treated as expenses in the period in which the related products *are sold*.

Period Costs

Period costs are all the costs that are not product costs. *All selling and administrative expenses are treated as period costs*. For example, sales commissions, advertising, executive salaries, public relations, and the rental costs of administrative offices are all period costs. Period costs are not included as part of the cost of either purchased or manufactured goods; instead, period costs are expensed on the income statement in the period in which they are incurred using the usual rules of accrual accounting. Keep in mind that the period in which a cost is incurred is not necessarily the period in which cash changes hands. For example, as discussed earlier, the costs of liability insurance are spread across the periods that benefit from the insurance—regardless of the period in which the insurance premium is paid.

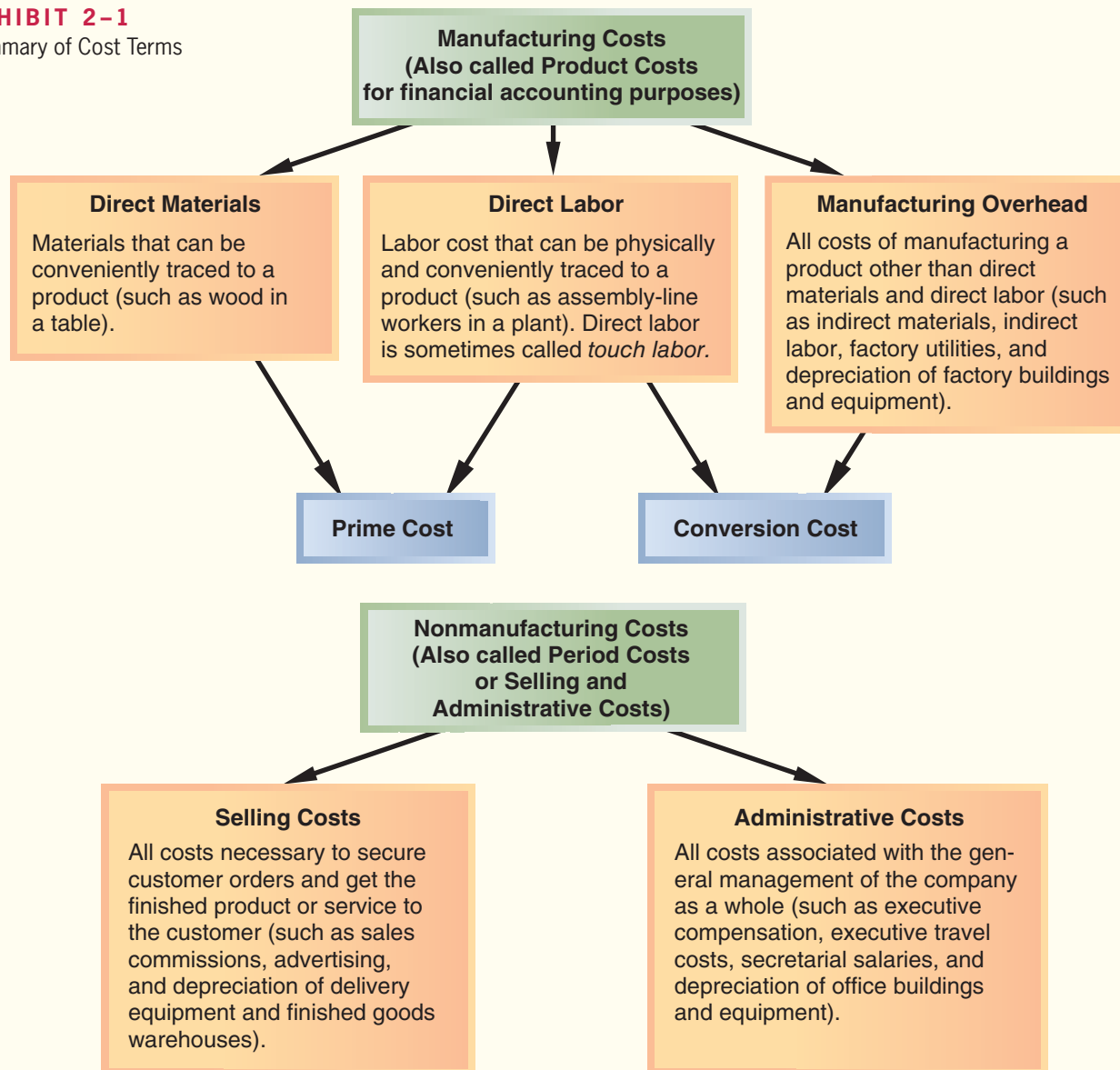
Prime Cost and Conversion Cost

Two more cost categories are often used in discussions of manufacturing costs—*prime cost* and *conversion cost*. **Prime cost** is the sum of direct materials cost and direct labor cost. **Conversion cost** is the sum of direct labor cost and manufacturing overhead cost. The term *conversion cost* is used to describe direct labor and manufacturing overhead because these costs are incurred to convert materials into the finished product.

Exhibit 2–1 contains a summary of the cost terms that we have introduced so far.

EXHIBIT 2-1

Summary of Cost Terms

**IN BUSINESS****THE CHALLENGES OF MANAGING CHARITABLE ORGANIZATIONS**

Charitable organizations, such as **Harlem Children's Zone**, **Sports4Kids**, and **Citizen Schools**, are facing a difficult situation. Many donors—aware of stories involving charities that spent excessively on themselves while losing sight of their mission—have started prohibiting their charity of choice from using donated funds to pay for administrative costs. However, even the most efficient charitable organizations find it difficult to expand without making additions to their infrastructure. For example, Sports4Kids' nationwide expansion of its sports programs drove up administrative costs from 5.6% to 14.7% of its total budget. The organization claims that this cost increase was necessary to build a more experienced management team to oversee the dramatically increased scale of operations.

Many charitable organizations are starting to seek gifts explicitly to fund administrative expenses. Their argument is simple—they cannot do good deeds for other people without incurring such costs.

Source: Rachel Emma Silverman and Sally Beatty, "Save the Children (But Pay the Bills, Too)," *The Wall Street Journal*, December 26, 2006, pp. D1–D2.

Cost Classifications for Predicting Cost Behavior

It is often necessary to predict how a certain cost will behave in response to a change in activity. For example, a manager at **Qwest**, a telephone company, may want to estimate the impact a 5 percent increase in long-distance calls by customers would have on Qwest's total electric bill. **Cost behavior** refers to how a cost reacts to changes in the level of activity. As the activity level rises and falls, a particular cost may rise and fall as well—or it may remain constant. For planning purposes, a manager must be able to anticipate which of these will happen; and if a cost can be expected to change, the manager must be able to estimate how much it will change. To help make such distinctions, costs are often categorized as *variable*, *fixed*, or *mixed*. The relative proportion of each type of cost in an organization is known as its **cost structure**. For example, an organization might have many fixed costs but few variable or mixed costs. Alternatively, it might have many variable costs but few fixed or mixed costs.

LEARNING OBJECTIVE 3

Understand cost behavior patterns including variable costs, fixed costs, and mixed costs.

Variable Cost

A **variable cost** varies, in total, in direct proportion to changes in the level of activity. Common examples of variable costs include cost of goods sold for a merchandising company, direct materials, direct labor, variable elements of manufacturing overhead, such as indirect materials, supplies, and power, and variable elements of selling and administrative expenses, such as commissions and shipping costs.¹

For a cost to be variable, it must be variable *with respect to something*. That “something” is its *activity base*. An **activity base** is a measure of whatever causes the incurrence of a variable cost. An activity base is sometimes referred to as a *cost driver*. Some of the most common activity bases are direct labor-hours, machine-hours, units produced, and units sold. Other examples of activity bases (cost drivers) include the number of miles driven by salespersons, the number of pounds of laundry cleaned by a hotel, the number of calls handled by technical support staff at a software company, and the number of beds occupied in a hospital. *While there are many activity bases within organizations, throughout this textbook, unless stated otherwise, you should assume that the activity base under consideration is the total volume of goods and services provided by the organization. We will specify the activity base only when it is something other than total output.*

IN BUSINESS

COST DRIVERS IN THE ELECTRONICS INDUSTRY

Accenture Ltd. estimates that the U.S. electronics industry spends \$13.8 billion annually to rebox, restock, and resell returned products. Conventional wisdom is that customers only return products when they are defective, but the data shows that this explanation only accounts for 5% of customer returns. The biggest cost drivers that cause product returns are that customers often inadvertently buy the wrong products and that they cannot understand how to use the products that they have purchased. Television manufacturer **Vizio Inc.** has started including more information on its packaging to help customers avoid buying the wrong product. **Seagate Technologies** is replacing thick instruction manuals with simpler guides that make it easier for customers to begin using their products.

Source: Christopher Lawton, “The War on Returns,” *The Wall Street Journal*, May 8, 2008, pp. D1 and D6.

To provide an example of a variable cost, consider Nooksack Expeditions, a small company that provides daylong whitewater rafting excursions on rivers in the North Cascade Mountains. The company provides all of the necessary equipment and experienced

¹ Direct labor costs often can be fixed instead of variable for a variety of reasons. For example, in some countries, such as France, Germany, and Japan, labor regulations and cultural norms may limit management's ability to adjust the labor force in response to changes in activity. In this textbook, always assume that direct labor is a variable cost unless you are explicitly told otherwise.

guides, and it serves gourmet meals to its guests. The meals are purchased from a caterer for \$30 a person for a daylong excursion. The behavior of this variable cost, on both a per unit and a total basis, is shown below:

Number of Guests	Cost of Meals per Guest	Total Cost of Meals
250	\$30	\$7,500
500	\$30	\$15,000
750	\$30	\$22,500
1,000	\$30	\$30,000

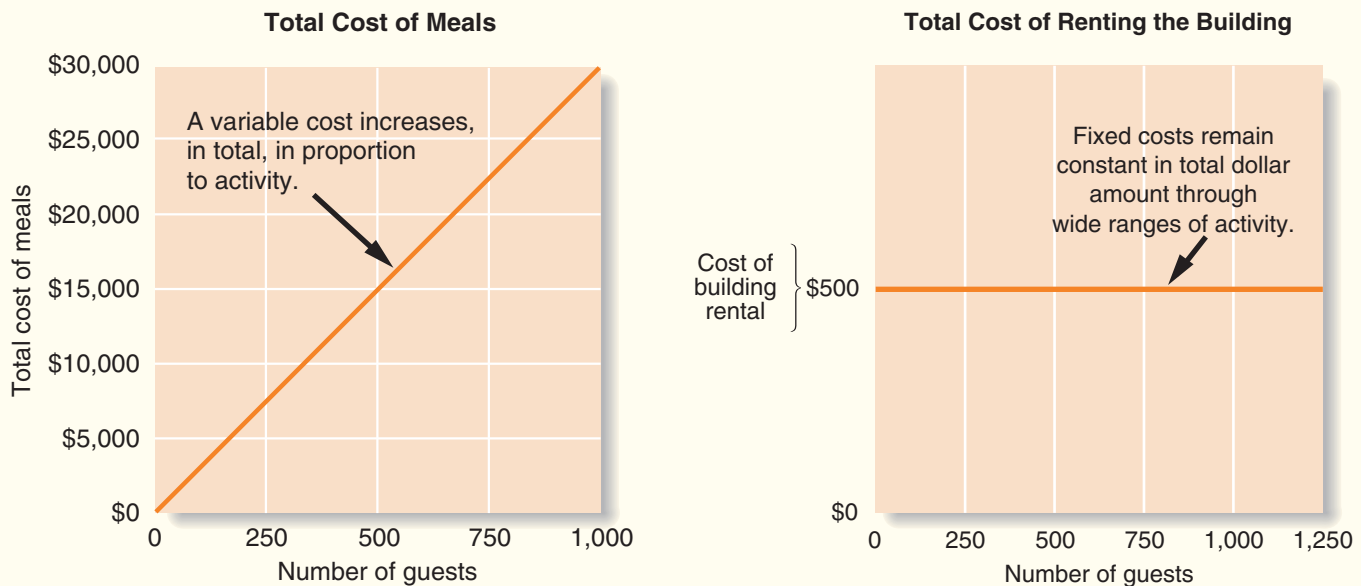
While total variable costs change as the activity level changes, it is important to note that a variable cost is constant if expressed on a *per unit* basis. For example, the per unit cost of the meals remains constant at \$30 even though the total cost of the meals increases and decreases with activity. The graph on the left-hand side of Exhibit 2–2 illustrates that the total variable cost rises and falls as the activity level rises and falls. At an activity level of 250 guests, the total meal cost is \$7,500. At an activity level of 1,000 guests, the total meal cost rises to \$30,000.

Fixed Cost

A **fixed cost** is a cost that remains constant, in total, regardless of changes in the level of activity. Examples of fixed costs include straight-line depreciation, insurance, property taxes, rent, supervisory salaries, administrative salaries, and advertising. Unlike variable costs, fixed costs are not affected by changes in activity. Consequently, as the activity level rises and falls, total fixed costs remain constant unless influenced by some outside force, such as a landlord increasing your monthly rental expense. To continue the Nooksack Expeditions example, assume the company rents a building for \$500 per month to store its equipment. The total amount of rent paid is the same regardless of the number of guests the company takes on its expeditions during any given month. The concept of a fixed cost is shown graphically on the right-hand side of Exhibit 2–2.

EXHIBIT 2–2

Variable and Fixed Cost Behavior



IN BUSINESS

FOOD COSTS AT A LUXURY HOTEL

The **Sporthotel Theresa** (<http://www.theresa.at/>), owned and operated by the Egger family, is a four star hotel located in Zell im Zillertal, Austria. The hotel features access to hiking, skiing, biking, and other activities in the Ziller alps as well as its own fitness facility and spa.

Three full meals a day are included in the hotel room charge. Breakfast and lunch are served buffet-style while dinner is a more formal affair with as many as six courses. The chef, Stefan Egger, believes that food costs are roughly proportional to the number of guests staying at the hotel; that is, they are a variable cost. He must order food from suppliers two or three days in advance, but he adjusts his purchases to the number of guests who are currently staying at the hotel and their consumption patterns. In addition, guests make their selections from the dinner menu early in the day, which helps Stefan plan which foodstuffs will be required for dinner. Consequently, he is able to prepare just enough food so that all guests are satisfied and yet waste is held to a minimum.

Source: Conversation with Stefan Egger, chef at the Sporthotel Theresa.



Because total fixed costs remain constant for large variations in the level of activity, the average fixed cost *per unit* becomes progressively smaller as the level of activity increases. If Nooksack Expeditions has only 250 guests in a month, the \$500 fixed rental cost would amount to an average of \$2 per guest. If there are 1,000 guests, the fixed rental cost would average only 50 cents per guest. The table below illustrates this aspect of the behavior of fixed costs. Note that as the number of guests increase, the average fixed cost per guest drops.

Monthly Rental Cost	Number of Guests	Average Cost per Guest
\$500	250	\$2.00
\$500	500	\$1.00
\$500	750	\$0.67
\$500	1,000	\$0.50

As a general rule, *we caution against expressing fixed costs on an average per unit basis in internal reports because it creates the false impression that fixed costs are like variable costs and that total fixed costs actually change as the level of activity changes.*

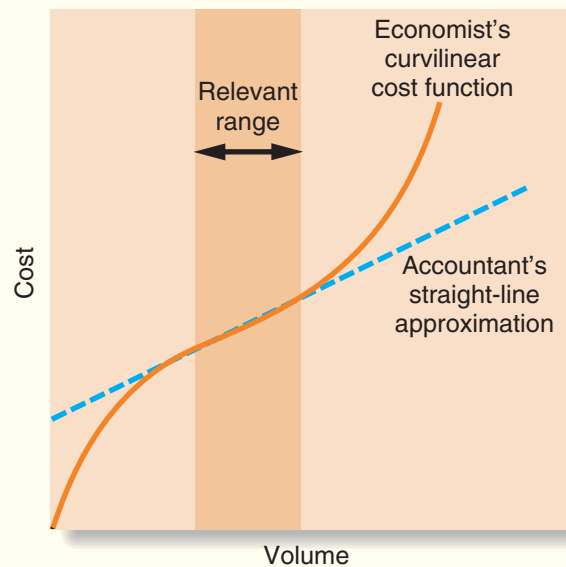
For planning purposes, fixed costs can be viewed as either *committed* or *discretionary*. **Committed fixed costs** represent organizational investments with a *multiyear* planning horizon that can't be significantly reduced even for short periods of time without making fundamental changes. Examples include investments in facilities and equipment, as well as real estate taxes, insurance expenses, and salaries of top management. Even if operations are interrupted or cut back, committed fixed costs remain largely unchanged in the short term because the costs of restoring them later are likely to be far greater than any short-run savings that might be realized. **Discretionary fixed costs** (often referred to as *managed fixed costs*) usually arise from *annual* decisions by management to spend on certain fixed cost items. Examples of discretionary fixed costs include advertising, research, public relations, management development programs, and internships for students. Discretionary fixed costs can be cut for short periods of time with minimal damage to the long-run goals of the organization.

The Linearity Assumption and the Relevant Range

Management accountants ordinarily assume that costs are strictly linear; that is, the relation between cost on the one hand and activity on the other can be represented by a straight line. Economists point out that many costs are actually curvilinear; that is, the

EXHIBIT 2-3

Curvilinear Costs and the Relevant Range

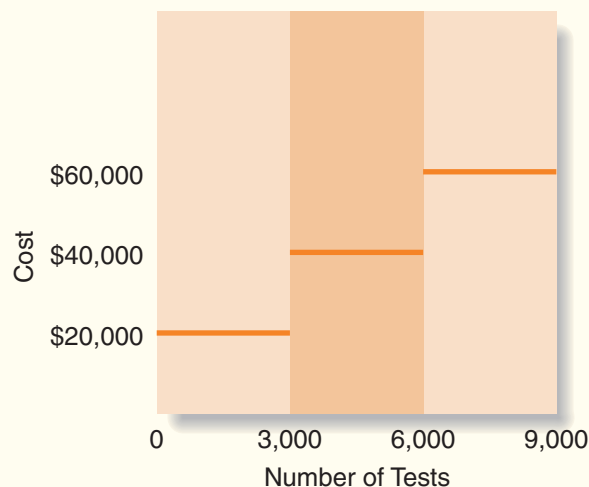


relation between cost and activity is a curve. Nevertheless, even if a cost is not strictly linear, it can be approximated within a narrow band of activity known as the *relevant range* by a straight line as illustrated in Exhibit 2-3. The **relevant range** is the range of activity within which the assumption that cost behavior is strictly linear is reasonably valid. Outside of the relevant range, a fixed cost may no longer be strictly fixed or a variable cost may not be strictly variable. Managers should always keep in mind that assumptions made about cost behavior may be invalid if activity falls outside of the relevant range.

The concept of the relevant range is important in understanding fixed costs. For example, suppose the Mayo Clinic rents a machine for \$20,000 per month that tests blood samples for the presence of leukemia cells. Furthermore, suppose that the capacity of the leukemia diagnostic machine is 3,000 tests per month. The assumption that the rent for the diagnostic machine is \$20,000 per month is only valid within the relevant range of 0 to 3,000 tests per month. If the Mayo Clinic needed to test 5,000 blood samples per month, then it would need to rent another machine for an additional \$20,000 per month. It would be difficult to rent half of a diagnostic machine; therefore, the step pattern depicted in Exhibit 2-4 is typical for such costs. This exhibit shows that the fixed rental expense

EXHIBIT 2-4

Fixed Costs and the Relevant Range



Cost	Behavior of the Cost (within the relevant range)	
	In Total	Per Unit
Variable cost	Total variable cost increases and decreases in proportion to changes in the activity level.	Variable cost per unit remains constant.
Fixed cost	Total fixed cost is not affected by changes in the activity level within the relevant range.	Fixed cost per unit decreases as the activity level rises and increases as the activity level falls.

EXHIBIT 2-5

Summary of Variable and Fixed Cost Behavior

is \$20,000 for a relevant range of 0 to 3,000 tests. The fixed rental expense increases to \$40,000 within the relevant range of 3,001 to 6,000 tests. The rental expense increases in discrete steps or increments of 3,000 tests, rather than increasing in a linear fashion per test.

This step-oriented cost behavior pattern can also be used to describe other costs, such as some labor costs. For example, salaried employee expenses can be characterized using a step pattern. Salaried employees are paid a fixed amount, such as \$40,000 per year, for providing the capacity to work a prespecified amount of time, such as 40 hours per week for 50 weeks a year (= 2,000 hours per year). In this example, the total salaried employee expense is \$40,000 within a relevant range of 0 to 2,000 hours of work. The total salaried employee expense increases to \$80,000 (or two employees) if the organization's work requirements expand to a relevant range of 2,001 to 4,000 hours of work. Cost behavior patterns such as salaried employees are often called *step-variable costs*. Step-variable costs can often be adjusted quickly as conditions change. Furthermore, the width of the steps for step-variable costs is generally so narrow that these costs can be treated essentially as variable costs for most purposes. The width of the steps for fixed costs, on the other hand, is so wide that these costs should be treated as entirely fixed within the relevant range.

Exhibit 2-5 summarizes four key concepts related to variable and fixed costs. Study it carefully before reading further.

HOW MANY GUIDES?

Majestic Ocean Kayaking, of Ucluelet, British Columbia, is owned and operated by Tracy Morben-Eeftink. The company offers a number of guided kayaking excursions ranging from three-hour tours of the Ucluelet harbor to six-day kayaking and camping trips in Clayoquot Sound. One of the company's excursions is a four-day kayaking and camping trip to The Broken Group Islands in the Pacific Rim National Park. Special regulations apply to trips in the park—including a requirement that one certified guide must be assigned for every five guests or fraction thereof. For example, a trip with 12 guests must have at least three certified guides. Guides are not salaried and are paid on a per-day basis. Therefore, the cost to the company of the guides for a trip is a step-variable cost rather than a fixed cost or a strictly variable cost. One guide is needed for 1 to 5 guests, two guides for 6 to 10 guests, three guides for 11 to 15 guests, and so on.

Sources: Tracy Morben-Eeftink, owner, Majestic Ocean Kayaking. For more information about the company, see www.oceankayaking.com.

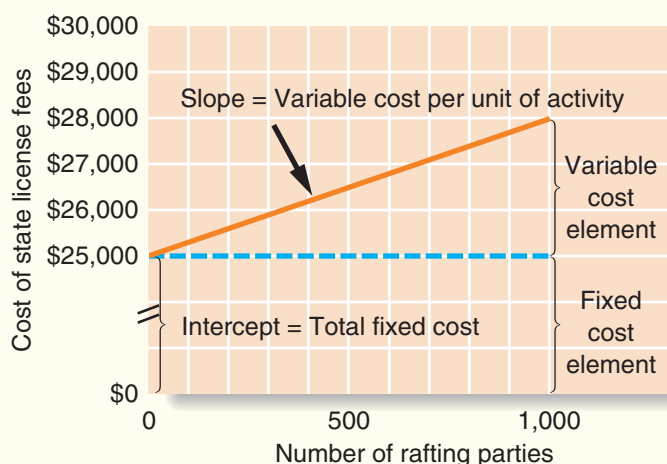
IN BUSINESS



Mixed Costs

A **mixed cost** contains both variable and fixed cost elements. Mixed costs are also known as semivariable costs. To continue the Nooksack Expeditions example, the company incurs a mixed cost called fees paid to the state. It includes a license fee of \$25,000 per year plus \$3 per rafting party paid to the state's Department of Natural Resources. If the

EXHIBIT 2-6
Mixed Cost Behavior



company runs 1,000 rafting parties this year, then the total fees paid to the state would be \$28,000, made up of \$25,000 in fixed cost plus \$3,000 in variable cost. Exhibit 2-6 depicts the behavior of this mixed cost.

Even if Nooksack fails to attract any customers, the company will still have to pay the license fee of \$25,000. This is why the cost line in Exhibit 2-6 intersects the vertical cost axis at the \$25,000 point. For each rafting party the company organizes, the total cost of the state fees will increase by \$3. Therefore, the total cost line slopes upward as the variable cost of \$3 per party is added to the fixed cost of \$25,000 per year.

Because the mixed cost in Exhibit 2-6 is represented by a straight line, the following equation for a straight line can be used to express the relationship between a mixed cost and the level of activity:

$$Y = a + bX$$

In this equation,

Y = The total mixed cost

a = The total fixed cost (the vertical intercept of the line)

b = The variable cost per unit of activity (the slope of the line)

X = The level of activity

Because the variable cost per unit equals the slope of the straight line, the steeper the slope, the higher the variable cost per unit.

In the case of the state fees paid by Nooksack Expeditions, the equation is written as follows:

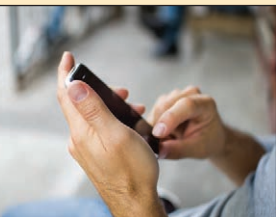
$$Y = \$25,000 + \$3.00X$$

↑ ↑ ↑ ↘
 Total Total Variable Activity
 mixed fixed cost per level
 cost cost unit of
 activity

This equation makes it easy to calculate the total mixed cost for any level of activity within the relevant range. For example, suppose that the company expects to organize 800 rafting parties in the next year. The total state fees would be calculated as follows:

$$\begin{aligned}
 Y &= \$25,000 + (\$3.00 \text{ per rafting party} \times 800 \text{ rafting parties}) \\
 &= \$27,400
 \end{aligned}$$

IN BUSINESS



THE ZIPCAR COMES TO COLLEGE CAMPUSES

Zipcar is a car sharing service based in Cambridge, Massachusetts. The company serves 13 cities and 120 university campuses. Members pay a \$50 annual fee plus \$7 an hour to rent a car. They can use their iPhones to rent a car, locate it in the nearest Zipcar parking lot, unlock it using an access code, and drive it off the lot. This mixed cost arrangement is attractive to customers who need a car infrequently and wish to avoid the large cash outlay that comes with buying or leasing a vehicle.

Source: Jefferson Graham, "An iPhone Gets Zipcar Drivers on Their Way," *USA Today*, September 30, 2009, p. 3B.

Traditional and Contribution Format Income Statements

LEARNING OBJECTIVE 5

Prepare income statements for a merchandising company using the traditional and contribution formats.

In this section of the chapter, we discuss how to prepare traditional and contribution format income statements for a merchandising company.² Merchandising companies do not manufacture the products that they sell to customers. For example, **Walmart** is a merchandising company because it buys finished products from manufacturers and then resells them to end consumers.

The Traditional Format Income Statement

Traditional income statements are prepared primarily for external reporting purposes. The left-hand side of Exhibit 2–12 shows a traditional income statement format for merchandising companies. This type of income statement organizes costs into two categories—cost of goods sold and selling and administrative expenses. Sales minus cost of goods sold equals the *gross margin*. The gross margin minus selling and administrative expenses equals net operating income.

The cost of goods sold reports the *product costs* attached to the merchandise sold during the period. The selling and administrative expenses report all *period costs* that

EXHIBIT 2-12

Comparing Traditional and Contribution Format Income Statements for Merchandising Companies (all numbers are given)

Traditional Format		Contribution Format	
Sales	\$12,000	Sales	\$12,000
Cost of goods sold*	6,000	Variable expenses:	
Gross margin	6,000	Cost of goods sold	\$6,000
Selling and administrative expenses:		Variable selling	600
Selling	\$3,100	Variable administrative	400
Administrative	1,900		7,000
Net operating income	\$ 1,000	Contribution margin	5,000
		Fixed expenses:	
		Fixed selling	2,500
		Fixed administrative	1,500
			4,000
		Net operating income	\$ 1,000

*For a manufacturing company, the cost of goods sold would include some variable costs, such as direct materials, direct labor, and variable overhead, and some fixed costs, such as fixed manufacturing overhead. Income statement formats for manufacturing companies will be explored in greater detail in a subsequent chapter.

² Subsequent chapters discuss the cost classifications used on the financial statements of manufacturing companies.

have been expensed as incurred. The cost of goods sold for a merchandising company can be computed directly by multiplying the number of units sold by their unit cost or indirectly using the equation below:

$$\text{Cost of goods sold} = \text{Beginning merchandise inventory} + \text{Purchases} - \text{Ending merchandise inventory}$$

For example, let's assume that the company depicted in Exhibit 2–12 purchased \$3,000 of merchandise inventory during the period and had beginning and ending merchandise inventory balances of \$7,000 and \$4,000, respectively. The equation above could be used to compute the cost of goods sold as follows:

$$\begin{aligned} \text{Cost of goods sold} &= \text{Beginning merchandise inventory} + \text{Purchases} - \text{Ending merchandise inventory} \\ &= \$7,000 + \$3,000 - \$4,000 \\ &= \$6,000 \end{aligned}$$

Although the traditional income statement is useful for external reporting purposes, it has serious limitations when used for internal purposes. It does not distinguish between fixed and variable costs. For example, under the heading “Selling and administrative expenses,” both variable administrative costs (\$400) and fixed administrative costs (\$1,500) are lumped together (\$1,900). Internally, managers need cost data organized by cost behavior to aid in planning, controlling, and decision making. The contribution format income statement has been developed in response to these needs.

The Contribution Format Income Statement

The crucial distinction between fixed and variable costs is at the heart of the **contribution approach** to constructing income statements. The unique thing about the contribution approach is that it provides managers with an income statement that clearly distinguishes between fixed and variable costs and therefore aids planning, controlling, and decision making. The right-hand side of Exhibit 2–12 shows a contribution format income statement for merchandising companies.

The contribution approach separates costs into fixed and variable categories, first deducting variable expenses from sales to obtain the *contribution margin*. For a merchandising company, cost of goods sold is a variable cost that gets included in the “Variable expenses” portion of the contribution format income statement. The **contribution margin** is the amount remaining from sales revenues after variable expenses have been deducted. This amount *contributes* toward covering fixed expenses and then toward profits for the period.

The contribution format income statement is used as an internal planning and decision-making tool. Its emphasis on cost behavior aids cost-volume-profit analysis (such as we shall be doing in a subsequent chapter), management performance appraisals, and budgeting. Moreover, the contribution approach helps managers organize data pertinent to numerous decisions such as product-line analysis, pricing, use of scarce resources, and make or buy analysis. All of these topics are covered in later chapters.

Cost Classifications for Assigning Costs to Cost Objects

Costs are assigned to cost objects for a variety of purposes including pricing, preparing profitability studies, and controlling spending. A **cost object** is anything for which cost data are desired—including products, customers, jobs, and organizational subunits. For purposes of assigning costs to cost objects, costs are classified as either *direct* or *indirect*.

LEARNING OBJECTIVE 6

Understand the differences between direct and indirect costs.

Direct Cost

A **direct cost** is a cost that can be easily and conveniently traced to a specified cost object. The concept of direct cost extends beyond just direct materials and direct labor. For example, if **Reebok** is assigning costs to its various regional and national sales offices, then the salary of the sales manager in its Tokyo office would be a direct cost of that office.

Indirect Cost

An **indirect cost** is a cost that cannot be easily and conveniently traced to a specified cost object. For example, a **Campbell Soup** factory may produce dozens of varieties of canned soups. The factory manager's salary would be an indirect cost of a particular variety such as chicken noodle soup. The reason is that the factory manager's salary is incurred as a consequence of running the entire factory—it is not incurred to produce any one soup variety. *To be traced to a cost object such as a particular product, the cost must be caused by the cost object.* The factory manager's salary is called a *common cost* of producing the various products of the factory. A **common cost** is a cost that is incurred to support a number of cost objects but cannot be traced to them individually. A common cost is a type of indirect cost.

A particular cost may be direct or indirect, depending on the cost object. While the Campbell Soup factory manager's salary is an *indirect* cost of manufacturing chicken noodle soup, it is a *direct* cost of the manufacturing division. In the first case, the cost object is chicken noodle soup. In the second case, the cost object is the entire manufacturing division.

Cost Classifications for Decision Making

LEARNING OBJECTIVE 7

Understand cost classifications used in making decisions: differential costs, opportunity costs, and sunk costs.

Costs are an important feature of many business decisions. In making decisions, it is essential to have a firm grasp of the concepts *differential cost*, *opportunity cost*, and *sunk cost*.

Differential Cost and Revenue

Decisions involve choosing between alternatives. In business decisions, each alternative will have costs and benefits that must be compared to the costs and benefits of the other available alternatives. A difference in costs between any two alternatives is known as a **differential cost**. A difference in revenues between any two alternatives is known as **differential revenue**.

A differential cost is also known as an **incremental cost**, although technically an incremental cost should refer only to an increase in cost from one alternative to another; decreases in cost should be referred to as *decremental costs*. Differential cost is a broader term, encompassing both cost increases (incremental costs) and cost decreases (decremental costs) between alternatives.

The accountant's differential cost concept can be compared to the economist's marginal cost concept. In speaking of changes in cost and revenue, the economist uses the terms *marginal cost* and *marginal revenue*. The revenue that can be obtained from selling one more unit of product is called marginal revenue, and the cost involved in producing one more unit of product is called marginal cost. The economist's marginal concept is basically the same as the accountant's differential concept applied to a single unit of output.

Differential costs can be either fixed or variable. To illustrate, assume that **Nature Way Cosmetics, Inc.**, is thinking about changing its marketing method from distribution

through retailers to distribution by a network of neighborhood sales representatives. Present costs and revenues are compared to projected costs and revenues in the following table:

	Retailer Distribution (present)	Sales Representatives (proposed)	Differential Costs and Revenues
Revenues (Variable)	\$700,000	\$800,000	\$100,000
Cost of goods sold (Variable)	350,000	400,000	50,000
Advertising (Fixed)	80,000	45,000	(35,000)
Commissions (Variable)	0	40,000	40,000
Warehouse depreciation (Fixed)	50,000	80,000	30,000
Other expenses (Fixed)	60,000	60,000	0
Total expenses	540,000	625,000	85,000
Net operating income	\$160,000	\$175,000	\$ 15,000

According to the above analysis, the differential revenue is \$100,000 and the differential costs total \$85,000, leaving a positive differential net operating income of \$15,000 under the proposed marketing plan.

The decision of whether Nature Way Cosmetics should stay with the present retail distribution or switch to sales representatives could be made on the basis of the net operating incomes of the two alternatives. As we see in the above analysis, the net operating income under the present distribution method is \$160,000, whereas the net operating income with sales representatives is estimated to be \$175,000. Therefore, using sales representatives is preferred because it would result in \$15,000 higher net operating income. Note that we would have arrived at exactly the same conclusion by simply focusing on the differential revenues, differential costs, and differential net operating income, which also show a \$15,000 advantage for sales representatives.

In general, only the differences between alternatives are relevant in decisions. Those items that are the same under all alternatives and that are not affected by the decision can be ignored. For example, in the Nature Way Cosmetics example above, the “Other expenses” category, which is \$60,000 under both alternatives, can be ignored because it has no effect on the decision. If it were removed from the calculations, the sales representatives would still be preferred by \$15,000. This is an extremely important principle in management accounting that we will revisit in later chapters.

Opportunity Cost

Opportunity cost is the potential benefit that is given up when one alternative is selected over another. To illustrate this important concept, consider the following examples:

Example 1 Vicki has a part-time job that pays \$200 per week while attending college. She would like to spend a week at the beach during spring break, and her employer has agreed to give her the time off, but without pay. The \$200 in lost wages would be an opportunity cost of taking the week off to be at the beach.

Example 2 Suppose that **Neiman Marcus** is considering investing a large sum of money in land that may be a site for a future store. Rather than invest the funds in land, the company could invest the funds in high-grade securities. The opportunity cost of buying the land is the investment income that could have been realized by purchasing the securities instead.



Understanding how products and services are costed is vital to managers because the way in which these costs are determined can have a substantial impact on reported profits, as well as on key management decisions.

A managerial costing system should provide cost data to help managers plan, control, and make decisions. Nevertheless, external financial reporting and tax reporting requirements often heavily influence how costs are accumulated and summarized on managerial reports. This is true of product costing. In this chapter we use *absorption costing* to determine product costs. In **absorption costing**, all manufacturing costs, both fixed and variable, are assigned to units of product—units are said to *fully absorb manufacturing costs*. In later chapters we look at alternatives to absorption costing such as variable costing and activity-based costing.

Most countries—including the United States—require some form of absorption costing for both external financial reports and for tax reports. In addition, the vast majority of companies throughout the world also use absorption costing in their management reports. Because absorption costing is the most common approach to product costing throughout the world, we discuss it first and then discuss the alternatives in subsequent chapters.

Job-Order Costing—An Overview

Under absorption costing, product costs include all manufacturing costs. Some manufacturing costs, such as direct materials, can be directly traced to particular products. For example, the cost of the airbags installed in a **Toyota** Camry can be easily traced to that particular auto. But what about manufacturing costs like factory rent? Such costs do not change from month to month, whereas the number and variety of products made in the factory may vary dramatically from one month to the next. Because these costs remain unchanged from month to month regardless of what products are made, they are clearly not caused by—and cannot be directly traced to—any particular product. Therefore, these types of costs are assigned to products and services by averaging across time and across products. The type of production process influences how this averaging is done.

Job-order costing is used in situations where many *different* products are produced each period. For example, a **Levi Strauss** clothing factory would typically make many different types of jeans for both men and women during a month. A particular order might consist of 1,000 boot-cut men's blue denim jeans, style number A312. This order of 1,000 jeans is called a *job*. In a job-order costing system, costs are traced and allocated to jobs and then the costs of the job are divided by the number of units in the job to arrive at an average cost per unit.

Other examples of situations where job-order costing would be used include large-scale construction projects managed by **Bechtel International**, commercial aircraft produced by **Boeing**, greeting cards designed and printed by **Hallmark**, and airline meals prepared by **LSG SkyChefs**. All of these examples are characterized by diverse outputs. Each Bechtel project is unique and different from every other—the company may be simultaneously constructing a dam in Zaire and a bridge in Indonesia. Likewise, each airline orders a different type of meal from LSG SkyChefs' catering service.

Job-order costing is also used extensively in service industries. For example, hospitals, law firms, movie studios, accounting firms, advertising agencies, and repair shops all use a variation of job-order costing to accumulate costs. Although the detailed example of job-order costing provided in the following section deals with a manufacturing company, the same basic concepts and procedures are used by many service organizations.



IS THIS REALLY A JOB?

VBT Bicycling Vacations of Bristol, Vermont, offers deluxe bicycling vacations in the United States, Canada, Europe, and other locations throughout the world. For example, the company offers a 10-day tour of the Puglia region of Italy—the “heel of the boot.” The tour price includes international airfare, 10 nights of lodging, most meals, use of a bicycle, and ground transportation as needed. Each tour is led by at least two local tour leaders, one of whom rides with the guests along the tour route. The other tour leader drives a “sag wagon” that carries extra water, snacks, and bicycle repair equipment and is available for a shuttle back to the hotel or up a hill. The sag wagon also transports guests’ luggage from one hotel to another.

Each specific tour can be considered a job. For example, Giuliano Astore and Debora Trippetti, two natives of Puglia, led a VBT tour with 17 guests over 10 days in late April. At the end of the tour, Giuliano submitted a report, a sort of job cost sheet, to VBT headquarters. This report detailed the on the ground expenses incurred for this specific tour, including fuel and operating costs for the van, lodging costs for the guests, the costs of meals provided to guests, the costs of snacks, the cost of hiring additional ground transportation as needed, and the wages of the tour leaders. In addition to these costs, some costs are paid directly by VBT in Vermont to vendors. The total cost incurred for the tour is then compared to the total revenue collected from guests to determine the gross profit for the tour.

Sources: Giuliano Astore and Gregg Marston, President, VBT Bicycling Vacations. For more information about VBT, see www.vbt.com.

IN BUSINESS



Job-Order Costing—An Example

To introduce job-order costing, we will follow a specific job as it progresses through the manufacturing process. This job consists of two experimental couplings that Yost Precision Machining has agreed to produce for Loops Unlimited, a manufacturer of roller coasters. Couplings connect the cars on the roller coaster and are a critical component in the performance and safety of the ride. Before we begin our discussion, recall from the previous chapter that companies generally classify manufacturing costs into three broad categories: (1) direct materials, (2) direct labor, and (3) manufacturing overhead. As we study the operation of a job-order costing system, we will see how each of these three types of costs is recorded and accumulated.

Yost Precision Machining is a small company in Michigan that specializes in fabricating precision metal parts that are used in a variety of applications ranging from deep-sea exploration vehicles to the inertial triggers in automobile air bags. The company’s top managers gather every morning at 8:00 A.M. in the company’s conference room for the daily planning meeting. Attending the meeting this morning are: Jean Yost, the company’s president; David Cheung, the marketing manager; Debbie Turner, the production manager; and Marc White, the company controller. The president opened the meeting:

Jean: The production schedule indicates we’ll be starting Job 2B47 today. Isn’t that the special order for experimental couplings, David?

David: That’s right. That’s the order from Loops Unlimited for two couplings for their new roller coaster ride for Magic Mountain.

Debbie: Why only two couplings? Don’t they need a coupling for every car?

David: Yes. But this is a completely new roller coaster. The cars will go faster and will be subjected to more twists, turns, drops, and loops than on any other existing roller coaster. To hold up under these stresses, Loops Unlimited’s engineers completely redesigned the cars and couplings. They want us to make just two of these new couplings for testing purposes. If the design works, then we’ll have the inside track on the order to supply couplings for the whole ride.

**MANAGERIAL
ACCOUNTING IN
ACTION**
The Issue



EXHIBIT 3-1

Materials Requisition Form

Materials Requisition Number		14873	Date	March 2
Job Number to Be Charged		2B47		
Department		Milling		
Description	Quantity	Unit Cost	Total Cost	
M46 Housing	2	\$124	\$248	
G7 Connector	4	\$103	412	
			\$660	

Jean: We agreed to take on this initial order at our cost just to get our foot in the door. Marc, will there be any problem documenting our cost so we can get paid?

Marc: No problem. The contract with Loops stipulates that they will pay us an amount equal to our cost of goods sold. With our job-order costing system, I can tell you the cost on the day the job is completed.

Jean: Good. Is there anything else we should discuss about this job at this time? No? Well then let's move on to the next item of business.

Measuring Direct Materials Cost

The blueprints submitted by Loops Unlimited indicate that each experimental coupling will require three parts that are classified as direct materials: two G7 Connectors and one M46 Housing. Each coupling requires two connectors and one housing, so to make two couplings, four connectors and two housings are required. This is a custom product that is being made for the first time, but if this were one of the company's standard products, it would have an established *bill of materials*. A **bill of materials** is a document that lists the type and quantity of each type of direct material needed to complete a unit of product.

When an agreement has been reached with the customer concerning the quantities, prices, and shipment date for the order, a *production order* is issued. The Production Department then prepares a *materials requisition form* similar to the form in Exhibit 3-1. The **materials requisition form** is a document that specifies the type and quantity of materials to be drawn from the storeroom and identifies the job that will be charged for the cost of the materials. The form is used to control the flow of materials into production and also for making entries in the accounting records.

The Yost Precision Machining materials requisition form in Exhibit 3-1 shows that the company's Milling Department has requisitioned two M46 Housings and four G7 Connectors for the Loops Unlimited job, which has been designated as Job 2B47.

Job Cost Sheet

After a production order has been issued, the Accounting Department's job-order costing software system automatically generates a *job cost sheet* like the one presented in Exhibit 3-2. A **job cost sheet** records the materials, labor, and manufacturing overhead costs charged to that job.

EXHIBIT 3-2

Job Cost Sheet

Measuring Direct Labor Cost

Today many companies rely on computerized systems (rather than paper and pencil) to maintain employee *time tickets*. A completed **time ticket** is an hour-by-hour summary of the employee's activities throughout the day. One computerized approach to creating time tickets uses bar codes to capture data. Each employee and each job has a unique bar code. When beginning work on a job, the employee scans three bar codes using a handheld device much like the bar code readers at grocery store checkout stands. The first bar code indicates that a job is being started; the second is the unique bar code on the employee's identity badge; and the third is the unique bar code of the job itself. This information is fed automatically via an electronic network to a computer that notes the time and records all of the data. When the task is completed, the employee scans a bar code indicating the task is complete, the bar code on his or her identity badge, and the bar code attached to the job. This information is relayed to the computer that again notes the

EXHIBIT 3-3

Employee Time Ticket

Time Ticket No.	843	Date	March 3
Employee	Mary Holden	Station	4

Started	Ended	Time Completed	Rate	Amount	Job Number
7:00	12:00	5.0	\$9	\$45	2B47
12:30	2:30	2.0	9	18	2B50
2:30	3:30	1.0	9	9	Maintenance
Totals		8.0		\$72	

time, and a time ticket, such as the one shown in Exhibit 3-3, is automatically prepared. Because all of the source data is already in computer files, the labor costs can be automatically posted to job cost sheets. For example, Exhibit 3-3 shows \$45 of direct labor cost related to Job 2B47. This amount is automatically posted to the job cost sheet shown in Exhibit 3-2. The time ticket in Exhibit 3-3 also shows \$9 of indirect labor costs related to performing maintenance. This cost is treated as part of manufacturing overhead and does not get posted on a job cost sheet.

IN BUSINESS**BUCKING THE TREND: USING PEOPLE INSTEAD OF MACHINES**

For decades, overhead costs have been going up and labor costs have been going down as companies have replaced people with machines. However, at the French automaker **Renault**, the exact opposite has been happening with its no-frills vehicle called the Logan. The Logan was intentionally stripped of costly elements and unnecessary technology so that the car could be sold for \$6,000 in emerging Eastern European markets. The car's simplified design enables Renault's manufacturing plant in Romania to assemble the car almost entirely with people instead of robots. The monthly pay for a line worker at Renault's Romanian plant is \$324 versus an average of more than \$4,700 per worker in Western European countries. Thanks in part to low-cost labor, Logan's production costs are estimated to be just \$1,089 per unit.

The Logan is finding buyers not only in emerging markets but also in more advanced Western European nations where customers have been clamoring for the car. Renault expects sales for the Logan to climb to one million vehicles—adding \$341 million to its profits.

Source: Gail Edmondson and Constance Faivre d'Arcier, "Got 5,000 Euros? Need a New Car?" *BusinessWeek*, July 4, 2005, p. 49.

LEARNING OBJECTIVE 1

Compute a predetermined overhead rate.

Computing Predetermined Overhead Rates

Recall that product costs include manufacturing overhead as well as direct materials and direct labor. Therefore, manufacturing overhead also needs to be recorded on the job cost sheet. However, assigning manufacturing overhead to a specific job involves some difficulties. There are three reasons for this:

1. Manufacturing overhead is an *indirect cost*. This means that it is either impossible or difficult to trace these costs to a particular product or job.

2. Manufacturing overhead consists of many different items ranging from the grease used in machines to the annual salary of the production manager.
3. Because of the fixed costs in manufacturing overhead, total manufacturing overhead costs tend to remain relatively constant from one period to the next even though the number of units produced can fluctuate widely. Consequently, the average cost per unit will vary from one period to the next.

Given these problems, allocation is used to assign overhead costs to products. Allocation is accomplished by selecting an *allocation base* that is common to all of the company's products and services. An **allocation base** is a measure such as direct labor-hours (DLH) or machine-hours (MH) that is used to assign overhead costs to products and services. The most widely used allocation bases in manufacturing are direct labor-hours, direct labor cost, machine-hours and (where a company has only a single product) units of product.

Manufacturing overhead is commonly assigned to products using a *predetermined overhead rate*. The **predetermined overhead rate** is computed by dividing the total estimated manufacturing overhead cost for the period by the estimated total amount of the allocation base as follows:

$$\text{Predetermined overhead rate} = \frac{\text{Estimated total manufacturing overhead cost}}{\text{Estimated total amount of the allocation base}}$$

The predetermined overhead rate is computed before the period begins using a four-step process. The first step is to estimate the total amount of the allocation base (the denominator) that will be required for next period's estimated level of production. The second step is to estimate the total fixed manufacturing overhead cost for the coming period and the variable manufacturing overhead cost per unit of the allocation base. The third step is to use the cost formula shown below to estimate the total manufacturing overhead cost (the numerator) for the coming period:

$$Y = a + bX$$

Where,

Y = The estimated total manufacturing overhead cost

a = The estimated total fixed manufacturing overhead cost

b = The estimated variable manufacturing overhead cost per unit of the allocation base

X = The estimated total amount of the allocation base

The fourth step is to compute the predetermined overhead rate. Notice, the estimated amount of the allocation base is determined before estimating the total manufacturing overhead cost. This needs to be done because total manufacturing overhead cost includes variable overhead costs that depend on the amount of the allocation base.

Applying Manufacturing Overhead

To repeat, the predetermined overhead rate is computed *before* the period begins. The predetermined overhead rate is then used to apply overhead cost to jobs throughout the period. The process of assigning overhead cost to jobs is called **overhead application**. The formula for determining the amount of overhead cost to apply to a particular job is:

$$\text{Overhead applied to a particular job} = \text{Predetermined overhead rate} \times \text{Amount of the allocation base incurred by the job}$$

For example, if the predetermined overhead rate is \$8 per direct labor-hour, then \$8 of overhead cost is *applied* to a job for each direct labor-hour incurred on the job. When the allocation base is direct labor-hours, the formula becomes:

$$\text{Overhead applied to a particular job} = \text{Predetermined overhead rate} \times \text{Actual direct labor-hours charged to the job}$$

LEARNING OBJECTIVE 2

Apply overhead cost to jobs using a predetermined overhead rate.

Manufacturing Overhead—A Closer Look

To illustrate the steps involved in computing and using a predetermined overhead rate, let's return to Yost Precision Machining and make the following assumptions. In step one, the company estimated that 40,000 direct labor-hours would be required to support the production planned for the year. In step two, it estimated \$220,000 of total fixed manufacturing overhead cost for the coming year and \$2.50 of variable manufacturing overhead cost per direct labor-hour. Given these assumptions, in step three the company used the cost formula shown below to estimate its total manufacturing overhead cost for the year:

$$Y = a + bX$$

$$Y = \$220,000 + (\$2.50 \text{ per direct labor-hour} \times 40,000 \text{ direct labor-hours})$$

$$Y = \$220,000 + \$100,000$$

$$Y = \$320,000$$

In step four, Yost Precision Machining computed its predetermined overhead rate for the year of \$8 per direct labor-hour as shown below:

$$\begin{aligned} \text{Predetermined overhead rate} &= \frac{\text{Estimated total manufacturing overhead cost}}{\text{Estimated total amount of the allocation base}} \\ &= \frac{\$320,000}{40,000 \text{ direct labor-hours}} \\ &= \$8 \text{ per direct labor-hour} \end{aligned}$$

The job cost sheet in Exhibit 3–4 indicates that 27 direct labor-hours (i.e., DLHs) were charged to Job 2B47. Therefore, a total of \$216 of manufacturing overhead cost would be applied to the job:

$$\begin{aligned} \text{Overhead applied to Job 2B47} &= \text{Predetermined overhead rate} \times \text{Actual direct labor-hours charged to Job 2B47} \\ &= \$8 \text{ per DLH} \times 27 \text{ DLHs} \\ &= \$216 \text{ of overhead applied to Job 2B47} \end{aligned}$$

This amount of overhead has been entered on the job cost sheet in Exhibit 3–4. Note that this is *not* the actual amount of overhead caused by the job. Actual overhead costs are *not* assigned to jobs—if that could be done, the costs would be direct costs, not overhead. The overhead assigned to the job is simply a share of the total overhead that was estimated at the beginning of the year. A **normal cost system**, which we have been describing, applies overhead to jobs by multiplying a predetermined overhead rate by the actual amount of the allocation base incurred by the jobs.

The Need for a Predetermined Rate

Instead of using a predetermined rate based on estimates, why not base the overhead rate on the *actual* total manufacturing overhead cost and the *actual* total amount of the allocation base incurred on a monthly, quarterly, or annual basis? If an actual rate is computed monthly or quarterly, seasonal factors in overhead costs or in the allocation base can produce fluctuations in the overhead rate. For example, the costs of heating and cooling a factory in Illinois will be highest in the winter and summer months and lowest in the spring and fall. If the overhead rate is recomputed at the end of each month or each quarter based on actual costs and activity, the overhead rate would go up in the winter and summer and down in the spring and fall. As a result, two identical jobs, one completed in the winter and one completed in the spring, would be assigned different manufacturing overhead costs. Many managers believe that such fluctuations in product costs serve

JOB COST SHEET

Job Number	2B47	Date Initiated	March 2
Department	Milling	Date Completed	March 8
Item	Special order coupling	Units Completed	2
For Stock			

Direct Materials		Direct Labor		Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate
14873	\$ 660	843	5	\$ 45	27	\$8/DLH
14875	506	846	8	60		
14912	238	850	4	21		
	\$1,404	851	10	54		
			27	\$180		

Cost Summary		Units Shipped		
		Date	Number	Balance
Direct Materials	\$ 1,404	March 8	—	2
Direct Labor	\$ 180			
Manufacturing Overhead	\$ 216			
Total Product Cost	\$ 1,800			
Unit Product Cost	\$ 900*			

*\$1,800 ÷ 2 units = \$900 per unit.

EXHIBIT 3-4

A Completed Job Cost Sheet

no useful purpose. To avoid such fluctuations, actual overhead rates could be computed on an annual or less-frequent basis. However, if the overhead rate is computed annually based on the actual costs and activity for the year, the manufacturing overhead assigned to any particular job would not be known until the end of the year. For example, the cost of Job 2B47 at Yost Precision Machining would not be known until the end of the year, even though the job will be completed and shipped to the customer in March. For these reasons, most companies use predetermined overhead rates rather than actual overhead rates in their cost accounting systems.

Choice of an Allocation Base for Overhead Cost

Ideally, the allocation base in the predetermined overhead rate should *drive* the overhead cost. A **cost driver** is a factor, such as machine-hours, beds occupied, computer time, or flight-hours, that causes overhead costs. If the base in the predetermined overhead rate does not “drive” overhead costs, product costs will be distorted. For example, if direct labor-hours is used to allocate overhead, but in reality overhead has little to do with direct labor-hours, then products with high direct labor-hour requirements will be overcosted.

Most companies use direct labor-hours or direct labor cost as the allocation base for manufacturing overhead. In the past, direct labor accounted for up to 60% of the cost of many products, with overhead cost making up only a portion of the remainder. This situation has changed for two reasons. First, sophisticated automated equipment has taken over functions that used to be performed by direct labor workers. Because the costs of acquiring and maintaining such equipment are classified as overhead, this increases overhead while decreasing direct labor. Second, products are becoming more

sophisticated and complex and are changed more frequently. This increases the need for highly skilled indirect workers such as engineers. As a result of these two trends, direct labor has decreased relative to overhead as a component of product costs.

In companies where direct labor and overhead costs have been moving in opposite directions, it would be difficult to argue that direct labor “drives” overhead costs. Accordingly, managers in some companies use *activity-based costing* principles to redesign their cost accounting systems. Activity-based costing is designed to more accurately reflect the demands that products, customers, and other cost objects make on overhead resources. The activity-based approach is discussed in more detail in Chapter 7.

Although direct labor may not be an appropriate allocation base in some industries, in others it continues to be a significant driver of manufacturing overhead. Indeed, most manufacturing companies in the United States continue to use direct labor as the primary or secondary allocation base for manufacturing overhead. The key point is that the allocation base used by the company should really drive, or cause, overhead costs, and direct labor is not always the most appropriate allocation base.

IN BUSINESS



REDUCING HEALTH-DAMAGING BEHAVIORS

Cianbro is an industrial construction company headquartered in Pittsfield, Maine, whose goal is “To be the healthiest company in America.” It introduced a corporate wellness program to attack employee behaviors that drive up health-care costs. The table below summarizes the number of employees in five health risk categories as of 2003 and 2005. The decreases in the number of employees in these high-risk categories are evidence that the wellness program was effective in helping employees make positive lifestyle changes. This should result in reduced health-care costs for the company.

Health Risk Category	Number of Employees		
	January 2003	March 2005	Decrease
Obesity	432	353	79
High cholesterol	637	515	122
Tobacco use	384	274	110
Inactivity	354	254	100
High blood pressure	139	91	48

Source: Cianbro, WELCOA's *Absolute Advantage* Magazine, 2006.

LEARNING OBJECTIVE 3

Compute the total cost and average cost per unit of a job.

Computation of Unit Costs

With the application of Yost Precision Machining’s \$216 of manufacturing overhead to the job cost sheet in Exhibit 3–4, the job cost sheet is complete except for two final steps. First, the totals for direct materials, direct labor, and manufacturing overhead are transferred to the Cost Summary section of the job cost sheet and added together to obtain the total cost for the job.¹ Then the total product cost (\$1,800) is divided by the number of units (2) to obtain the unit product cost (\$900). This unit product cost information is used for valuing unsold units in ending inventory and for determining cost of goods sold. As indicated earlier, *this unit product cost is an average cost and should not be interpreted as the cost that would actually be incurred if another unit were produced*. The incremental cost of an additional unit is something less than the average unit cost of \$900 because much of the actual overhead costs would not change if another unit were produced.

¹ Notice, we are assuming that Job 2B47 required direct materials and direct labor beyond the charges shown in Exhibits 3–1 and 3–3.

In the 8:00 A.M. daily planning meeting on March 9, Jean Yost, the president of Yost Precision Machining, once again drew attention to Job 2B47, the experimental couplings:

Jean: I see Job 2B47 is completed. Let's get those couplings shipped immediately to Loops Unlimited so they can get their testing program under way. Marc, how much are we going to bill Loops for those two units?

Marc: Because we agreed to sell the experimental couplings at cost, we will be charging Loops Unlimited just \$900 a unit.

Jean: Fine. Let's hope the couplings work out and we make some money on the big order later.

**MANAGERIAL
ACCOUNTING IN
ACTION**
The Wrap-up



IN BUSINESS

ONE-OF-A-KIND MASTERPIECE

In a true job-order costing environment, every job is unique. For example, **Purdey** manufactures 80–90 shotguns per year with each gun being a specially commissioned one-of-a-kind masterpiece. The prices start at \$110,000 because every detail is custom built, engraved, assembled, and polished by a skilled craftsman. The hand engraving can take months to complete and may add as much as \$100,000 to the price. The guns are designed to shoot perfectly straight and their value increases over time even with heavy use. One Purdey gun collector said “when I shoot my Purdeys I feel like an orchestra conductor waving my baton.”

Source: Eric Arnold, “Aim High,” *Forbes*, December 28, 2009, p. 86.

Job-Order Costing—The Flow of Costs

We are now ready to discuss the flow of costs through a job-order costing system. Exhibit 3–5 provides a conceptual overview of these cost flows. It highlights the fact that *product costs* flow through inventories on the balance sheet and then on to cost of goods sold in the income statement. More specifically, raw materials purchases are recorded in the *Raw Materials* inventory account. **Raw materials** include any materials that go into the final product. When raw materials are used in production, their costs are transferred to the *Work in Process* inventory account as direct materials.² **Work in process** consists of units of product that are only partially complete and will require further work before they are ready for sale to the customer. Notice that direct labor costs are added directly to Work in Process—they do not flow through Raw Materials inventory. Manufacturing overhead costs are applied to Work in Process by multiplying the predetermined overhead rate by the actual quantity of the allocation base consumed by each job.³ When goods are completed, their costs are transferred from Work in Process to *Finished Goods*. **Finished goods** consist of completed units of product that have not yet been sold to customers. The amount transferred from Work in Process to Finished Goods is referred to as the *cost of goods manufactured*. The **cost of goods manufactured** includes the manufacturing costs associated with the goods that were finished during the period. As goods are sold, their costs are transferred from Finished Goods to Cost of Goods Sold. At this point, the various costs required to make the product are finally recorded as an expense. Until that point, these costs are in inventory accounts on the balance sheet. Period costs (or selling and administrative expenses) do not flow through inventories on the balance sheet. They are recorded as expenses on the income statement in the period incurred.

LEARNING OBJECTIVE 4

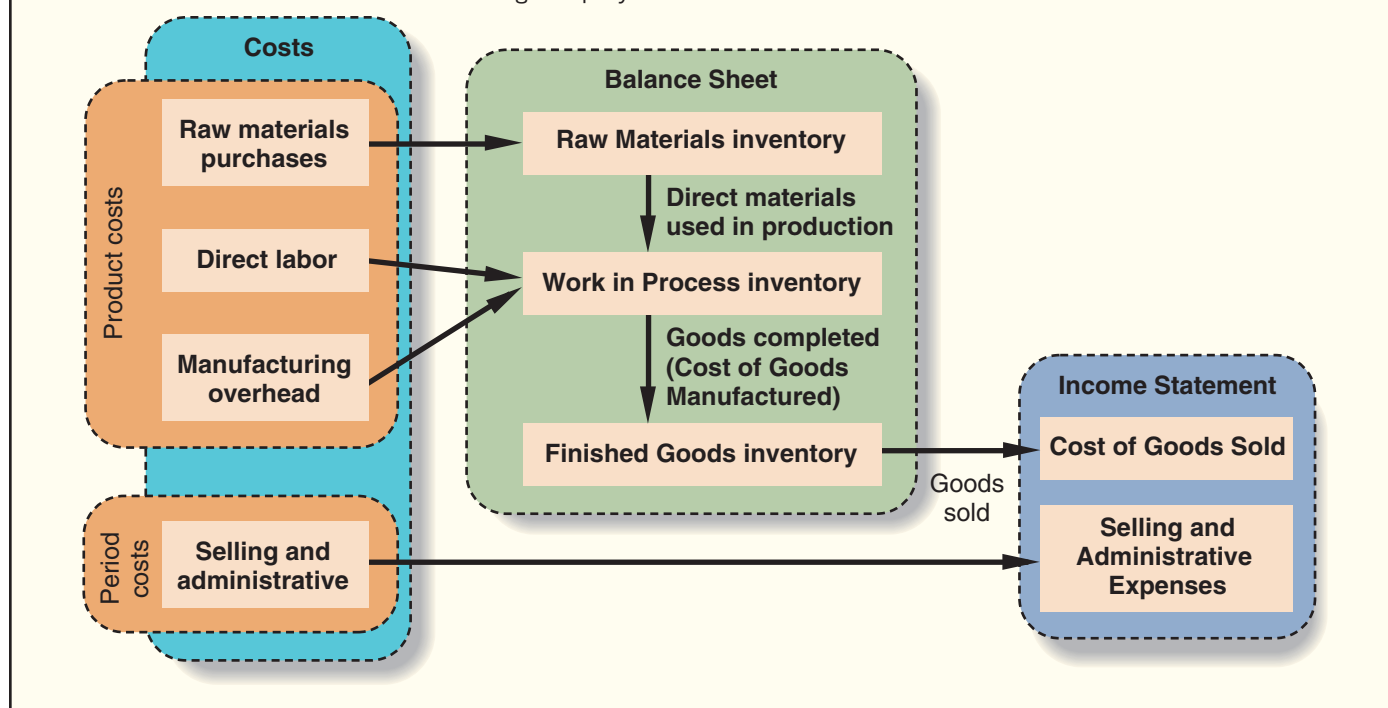
Understand the flow of costs in a job-order costing system and prepare appropriate journal entries to record costs.

² Indirect material costs are accounted for as part of manufacturing overhead.

³ For simplicity, Exhibit 3–5 assumes that Cost of Goods Sold does not need to be adjusted as discussed later in the chapter.

EXHIBIT 3-5

Cost Flows and Classifications in a Manufacturing Company



To illustrate the cost flows through a company's general ledger, we will consider a single month's activity at Ruger Corporation, a producer of gold and silver commemorative medallions. Ruger Corporation has two jobs in process during April, the first month of its fiscal year. Job A, a special minting of 1,000 gold medallions commemorating the invention of motion pictures, was started during March. By the end of March, \$30,000 in manufacturing costs had been recorded for the job. Job B, an order for 10,000 silver medallions commemorating the fall of the Berlin Wall, was started in April.

The Purchase and Issue of Materials

On April 1, Ruger Corporation had \$7,000 in raw materials on hand. During the month, the company purchased on account an additional \$60,000 in raw materials. The purchase is recorded in journal entry (1) below:

(1)		
Raw Materials	60,000	
Accounts Payable		60,000

As explained in the previous chapter, Raw Materials is an asset account. Thus, when raw materials are purchased, they are initially recorded as an asset—not as an expense.

Issue of Direct and Indirect Materials During April, \$52,000 in raw materials were requisitioned from the storeroom for use in production. These raw materials included \$50,000 of direct and \$2,000 of indirect materials. Entry (2) records issuing the materials to the production departments.

(2)		
Work in Process	50,000	
Manufacturing Overhead	2,000	
Raw Materials		52,000

The materials charged to Work in Process represent direct materials for specific jobs. These costs are also recorded on the appropriate job cost sheets. This point is illustrated in Exhibit 3–6, where \$28,000 of the \$50,000 in direct materials is charged to Job A's cost sheet and the remaining \$22,000 is charged to Job B's cost sheet. (In this example, all data are presented in summary form and the job cost sheet is abbreviated.)

The \$2,000 charged to Manufacturing Overhead in entry (2) represents indirect materials. Observe that the Manufacturing Overhead account is separate from the Work in Process account. The purpose of the Manufacturing Overhead account is to accumulate all manufacturing overhead costs as they are incurred during a period.

Before leaving Exhibit 3–6, we need to point out one additional thing. Notice from the exhibit that the job cost sheet for Job A contains a beginning balance of \$30,000. We stated earlier that this balance represents the cost of work done during March that has been carried forward to April. Also note that the Work in Process account contains the same \$30,000 balance. Thus, the Work in Process account summarizes all of the costs appearing on the job cost sheets of the jobs that are in process. Job A was the only job in process at the beginning of April, so the beginning balance in the Work in Process account equals Job A's beginning balance of \$30,000.

Labor Cost

In April, the employee time tickets included \$60,000 recorded for direct labor and \$15,000 for indirect labor. The following entry summarizes these costs:

(3)		
Work in Process	60,000	
Manufacturing Overhead	15,000	
Salaries and Wages Payable		75,000

Only the direct labor cost of \$60,000 is added to the Work in Process account. At the same time that direct labor costs are added to Work in Process, they are also added to the individual job cost sheets, as shown in Exhibit 3–7. During April, \$40,000 of direct labor cost was charged to Job A and the remaining \$20,000 was charged to Job B.

The labor costs charged to Manufacturing Overhead (\$15,000) represent the indirect labor costs of the period, such as supervision, janitorial work, and maintenance.

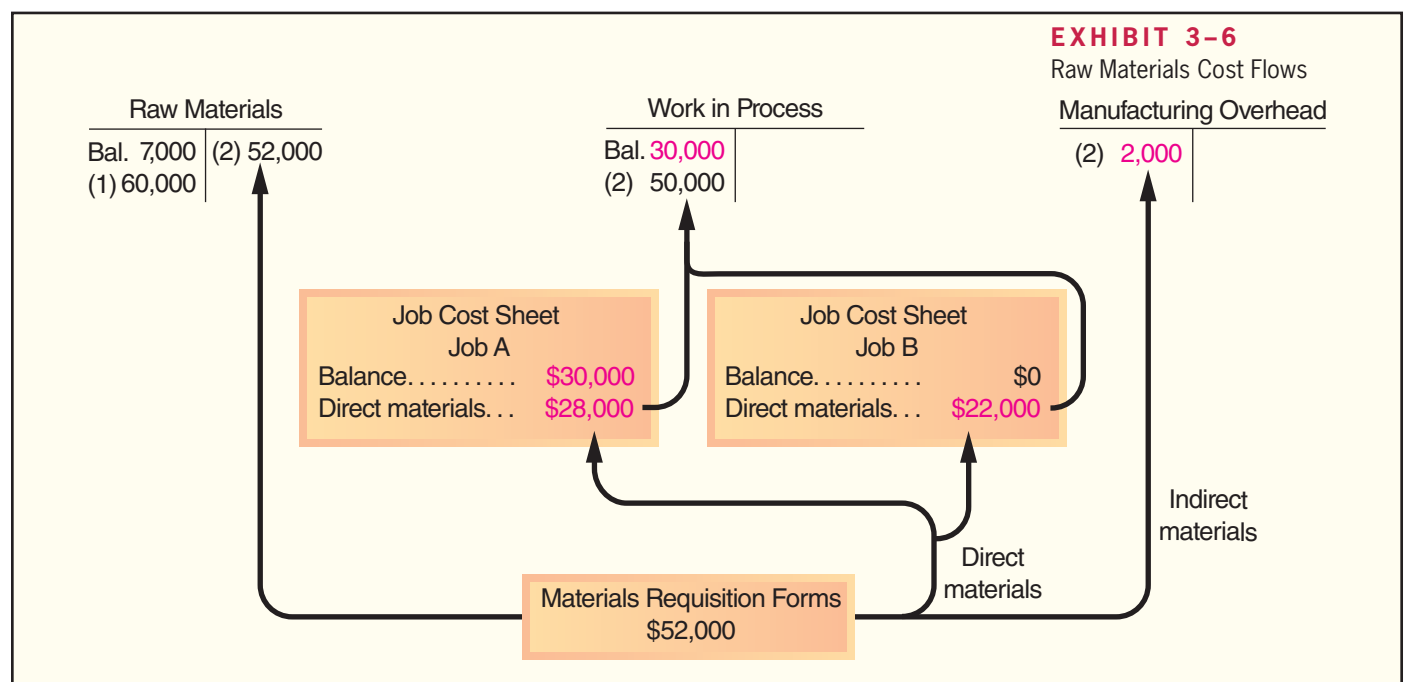
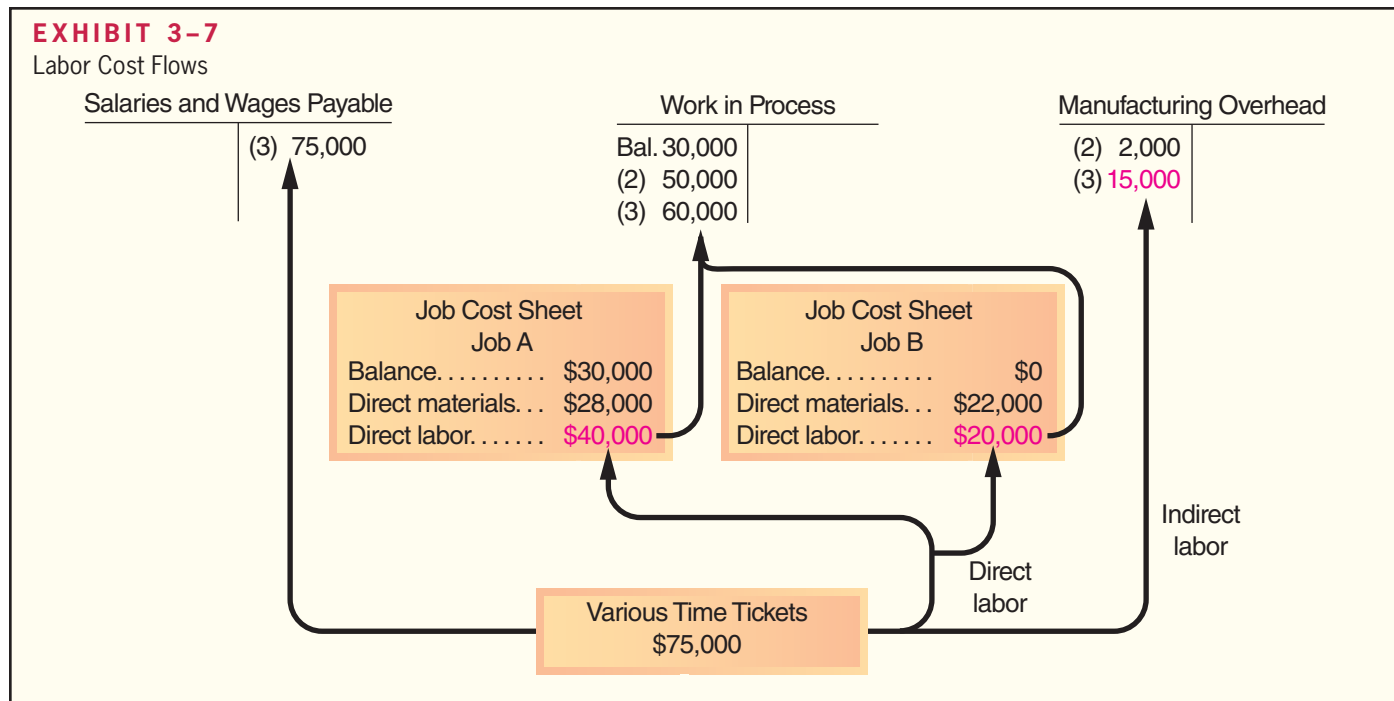


EXHIBIT 3-7

Labor Cost Flows

**Manufacturing Overhead Costs**

Recall that all manufacturing costs other than direct materials and direct labor are classified as manufacturing overhead costs. These costs are entered directly into the Manufacturing Overhead account as they are incurred. To illustrate, assume that Ruger Corporation incurred the following general factory costs during April:

Utilities (heat, water, and power)	\$21,000
Rent on factory equipment	16,000
Miscellaneous factory overhead costs	3,000
Total	\$40,000

The following entry records the incurrence of these costs:

(4)		
Manufacturing Overhead	40,000	
Accounts Payable*		40,000

*Accounts such as Cash may also be credited

In addition, assume that during April, Ruger Corporation recognized \$13,000 in accrued property taxes and that \$7,000 in prepaid insurance expired on factory buildings and equipment. The following entry records these items:

(5)		
Manufacturing Overhead	20,000	
Property Taxes Payable		13,000
Prepaid Insurance		7,000

Finally, assume that the company recognized \$18,000 in depreciation on factory equipment during April. The following entry records the accrual of this depreciation:

(6)		
Manufacturing Overhead	18,000	
Accumulated Depreciation		18,000

In short, manufacturing overhead costs are recorded directly into the Manufacturing Overhead account as they are incurred.

Applying Manufacturing Overhead

Because actual manufacturing costs are charged to the Manufacturing Overhead control account rather than to Work in Process, how are manufacturing overhead costs assigned to Work in Process? The answer is, by means of the predetermined overhead rate. Recall from our discussion earlier in the chapter that a predetermined overhead rate is established at the beginning of each year. The rate is calculated by dividing the estimated total manufacturing overhead cost for the year by the estimated total amount of the allocation base (measured in machine-hours, direct labor-hours, or some other base). The predetermined overhead rate is then used to apply overhead costs to jobs. For example, if machine-hours is the allocation base, overhead cost is applied to each job by multiplying the predetermined overhead rate by the number of machine-hours charged to the job.

To illustrate, assume that Ruger Corporation's predetermined overhead rate is \$6 per machine-hour. Also assume that during April, 10,000 machine-hours were worked on Job A and 5,000 machine-hours were worked on Job B (a total of 15,000 machine-hours). Thus, \$90,000 in overhead cost (\$6 per machine-hour $\times$ 15,000 machine-hours = \$90,000) would be applied to Work in Process. The following entry records the application of Manufacturing Overhead to Work in Process:

(7)	
Work in Process	90,000
Manufacturing Overhead	90,000

The flow of costs through the Manufacturing Overhead account is shown in Exhibit 3–8. The actual overhead costs on the debit side in the Manufacturing Overhead account in Exhibit 3–8 are the costs that were added to the account in entries (2)–(6). Observe that recording these actual overhead costs [entries (2)–(6)] and the application of overhead to Work in Process [entry (7)] represent two separate and entirely distinct processes.

The Concept of a Clearing Account The Manufacturing Overhead account operates as a clearing account. As we have noted, actual factory overhead costs are debited to the account as they are incurred throughout the year. When a job is completed (or at the end of an accounting period), overhead cost is applied to the job using the predetermined overhead rate, and Work in Process is debited and Manufacturing Overhead is credited. This sequence of events is illustrated below:

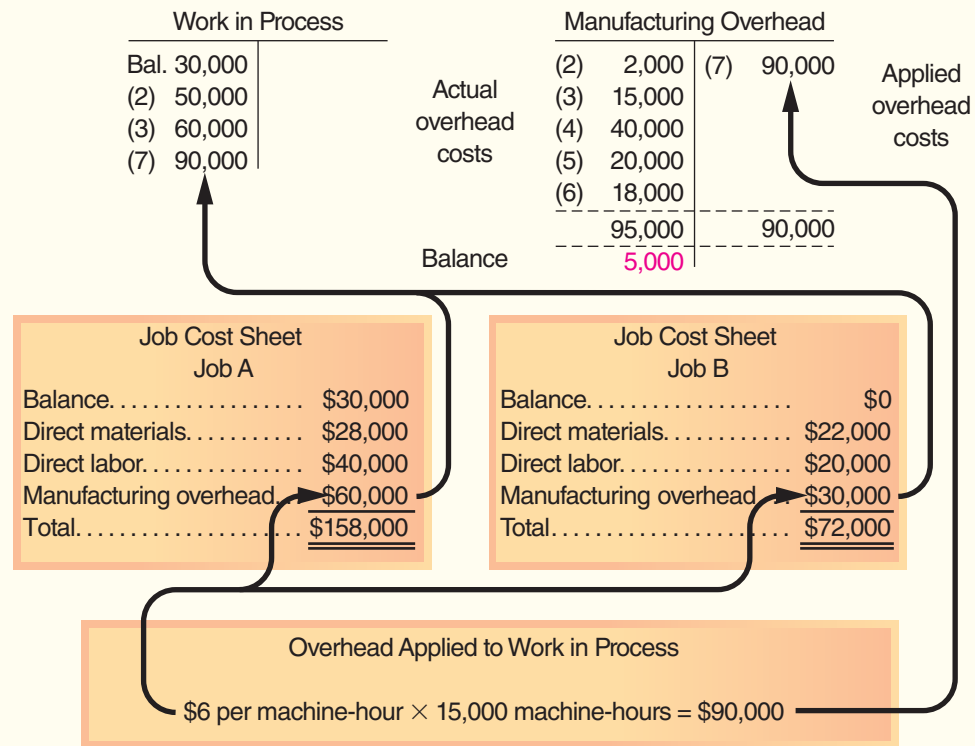
Manufacturing Overhead (a clearing account)	
Actual overhead costs are charged to this account as they are incurred throughout the period.	Overhead is applied to Work in Process using the predetermined overhead rate.

As we emphasized earlier, the predetermined overhead rate is based entirely on estimates of what the level of activity and overhead costs are *expected* to be, and it is established before the year begins. As a result, the overhead cost applied during a year will almost certainly turn out to be more or less than the actual overhead cost incurred. For example, notice from Exhibit 3–8 that Ruger Corporation's actual overhead costs for the period are \$5,000 greater than the overhead cost that has been applied to Work in Process, resulting in a **\$5,000** debit balance in the Manufacturing Overhead account. We will reserve discussion of what to do with this \$5,000 balance until later in the chapter.

For the moment, we can conclude from Exhibit 3–8 that the cost of a completed job consists of the actual direct materials cost of the job, the actual direct labor cost of the job, and the manufacturing overhead cost *applied* to the job. Pay particular attention to the following subtle but important point: *Actual overhead costs are not*

EXHIBIT 3-8

The Flow of Costs in Overhead Application



charged to jobs; actual overhead costs do not appear on the job cost sheet nor do they appear in the Work in Process account. Only the applied overhead cost, based on the predetermined overhead rate, appears on the job cost sheet and in the Work in Process account.

Nonmanufacturing Costs

In addition to manufacturing costs, companies also incur selling and administrative costs. These costs should be treated as period expenses and charged directly to the income statement. *Nonmanufacturing costs should not go into the Manufacturing Overhead account.* To illustrate the correct treatment of nonmanufacturing costs, assume that Ruger Corporation incurred \$30,000 in selling and administrative salary costs during April. The following entry summarizes the accrual of those salaries:

(8)			
Salaries Expense	30,000		
Salaries and Wages Payable		30,000	

Assume that depreciation on office equipment during April was \$7,000. The entry is as follows:

(9)			
Depreciation Expense	7,000		
Accumulated Depreciation		7,000	

Pay particular attention to the difference between this entry and entry (6) where we recorded depreciation on factory equipment. In journal entry (6), depreciation on factory equipment was debited to Manufacturing Overhead and is therefore a product cost. In journal entry (9) above, depreciation on office equipment is debited to Depreciation Expense. Depreciation on office equipment is a period expense rather than a product cost.

Finally, assume that advertising was \$42,000 and that other selling and administrative expenses in April totaled \$8,000. The following entry records these items:

(10)		
Advertising Expense	42,000	
Other Selling and Administrative Expense	8,000	
Accounts Payable*		50,000

*Other accounts, such as Cash may be credited.

The amounts in entries (8) through (10) are recorded directly into expense accounts—they have no effect on product costs. The same will be true of any other selling and administrative expenses incurred during April, including sales commissions, depreciation on sales equipment, rent on office facilities, insurance on office facilities, and related costs.

Cost of Goods Manufactured

When a job has been completed, the finished output is transferred from the production departments to the finished goods warehouse. By this time, the accounting department will have charged the job with direct materials and direct labor cost, and manufacturing overhead will have been applied using the predetermined overhead rate. A transfer of costs is made within the costing system that *parallels* the physical transfer of goods to the finished goods warehouse. The costs of the completed job are transferred out of the Work in Process account and into the Finished Goods account. The sum of all amounts transferred between these two accounts represents the cost of goods manufactured for the period.

In the case of Ruger Corporation, assume that Job A was completed during April. The following entry transfers the cost of Job A from Work in Process to Finished Goods:

(11)		
Finished Goods	158,000	
Work in Process		158,000

The \$158,000 represents the completed cost of Job A, as shown on the job cost sheet in Exhibit 3–8. Because Job A was the only job completed during April, the \$158,000 also represents the cost of goods manufactured for the month.

Job B was not completed by the end of the month, so its cost will remain in the Work in Process account and carry over to the next month. If a balance sheet is prepared at the end of April, the cost accumulated thus far on Job B will appear as the asset “Work in Process inventory.”

Cost of Goods Sold

As finished goods are shipped to customers, their accumulated costs are transferred from the Finished Goods account to the Cost of Goods Sold account. If an entire job is shipped at one time, then the entire cost appearing on the job cost sheet is transferred to the Cost of Goods Sold account. In most cases, however, only a portion of the units involved in a particular job will be immediately sold. In these situations, the unit product cost must be used to determine how much product cost should be removed from Finished Goods and charged to Cost of Goods Sold.

For Ruger Corporation, we will assume 750 of the 1,000 gold medallions in Job A were shipped to customers by the end of the month for total sales revenue of \$225,000. Because 1,000 units were produced and the total cost of the job from the job cost sheet was \$158,000, the unit product cost was \$158. The following journal entries would record the sale (all sales were on account):

(12)		
Accounts Receivable	225,000	
Sales		225,000

(13)		
Cost of Goods Sold	118,500	
Finished Goods		118,500
(750 units × \$158 per unit = \$118,500)		

LEARNING OBJECTIVE 5

Use T-accounts to show the flow of costs in a job-order costing system.

Entry (13) completes the flow of costs through the job-order costing system. To pull the entire Ruger Corporation example together, journal entries (1) through (13) are summarized in Exhibit 3–9. The flow of costs through the accounts is presented in T-account form in Exhibit 3–10.

EXHIBIT 3–9

Summary of Ruger Corporation
Journal Entries

(1)			
Raw Materials	60,000		
Accounts Payable		60,000	
(2)			
Work in Process	50,000		
Manufacturing Overhead	2,000		
Raw Materials		52,000	
(3)			
Work in Process	60,000		
Manufacturing Overhead	15,000		
Salaries and Wages Payable		75,000	
(4)			
Manufacturing Overhead	40,000		
Accounts Payable		40,000	
(5)			
Manufacturing Overhead	20,000		
Property Taxes Payable		13,000	
Prepaid Insurance		7,000	
(6)			
Manufacturing Overhead	18,000		
Accumulated Depreciation		18,000	
(7)			
Work in Process	90,000		
Manufacturing Overhead		90,000	
(8)			
Salaries Expense	30,000		
Salaries and Wages Payable		30,000	
(9)			
Depreciation Expense	7,000		
Accumulated Depreciation		7,000	
(10)			
Advertising Expense	42,000		
Other Selling and Administrative Expense	8,000		
Accounts Payable		50,000	
(11)			
Finished Goods	158,000		
Work in Process		158,000	
(12)			
Accounts Receivable	225,000		
Sales		225,000	
(13)			
Cost of Goods Sold	118,500		
Finished Goods		118,500	

EXHIBIT 3-10

Summary of Cost Flows—Ruger Corporation

Accounts Receivable		Accounts Payable		Capital Stock	
Bal.	XX		Bal.	XX	
(12)	225,000		(1)	60,000	
			(4)	40,000	
			(10)	50,000	
Prepaid Insurance		Salaries and Wages Payable		Retained Earnings	
Bal.	XX		Bal.	XX	
	(5)				
	7,000				
Raw Materials		Property Taxes Payable		Sales	
Bal.	7,000		Bal.	XX	
(1)	60,000		(3)	75,000	
	(2)		(8)	30,000	
	52,000				
Bal.	15,000				
Work in Process		Cost of Goods Sold		Salaries Expense	
Bal.	30,000		(13)	118,500	
(2)	50,000				
(3)	60,000				
(7)	90,000				
	(11)				
	158,000				
Bal.	72,000				
Finished Goods		Depreciation Expense		Advertising Expense	
Bal.	10,000		(9)	7,000	
(11)	158,000				
	(13)				
	118,500				
Bal.	49,500				
Accumulated Depreciation		Other Selling and Administrative Expense			
	Bal.		(10)	42,000	
	XX				
	(6)				
	18,000				
	(9)				
	7,000				
Manufacturing Overhead					
(2)	2,000				
(3)	15,000				
(4)	40,000				
(5)	20,000				
(6)	18,000				
	95,000				
	(7)				
	90,000				
Bal.	5,000				

Explanation of entries:

- (1) Raw materials purchased.
- (2) Direct and indirect materials issued into production.
- (3) Direct and indirect factory labor cost incurred.
- (4) Utilities and other factory costs incurred.
- (5) Property taxes and insurance incurred on the factory.
- (6) Depreciation recorded on factory assets.
- (7) Overhead cost applied to Work in Process.

- (8) Administrative salaries expense incurred.
- (9) Depreciation recorded on office equipment.
- (10) Advertising and other selling and administrative expense incurred.
- (11) Cost of goods manufactured transferred to finished goods.
- (12) Sale of Job A recorded.
- (13) Cost of goods sold recorded for Job A.

MANAGERIAL ACCOUNTING IN ACTION

The Issue



Cost-volume-profit (CVP) analysis is a powerful tool that helps managers understand the relationships among cost, volume, and profit. CVP analysis focuses on how profits are affected by the following five factors:

1. Selling prices.
2. Sales volume.
3. Unit variable costs.
4. Total fixed costs.
5. Mix of products sold.

Because CVP analysis helps managers understand how profits are affected by these key factors, it is a vital tool in many business decisions. These decisions include what products and services to offer, what prices to charge, what marketing strategy to use, and what cost structure to maintain. To help understand the role of CVP analysis in business decisions, consider the case of Acoustic Concepts, Inc., a company founded by Prem Narayan.

Prem, who was a graduate student in engineering at the time, started Acoustic Concepts to market a radical new speaker he had designed for automobile sound systems. The speaker, called the Sonic Blaster, uses an advanced microprocessor and proprietary software to boost amplification to awesome levels. Prem contracted with a Taiwanese electronics manufacturer to produce the speaker. With seed money provided by his family, Prem placed an order with the manufacturer and ran advertisements in auto magazines.

The Sonic Blaster was an almost immediate success, and sales grew to the point that Prem moved the company's headquarters out of his apartment and into rented quarters in a nearby industrial park. He also hired a receptionist, an accountant, a sales manager, and a small sales staff to sell the speakers to retail stores. The accountant, Bob Luchinni, had worked for several small companies where he had acted as a business advisor as well as accountant and bookkeeper. The following discussion occurred soon after Bob was hired:

Prem: Bob, I've got a lot of questions about the company's finances that I hope you can help answer.

Bob: We're in great shape. The loan from your family will be paid off within a few months.

Prem: I know, but I am worried about the risks I've taken on by expanding operations. What would happen if a competitor entered the market and our sales slipped? How far could sales drop without putting us into the red? Another question I've been trying to resolve is how much our sales would have to increase to justify the big marketing campaign the sales staff is pushing for.

Bob: Marketing always wants more money for advertising.

Prem: And they are always pushing me to drop the selling price on the speaker. I agree with them that a lower price will boost our sales volume, but I'm not sure the increased volume will offset the loss in revenue from the lower price.

Bob: It sounds like these questions are all related in some way to the relationships among our selling prices, our costs, and our volume. I shouldn't have a problem coming up with some answers.

Prem: Can we meet again in a couple of days to see what you have come up with?

Bob: Sounds good. By then I'll have some preliminary answers for you as well as a model you can use for answering similar questions in the future.

The Basics of Cost-Volume-Profit (CVP) Analysis

Bob Luchinni's preparation for his forthcoming meeting with Prem begins with the contribution income statement. The contribution income statement emphasizes the behavior of costs and therefore is extremely helpful to managers in judging the impact on profits

of changes in selling price, cost, or volume. Bob will base his analysis on the following contribution income statement he prepared last month:

Acoustic Concepts, Inc. Contribution Income Statement For the Month of June		
	Total	Per Unit
Sales (400 speakers)	\$100,000	\$250
Variable expenses	60,000	150
Contribution margin	40,000	\$100
Fixed expenses	35,000	
Net operating income	\$ 5,000	

Notice that sales, variable expenses, and contribution margin are expressed on a per unit basis as well as in total on this contribution income statement. The per unit figures will be very helpful to Bob in some of his calculations. Note that this contribution income statement has been prepared for management's use inside the company and would not ordinarily be made available to those outside the company.

Contribution Margin

Contribution margin is the amount remaining from sales revenue after variable expenses have been deducted. Thus, it is the amount available to cover fixed expenses and then to provide profits for the period. Notice the sequence here—contribution margin is used *first* to cover the fixed expenses, and then whatever remains goes toward profits. If the contribution margin is not sufficient to cover the fixed expenses, then a loss occurs for the period. To illustrate with an extreme example, assume that Acoustic Concepts sells only one speaker during a particular month. The company's income statement would appear as follows:

LEARNING OBJECTIVE 1

Explain how changes in activity affect contribution margin and net operating income.

Contribution Income Statement Sales of 1 Speaker		
	Total	Per Unit
Sales (1 speaker)	\$ 250	\$250
Variable expenses	150	150
Contribution margin	100	\$100
Fixed expenses	35,000	
Net operating loss	\$(34,900)	

For each additional speaker the company sells during the month, \$100 more in contribution margin becomes available to help cover the fixed expenses. If a second speaker is sold, for example, then the total contribution margin will increase by \$100 (to a total of \$200) and the company's loss will decrease by \$100, to \$34,800:

Contribution Income Statement Sales of 2 Speakers		
	Total	Per Unit
Sales (2 speakers)	\$ 500	\$250
Variable expenses	300	150
Contribution margin	200	\$100
Fixed expenses	35,000	
Net operating loss	\$(34,800)	

If enough speakers can be sold to generate \$35,000 in contribution margin, then all of the fixed expenses will be covered and the company will *break even* for the month—that is, it will show neither profit nor loss but just cover all of its costs. To reach the break-even point, the company will have to sell 350 speakers in a month because each speaker sold yields \$100 in contribution margin:

Contribution Income Statement Sales of 350 Speakers		
	Total	Per Unit
Sales (350 speakers)	\$87,500	\$250
Variable expenses	52,500	150
Contribution margin	35,000	\$100
Fixed expenses	35,000	
Net operating income	\$ 0	

Computation of the break-even point is discussed in detail later in the chapter; for the moment, note that the **break-even point** is the level of sales at which profit is zero.

Once the break-even point has been reached, net operating income will increase by the amount of the unit contribution margin for each additional unit sold. For example, if 351 speakers are sold in a month, then the net operating income for the month will be \$100 because the company will have sold 1 speaker more than the number needed to break even:

Contribution Income Statement Sales of 351 Speakers		
	Total	Per Unit
Sales (351 speakers)	\$87,750	\$250
Variable expenses	52,650	150
Contribution margin	35,100	\$100
Fixed expenses	35,000	
Net operating income	\$ 100	

If 352 speakers are sold (2 speakers above the break-even point), the net operating income for the month will be \$200. If 353 speakers are sold (3 speakers above the break-even point), the net operating income for the month will be \$300, and so forth. To estimate the profit at any sales volume above the break-even point, multiply the number of units sold in excess of the break-even point by the unit contribution margin. The result represents the anticipated profits for the period. Or, to estimate the effect of a planned increase in sales on profits, simply multiply the increase in units sold by the unit contribution margin. The result will be the expected increase in profits. To illustrate, if Acoustic Concepts is currently selling 400 speakers per month and plans to increase sales to 425 speakers per month, the anticipated impact on profits can be computed as follows:

Increased number of speakers to be sold	25
Contribution margin per speaker	× \$100
Increase in net operating income	\$2,500

These calculations can be verified as follows:

	Sales Volume		Difference (25 Speakers)	Per Unit
	400 Speakers	425 Speakers		
Sales (@ \$250 per speaker)	\$100,000	\$106,250	\$6,250	\$250
Variable expenses (@ \$150 per speaker)	60,000	63,750	3,750	150
Contribution margin	40,000	42,500	2,500	\$100
Fixed expenses	35,000	35,000	0	
Net operating income	\$ 5,000	\$ 7,500	\$2,500	

To summarize, if sales are zero, the company's loss would equal its fixed expenses. Each unit that is sold reduces the loss by the amount of the unit contribution margin. Once the break-even point has been reached, each additional unit sold increases the company's profit by the amount of the unit contribution margin.

CVP Relationships in Equation Form

The contribution format income statement can be expressed in equation form as follows:

$$\text{Profit} = (\text{Sales} - \text{Variable expenses}) - \text{Fixed expenses}$$

For brevity, we use the term *profit* to stand for net operating income in equations.

When a company has only a *single* product, as at Acoustic Concepts, we can further refine the equation as follows:

$$\text{Sales} = \text{Selling price per unit} \times \text{Quantity sold} = P \times Q$$

$$\text{Variable expenses} = \text{Variable expenses per unit} \times \text{Quantity sold} = V \times Q$$

$$\text{Profit} = (P \times Q - V \times Q) - \text{Fixed expenses}$$

We can do all of the calculations of the previous section using this simple equation. For example, on the previous page we computed that the net operating income (profit) at sales of 351 speakers would be \$100. We can arrive at the same conclusion using the above equation as follows:

$$\text{Profit} = (P \times Q - V \times Q) - \text{Fixed expenses}$$

$$\text{Profit} = (\$250 \times 351 - \$150 \times 351) - \$35,000$$

$$= (\$250 - \$150) \times 351 - \$35,000$$

$$= (\$100) \times 351 - \$35,000$$

$$= \$35,100 - \$35,000 = \$100$$

It is often useful to express the simple profit equation in terms of the unit contribution margin (Unit CM) as follows:

$$\text{Unit CM} = \text{Selling price per unit} - \text{Variable expenses per unit} = P - V$$

$$\text{Profit} = (P \times Q - V \times Q) - \text{Fixed expenses}$$

$$\text{Profit} = (P - V) \times Q - \text{Fixed expenses}$$

$$\text{Profit} = \text{Unit CM} \times Q - \text{Fixed expenses}$$

We could also have used this equation to determine the profit at sales of 351 speakers as follows:

$$\begin{aligned} \text{Profit} &= \text{Unit CM} \times Q - \text{Fixed expenses} \\ &= \$100 \times 351 - \$35,000 \\ &= \$35,100 - \$35,000 = \$100 \end{aligned}$$

For those who are comfortable with algebra, the quickest and easiest approach to solving the problems in this chapter may be to use the simple profit equation in one of its forms.

CVP Relationships in Graphic Form

LEARNING OBJECTIVE 2
Prepare and interpret a cost-volume-profit (CVP) graph and a profit graph.

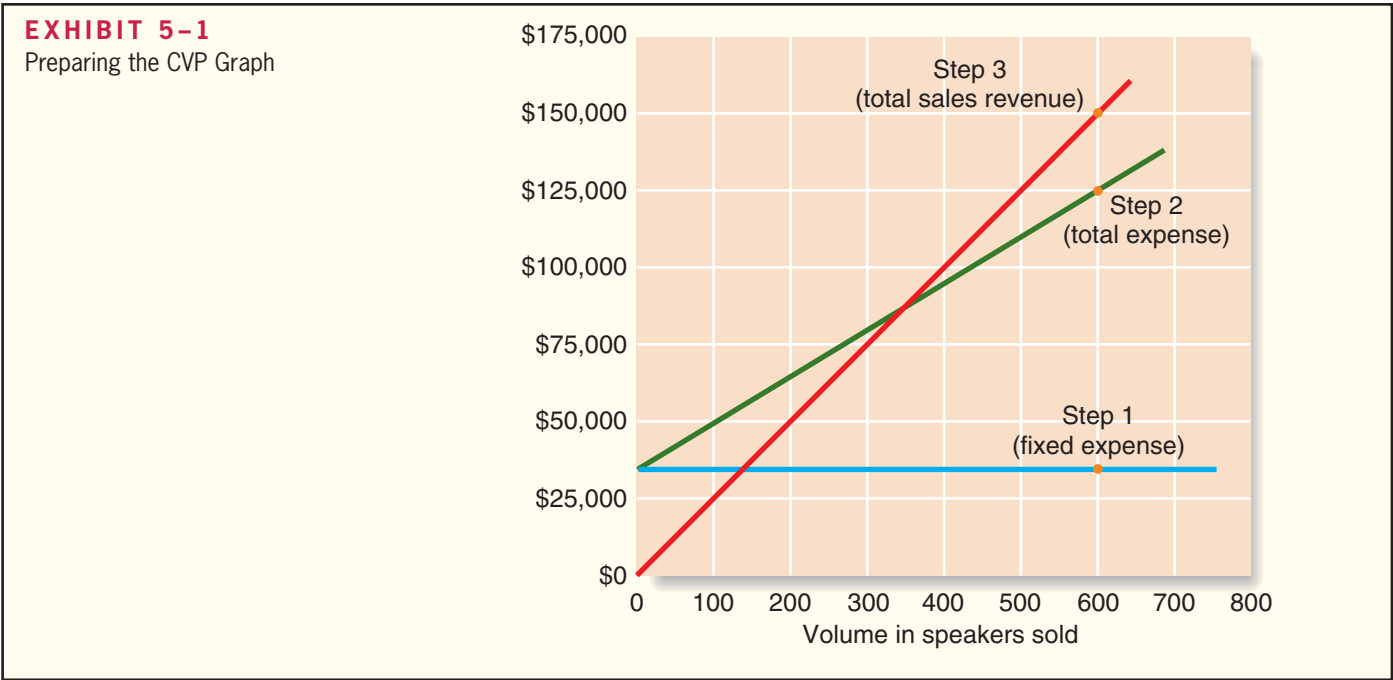
The relationships among revenue, cost, profit, and volume are illustrated on a **cost-volume-profit (CVP) graph**. A CVP graph highlights CVP relationships over wide ranges of activity. To help explain his analysis to Prem Narayan, Bob Luchinni prepared a CVP graph for Acoustic Concepts.

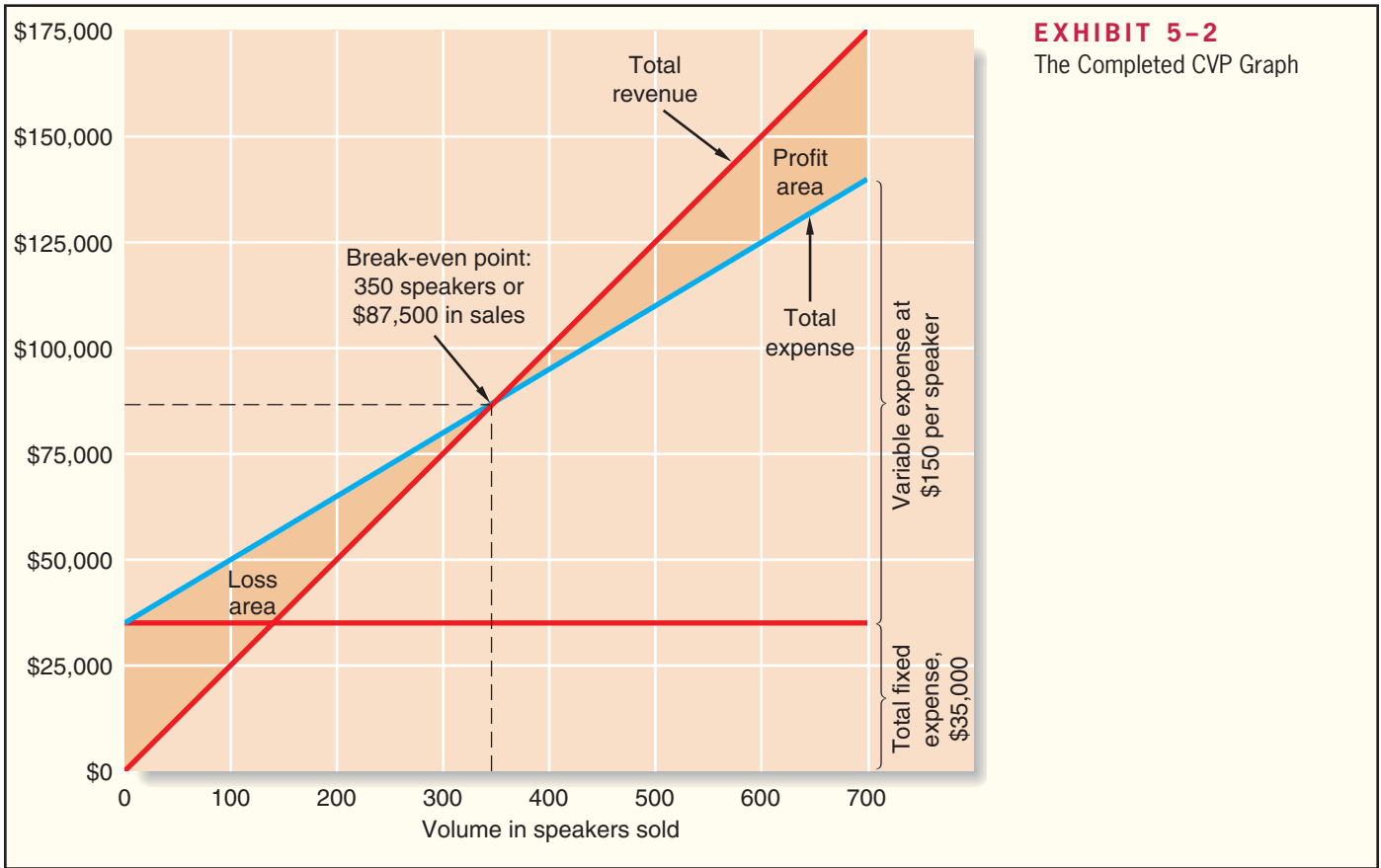
Preparing the CVP Graph In a CVP graph (sometimes called a *break-even chart*), unit volume is represented on the horizontal (X) axis and dollars on the vertical (Y) axis. Preparing a CVP graph involves the three steps depicted in Exhibit 5–1:

- 1. Draw a line parallel to the volume axis to represent total fixed expense. For Acoustic Concepts, total fixed expenses are \$35,000.
- 2. Choose some volume of unit sales and plot the point representing total expense (fixed and variable) at the sales volume you have selected. In Exhibit 5–1, Bob Luchinni chose a volume of 600 speakers. Total expense at that sales volume is:

Fixed expense	\$ 35,000
Variable expense (600 speakers × \$150 per speaker)	90,000
Total expense	<u>\$125,000</u>

After the point has been plotted, draw a line through it back to the point where the fixed expense line intersects the dollars axis.





- Again choose some sales volume and plot the point representing total sales dollars at the activity level you have selected. In Exhibit 5-1, Bob Luchinni again chose a volume of 600 speakers. Sales at that volume total \$150,000 (600 speakers $\times$ \$250 per speaker). Draw a line through this point back to the origin.

The interpretation of the completed CVP graph is given in Exhibit 5-2. The anticipated profit or loss at any given level of sales is measured by the vertical distance between the total revenue line (sales) and the total expense line (variable expense plus fixed expense).

The break-even point is where the total revenue and total expense lines cross. The break-even point of 350 speakers in Exhibit 5-2 agrees with the break-even point computed earlier.

As discussed earlier, when sales are below the break-even point—in this case, 350 units—the company suffers a loss. Note that the loss (represented by the vertical distance between the total expense and total revenue lines) gets bigger as sales decline. When sales are above the break-even point, the company earns a profit and the size of the profit (represented by the vertical distance between the total revenue and total expense lines) increases as sales increase.

An even simpler form of the CVP graph, which we call a profit graph, is presented in Exhibit 5-3. That graph is based on the following equation:

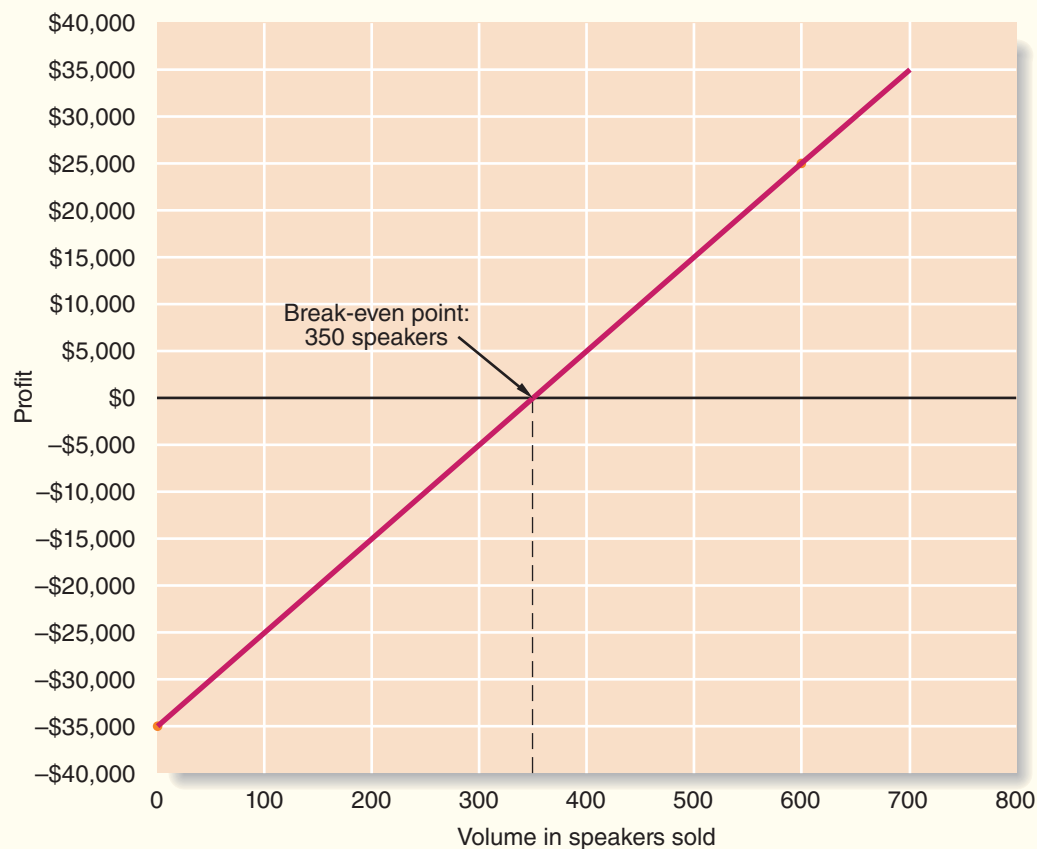
$$\text{Profit} = \text{Unit CM} \times Q - \text{Fixed expenses}$$

In the case of Acoustic Concepts, the equation can be expressed as:

$$\text{Profit} = \$100 \times Q - \$35,000$$

EXHIBIT 5-3

The Profit Graph



Because this is a linear equation, it plots as a single straight line. To plot the line, compute the profit at two different sales volumes, plot the points, and then connect them with a straight line. For example, when the sales volume is zero (i.e., $Q = 0$), the profit is $-\$35,000$ ($= \$100 \times 0 - \$35,000$). When Q is 600, the profit is $\$25,000$ ($= \$100 \times 600 - \$35,000$). These two points are plotted in Exhibit 5-3 and a straight line has been drawn through them.

The break-even point on the profit graph is the volume of sales at which profit is zero and is indicated by the dashed line on the graph. Note that the profit steadily increases to the right of the break-even point as the sales volume increases and that the loss becomes steadily worse to the left of the break-even point as the sales volume decreases.

Contribution Margin Ratio (CM Ratio)

LEARNING OBJECTIVE 3

Use the contribution margin ratio (CM ratio) to compute changes in contribution margin and net operating income resulting from changes in sales volume.

In the previous section, we explored how cost-volume-profit relationships can be visualized. In this section, we show how the *contribution margin ratio* can be used in cost-volume-profit calculations. As the first step, we have added a column to Acoustic Concepts' contribution format income statement in which sales revenues, variable expenses, and contribution margin are expressed as a percentage of sales:

	Total	Per Unit	Percent of Sales
Sales (400 speakers)	\$100,000	\$250	100%
Variable expenses	60,000	150	60%
Contribution margin	40,000	\$100	40%
Fixed expenses	35,000		
Net operating income	\$ 5,000		

The contribution margin as a percentage of sales is referred to as the **contribution margin ratio (CM ratio)**. This ratio is computed as follows:

$$\text{CM ratio} = \frac{\text{Contribution margin}}{\text{Sales}}$$

For Acoustic Concepts, the computations are:

$$\text{CM ratio} = \frac{\text{Total contribution margin}}{\text{Total sales}} = \frac{\$40,000}{\$100,000} = 40\%$$

In a company such as Acoustic Concepts that has only one product, the CM ratio can also be computed on a per unit basis as follows:

$$\text{CM ratio} = \frac{\text{Unit contribution margin}}{\text{Unit selling price}} = \frac{\$100}{\$250} = 40\%$$

The CM ratio shows how the contribution margin will be affected by a change in total sales. Acoustic Concepts' CM ratio of 40% means that for each dollar increase in sales, total contribution margin will increase by 40 cents (\$1 sales $\times$ CM ratio of 40%). Net operating income will also increase by 40 cents, assuming that fixed costs are not affected by the increase in sales. Generally, the effect of a change in sales on the contribution margin is expressed in equation form as:

$$\text{Change in contribution margin} = \text{CM ratio} \times \text{Change in sales}$$

As this illustration suggests, *the impact on net operating income of any given dollar change in total sales can be computed by applying the CM ratio to the dollar change*. For example, if Acoustic Concepts plans a \$30,000 increase in sales during the coming month, the contribution margin should increase by \$12,000 (\$30,000 increase in sales $\times$ CM ratio of 40%). As we noted above, net operating income will also increase by \$12,000 if fixed costs do not change. This is verified by the following table:

	Sales Volume			Percent of Sales
	Present	Expected	Increase	
Sales	\$100,000	\$130,000	\$30,000	100%
Variable expenses	60,000	78,000*	18,000	60%
Contribution margin	40,000	52,000	12,000	40%
Fixed expenses	35,000	35,000	0	
Net operating income	\$ 5,000	\$ 17,000	\$12,000	

*\$130,000 expected sales $\div$ \$250 per unit = 520 units. 520 units $\times$ \$150 per unit = \$78,000.

The relation between profit and the CM ratio can also be expressed using the following equation:

$$\text{Profit} = \text{CM ratio} \times \text{Sales} - \text{Fixed expenses}^1$$

For example, at sales of \$130,000, the profit is expected to be \$17,000 as shown below:

$$\begin{aligned} \text{Profit} &= \text{CM ratio} \times \text{Sales} - \text{Fixed expenses} \\ &= 0.40 \times \$130,000 - \$35,000 \\ &= \$52,000 - \$35,000 = \$17,000 \end{aligned}$$

¹ This equation can be derived using the basic profit equation and the definition of the CM ratio as follows:

$$\text{Profit} = (\text{Sales} - \text{Variable expenses}) - \text{Fixed expenses}$$

$$\text{Profit} = \text{Contribution margin} - \text{Fixed expenses}$$

$$\text{Profit} = \frac{\text{Contribution margin}}{\text{Sales}} \times \text{Sales} - \text{Fixed expenses}$$

$$\text{Profit} = \text{CM ratio} \times \text{Sales} - \text{Fixed expenses}$$

Again, if you are comfortable with algebra, this approach will often be quicker and easier than constructing contribution format income statements.

The CM ratio is particularly valuable in situations where the dollar sales of one product must be traded off against the dollar sales of another product. In this situation, products that yield the greatest amount of contribution margin per dollar of sales should be emphasized.

Some Applications of CVP Concepts

LEARNING OBJECTIVE 4

Show the effects on net operating income of changes in variable costs, fixed costs, selling price, and volume.

Bob Luchinni, the accountant at Acoustic Concepts, wanted to demonstrate to the company's president Prem Narayan how the concepts developed on the preceding pages can be used in planning and decision making. Bob gathered the following basic data:

	Per Unit	Percent of Sales
Selling price	\$250	100%
Variable expenses	150	60%
Contribution margin	<u>\$100</u>	<u>40%</u>

Recall that fixed expenses are \$35,000 per month. Bob Luchinni will use these data to show the effects of changes in variable costs, fixed costs, sales price, and sales volume on the company's profitability in a variety of situations.

Before proceeding further, however, we need to introduce another concept—the *variable expense ratio*. The **variable expense ratio** is the ratio of variable expenses to sales. It can be computed by dividing the total variable expenses by the total sales, or in a single product analysis, it can be computed by dividing the variable expenses per unit by the unit selling price. In the case of Acoustic Concepts, the variable expense ratio is 0.60; that is, variable expense is 60% of sales. Expressed as an equation, the definition of the variable expense ratio is:

$$\text{Variable expense ratio} = \frac{\text{Variable expenses}}{\text{Sales}}$$

This leads to a useful equation that relates the CM ratio to the variable expense ratio as follows:

$$\text{CM ratio} = \frac{\text{Contribution margin}}{\text{Sales}}$$

$$\text{CM ratio} = \frac{\text{Sales} - \text{Variable expenses}}{\text{Sales}}$$

$$\text{CM ratio} = 1 - \text{Variable expense ratio}$$

Change in Fixed Cost and Sales Volume Acoustic Concepts is currently selling 400 speakers per month at \$250 per speaker for total monthly sales of \$100,000. The sales manager feels that a \$10,000 increase in the monthly advertising budget would increase monthly sales by \$30,000 to a total of 520 units. Should the advertising budget be increased? The table on the next page shows the financial impact of the proposed change in the monthly advertising budget.

	Current Sales	Sales with Additional Advertising Budget	Difference	Percent of Sales
Sales	\$100,000	\$130,000	\$30,000	100%
Variable expenses	60,000	78,000*	18,000	60%
Contribution margin	40,000	52,000	12,000	40%
Fixed expenses	35,000	45,000†	10,000	
Net operating income	\$ 5,000	\$ 7,000	\$ 2,000	

*520 units × \$150 per unit = \$78,000.
†\$35,000 + additional \$10,000 monthly advertising budget = \$45,000.

Assuming no other factors need to be considered, the increase in the advertising budget should be approved because it would increase net operating income by \$2,000. There are two shorter ways to arrive at this solution. The first alternative solution follows:

Alternative Solution 1

Expected total contribution margin:	
\$130,000 × 40% CM ratio	\$52,000
Present total contribution margin:	
\$100,000 × 40% CM ratio	40,000
Incremental contribution margin	12,000
Change in fixed expenses:	
Less incremental advertising expense	10,000
Increased net operating income	\$ 2,000

Because in this case only the fixed costs and the sales volume change, the solution can also be quickly derived as follows:

Alternative Solution 2

Incremental contribution margin:	
\$30,000 × 40% CM ratio	\$12,000
Less incremental advertising expense	10,000
Increased net operating income	\$ 2,000

Notice that this approach does not depend on knowledge of previous sales. Also note that it is unnecessary under either shorter approach to prepare an income statement. Both of the alternative solutions involve **incremental analysis**—they consider only the revenue, cost, and volume that will change if the new program is implemented. Although in each case a new income statement could have been prepared, the incremental approach is simpler and more direct and focuses attention on the specific changes that would occur as a result of the decision.

Change in Variable Costs and Sales Volume Refer to the original data. Recall that Acoustic Concepts is currently selling 400 speakers per month. Prem is considering the use of higher-quality components, which would increase variable costs (and thereby reduce the contribution margin) by \$10 per speaker. However, the sales manager predicts that using higher-quality components would increase sales to 480 speakers per month. Should the higher-quality components be used?

The \$10 increase in variable costs would decrease the unit contribution margin by \$10—from \$100 down to \$90.

Solution

Expected total contribution margin with higher-quality components:	
480 speakers × \$90 per speaker	\$43,200
Present total contribution margin:	
400 speakers × \$100 per speaker	40,000
Increase in total contribution margin	<u>\$ 3,200</u>

According to this analysis, the higher-quality components should be used. Because fixed costs would not change, the \$3,200 increase in contribution margin shown above should result in a \$3,200 increase in net operating income.

IN BUSINESS

GROWING SALES AT AMAZON.COM

Amazon.com was deciding between two tactics for growing sales and profits. The first approach was to invest in television advertising. The second approach was to offer free shipping on larger orders. To evaluate the first option, Amazon.com invested in television ads in two markets—Minneapolis, Minnesota, and Portland, Oregon. The company quantified the profit impact of this choice by subtracting the increase in fixed advertising costs from the increase in contribution margin. The profit impact of television advertising paled in comparison to the free “super saver shipping” program, which the company introduced on orders over \$99. In fact, the free shipping option proved to be so popular and profitable that within two years Amazon.com dropped its qualifying threshold to \$49 and then again to a mere \$25. At each stage of this progression, Amazon.com used cost-volume-profit analysis to determine whether the extra volume from liberalizing the free shipping offer more than offset the associated increase in shipping costs.

Source: Rob Walker, “Because ‘Optimism is Essential,’ ” *Inc.* magazine, April 2004 pp. 149–150.

Change in Fixed Cost, Sales Price, and Sales Volume Refer to the original data and recall again that Acoustic Concepts is currently selling 400 speakers per month. To increase sales, the sales manager would like to cut the selling price by \$20 per speaker and increase the advertising budget by \$15,000 per month. The sales manager believes that if these two steps are taken, unit sales will increase by 50% to 600 speakers per month. Should the changes be made?

A decrease in the selling price of \$20 per speaker would decrease the unit contribution margin by \$20 down to \$80.

Solution

Expected total contribution margin with lower selling price:	
600 speakers × \$80 per speaker	\$48,000
Present total contribution margin:	
400 speakers × \$100 per speaker	40,000
Incremental contribution margin	8,000
Change in fixed expenses:	
Less incremental advertising expense	15,000
Reduction in net operating income	<u>\$ (7,000)</u>

According to this analysis, the changes should not be made. The \$7,000 reduction in net operating income that is shown above can be verified by preparing comparative income statements as shown on the next page.

	Present 400 Speakers per Month		Expected 600 Speakers per Month		Difference
	Total	Per Unit	Total	Per Unit	
Sales	\$100,000	\$250	\$138,000	\$230	\$38,000
Variable expenses	60,000	150	90,000	150	30,000
Contribution margin	40,000	\$100	48,000	\$ 80	8,000
Fixed expenses	35,000		50,000*		15,000
Net operating income (loss)	\$ 5,000		\$ (2,000)		\$ (7,000)

*35,000 + Additional monthly advertising budget of \$15,000 = \$50,000.

Change in Variable Cost, Fixed Cost, and Sales Volume Refer to Acoustic Concepts' original data. As before, the company is currently selling 400 speakers per month. The sales manager would like to pay salespersons a sales commission of \$15 per speaker sold, rather than the flat salaries that now total \$6,000 per month. The sales manager is confident that the change would increase monthly sales by 15% to 460 speakers per month. Should the change be made?

Solution Changing the sales staff's compensation from salaries to commissions would affect both fixed and variable expenses. Fixed expenses would decrease by \$6,000, from \$35,000 to \$29,000. Variable expenses per unit would increase by \$15, from \$150 to \$165, and the unit contribution margin would decrease from \$100 to \$85.

Expected total contribution margin with sales staff on commissions:	
460 speakers × \$85 per speaker	\$39,100
Present total contribution margin:	
400 speakers × \$100 per speaker	40,000
Decrease in total contribution margin	(900)
Change in fixed expenses:	
Add salaries avoided if a commission is paid	6,000
Increase in net operating income	\$ 5,100

According to this analysis, the changes should be made. Again, the same answer can be obtained by preparing comparative income statements:

	Present 400 Speakers per Month		Expected 460 Speakers per Month		Difference
	Total	Per Unit	Total	Per Unit	
Sales	\$100,000	\$250	\$115,000	\$250	\$15,000
Variable expenses	60,000	150	75,900	165	15,900
Contribution margin	40,000	\$100	39,100	\$ 85	900
Fixed expenses	35,000		29,000		(6,000)*
Net operating income ...	\$ 5,000		\$ 10,100		\$ 5,100

*Note: A *reduction* in fixed expenses has the effect of *increasing* net operating income.

Change in Selling Price Refer to the original data where Acoustic Concepts is currently selling 400 speakers per month. The company has an opportunity to make a bulk sale of 150 speakers to a wholesaler if an acceptable price can be negotiated. This sale would not disturb the company’s regular sales and would not affect the company’s total fixed expenses. What price per speaker should be quoted to the wholesaler if Acoustic Concepts is seeking a profit of \$3,000 on the bulk sale?

Solution

Variable cost per speaker	\$150
Desired profit per speaker:	
\$3,000 ÷ 150 speakers	20
Quoted price per speaker	<u>\$170</u>

Notice that fixed expenses are not included in the computation. This is because fixed expenses are not affected by the bulk sale, so all of the additional contribution margin increases the company’s profits.

IN BUSINESS



MANAGING RISK IN THE BOOK PUBLISHING INDUSTRY

Greenleaf Book Group is a book publishing company in Austin, Texas, that attracts authors who are willing to pay publishing costs and forgo up-front advances in exchange for a larger royalty rate on each book sold. For example, assume a typical publisher prints 10,000 copies of a new book that it sells for \$12.50 per unit. The publisher pays the author an advance of \$20,000 to write the book and then incurs \$60,000 of expenses to market, print, and edit the book. The publisher also pays the author a 20% royalty (or \$2.50 per unit) on each book sold above 8,000 units. In this scenario, the publisher must sell 6,400 books to break even (= \$80,000 in fixed costs ÷ \$12.50 per unit). If all 10,000 copies are sold, the author earns \$25,000 (= \$20,000 advance + 2,000 copies × \$2.50) and the publisher earns \$40,000 (= \$125,000 – \$60,000 – \$20,000 – \$5,000).

Greenleaf alters the financial arrangement described above by requiring the author to assume the risk of poor sales. It pays the author a 70% royalty on all units sold (or \$8.75 per unit), but the author forgoes the \$20,000 advance and pays Greenleaf \$60,000 to market, print, and edit the book. If the book flops, the author fails to recover her production costs. If all 10,000 units are sold, the author earns \$27,500 (= \$10,000 units × \$8.75 – \$60,000) and Greenleaf earns \$37,500 (= 10,000 units × (\$12.50 – \$8.75)).

Source: Christopher Steiner, “Book It,” *Forbes*, September 7, 2009, p. 58.

Target Profit and Break-Even Analysis

Target profit analysis and break-even analysis are used to answer questions such as how much would we have to sell to make a profit of \$10,000 per month or how much would we have to sell to avoid incurring a loss?

Target Profit Analysis

LEARNING OBJECTIVE 5
Determine the level of sales needed to achieve a desired target profit.

Target profit analysis is one of the key uses of CVP analysis. In **target profit analysis**, we estimate what sales volume is needed to achieve a specific target profit. For example, suppose that Prem Narayan of Acoustic Concepts would like to know what sales would have to be to attain a target profit of \$40,000 per month. To answer this question, we can proceed using the equation method or the formula method.

The Equation Method We can use a basic profit equation to find the sales volume required to attain a target profit. In the case of Acoustic Concepts, the company has only one product so we can use the contribution margin form of the equation. Remembering that the target profit is \$40,000, the unit contribution margin is \$100, and the fixed expense is \$35,000, we can solve as follows:

$$\begin{aligned}\text{Profit} &= \text{Unit CM} \times Q - \text{Fixed expense} \\ \$40,000 &= \$100 \times Q - \$35,000 \\ \$100 \times Q &= \$40,000 + \$35,000 \\ Q &= (\$40,000 + \$35,000) \div \$100 \\ Q &= 750\end{aligned}$$

Thus, the target profit can be achieved by selling 750 speakers per month.

The Formula Method The formula method is a short-cut version of the equation method. Note that in the next to the last line of the above solution, the sum of the target profit of \$40,000 and the fixed expense of \$35,000 is divided by the unit contribution margin of \$100. In general, in a single-product situation, we can compute the sales volume required to attain a specific target profit using the following formula:

$$\text{Unit sales to attain the target profit} = \frac{\text{Target profit} + \text{Fixed expenses}^2}{\text{Unit CM}}$$

In the case of Acoustic Concepts, the formula yields the following answer:

$$\begin{aligned}\text{Unit sales to attain the target profit} &= \frac{\text{Target profit} + \text{Fixed expenses}}{\text{Unit CM}} \\ &= \frac{\$40,000 + \$35,000}{\$100} \\ &= 750\end{aligned}$$

Note that this is the same answer we got when we used the equation method—and it always will be. The formula method simply skips a few steps in the equation method.

Target Profit Analysis in Terms of Sales Dollars Instead of unit sales, we may want to know what dollar sales are needed to attain the target profit. We can get this answer using several methods. First, we could solve for the unit sales to attain the target profit using the equation method or the formula method and then multiply the result by the selling price. In the case of Acoustic Concepts, the required sales volume using this approach would be computed as 750 speakers $\times$ \$250 per speaker or \$187,500 in total sales.

We can also solve for the required sales volume to attain the target profit of \$40,000 at Acoustic Concepts using the basic equation stated in terms of the contribution margin ratio:

$$\begin{aligned}\text{Profit} &= \text{CM ratio} \times \text{Sales} - \text{Fixed expenses} \\ \$40,000 &= 0.40 \times \text{Sales} - \$35,000 \\ 0.40 \times \text{Sales} &= \$40,000 + \$35,000 \\ \text{Sales} &= (\$40,000 + \$35,000) \div 0.40 \\ \text{Sales} &= \$187,500\end{aligned}$$

² This equation can be derived as follows:

$$\begin{aligned}\text{Profit} &= \text{Unit CM} \times Q - \text{Fixed expenses} \\ \text{Target profit} &= \text{Unit CM} \times Q - \text{Fixed expenses} \\ \text{Unit CM} \times Q &= \text{Target profit} + \text{Fixed expenses} \\ Q &= (\text{Target profit} + \text{Fixed expenses}) \div \text{Unit CM}\end{aligned}$$

Note that in the next to the last line of the previous solution, the sum of the target profit of \$40,000 and the fixed expense of \$35,000 is divided by the contribution margin ratio of 0.40. In general, we can compute dollar sales to attain a target profit as follows:

$$\text{Dollar sales to attain a target profit} = \frac{\text{Target profit} + \text{Fixed expenses}^3}{\text{CM ratio}}$$

At Acoustic Concepts, the formula yields the following answer:

$$\begin{aligned} \text{Dollar sales to attain a target profit} &= \frac{\text{Target profit} + \text{Fixed expenses}}{\text{CM ratio}} \\ &= \frac{\$40,000 + \$35,000}{0.40} \\ &= \$187,500 \end{aligned}$$

Again, you get exactly the same answer whether you use the equation method or just use the formula.

In companies with multiple products, sales volume is more conveniently expressed in terms of total sales dollars than in terms of unit sales. The contribution margin ratio approach to target profit analysis is particularly useful for such companies.

Break-Even Analysis

LEARNING OBJECTIVE 6
Determine the break-even point.

Earlier in the chapter we defined the break-even point as the level of sales at which the company's profit is zero. What we call *break-even analysis* is really just a special case of target profit analysis in which the target profit is zero. We can use either the equation method or the formula method to solve for the break-even point, but for brevity we will illustrate just the formula method. The equation method works exactly like it did in target profit analysis. The only difference is that the target profit is zero in break-even analysis.

Break-Even in Unit Sales In a single product situation, recall that the formula for the unit sales to attain a specific target profit is:

$$\text{Unit sales to attain the target profit} = \frac{\text{Target profit} + \text{Fixed expenses}}{\text{Unit CM}}$$

To compute the unit sales to break even, all we have to do is to set the target profit to zero in the above equation as follows:

$$\text{Unit sales to break even} = \frac{\$0 + \text{Fixed expenses}}{\text{Unit CM}}$$

$$\text{Unit sales to break even} = \frac{\text{Fixed expenses}}{\text{Unit CM}}$$

In the case of Acoustic Concepts, the break-even point can be computed as follows:

$$\begin{aligned} \text{Unit sales to break even} &= \frac{\text{Fixed expenses}}{\text{Unit CM}} \\ &= \frac{\$35,000}{\$100} \\ &= 350 \end{aligned}$$

Thus, as we determined earlier in the chapter, Acoustic Concepts breaks even at sales of 350 speakers per month.

³ This equation can be derived as follows:

$$\text{Profit} = \text{CM ratio} \times \text{Sales} - \text{Fixed expenses}$$

$$\text{Target profit} = \text{CM ratio} \times \text{Sales} - \text{Fixed expenses}$$

$$\text{CM ratio} \times \text{Sales} = \text{Target profit} + \text{Fixed expenses}$$

$$\text{Sales} = (\text{Target profit} + \text{Fixed expenses}) \div \text{CM ratio}$$

Break-Even in Sales Dollars We can find the break-even point in sales dollars using several methods. First, we could solve for the break-even point in unit sales using the equation method or the formula method and then multiply the result by the selling price. In the case of Acoustic Concepts, the break-even point in sales dollars using this approach would be computed as 350 speakers $\times$ \$250 per speaker or \$87,500 in total sales.

We can also solve for the break-even point in sales dollars at Acoustic Concepts using the basic profit equation stated in terms of the contribution margin ratio or we can use the formula for the target profit. Again, for brevity, we will use the formula.

$$\text{Dollar sales to attain a target profit} = \frac{\text{Target profit} + \text{Fixed expenses}}{\text{CM ratio}}$$

$$\text{Dollar sales to break even} = \frac{\$0 + \text{Fixed expenses}}{\text{CM ratio}}$$

$$\text{Dollar sales to break even} = \frac{\text{Fixed expenses}}{\text{CM ratio}}$$

The break-even point at Acoustic Concepts would be computed as follows:

$$\begin{aligned} \text{Dollar sales to break even} &= \frac{\text{Fixed expenses}}{\text{CM ratio}} \\ &= \frac{\$35,000}{0.40} \\ &= \$87,500 \end{aligned}$$

COST OVERRUNS INCREASE THE BREAK-EVEN POINT

When Airbus launched the A380 555-seat jetliner in 2000, the company said it would need to sell 250 units to break even on the project. By 2006, Airbus was admitting that more than \$3 billion of cost overruns had raised the project's break-even point to 420 airplanes. Although Airbus has less than 170 orders for the A380, the company remains optimistic that it will sell 751 units over the next 20 years. Given that Airbus rival Boeing predicts the total market size for all airplanes with more than 400 seats will not exceed 990 units, it remains unclear if Airbus will ever break even on its investment in the A380 aircraft.

Source: Daniel Michaels, "Embattled Airbus Lifts Sales Target for A380 to Profit," *The Wall Street Journal*, October 20, 2006, p. A6.

IN BUSINESS



The Margin of Safety

The **margin of safety** is the excess of budgeted or actual sales dollars over the break-even volume of sales dollars. It is the amount by which sales can drop before losses are incurred. The higher the margin of safety, the lower the risk of not breaking even and incurring a loss. The formula for the margin of safety is:

$$\text{Margin of safety in dollars} = \text{Total budgeted (or actual) sales} - \text{Break-even sales}$$

The margin of safety can also be expressed in percentage form by dividing the margin of safety in dollars by total dollar sales:

$$\text{Margin of safety percentage} = \frac{\text{Margin of safety in dollars}}{\text{Total budgeted (or actual) sales in dollars}}$$

The calculation of the margin of safety for Acoustic Concepts is:

Sales (at the current volume of 400 speakers) (a)	\$100,000
Break-even sales (at 350 speakers)	87,500
Margin of safety in dollars (b)	<u>\$ 12,500</u>
Margin of safety percentage, (b) $\div$ (a)	<u>12.5%</u>

LEARNING OBJECTIVE 7

Compute the margin of safety and explain its significance.

This margin of safety means that at the current level of sales and with the company's current prices and cost structure, a reduction in sales of \$12,500, or 12.5%, would result in just breaking even.

In a single-product company like Acoustic Concepts, the margin of safety can also be expressed in terms of the number of units sold by dividing the margin of safety in dollars by the selling price per unit. In this case, the margin of safety is 50 speakers ($\$12,500 \div \$250 \text{ per speaker} = 50 \text{ speakers}$).

IN BUSINESS

COMPUTING MARGIN OF SAFETY FOR A SMALL BUSINESS

Sam Calagione owns **Dogfish Head Craft Brewery**, a microbrewery in Rehobeth Beach, Delaware. He charges distributors as much as \$100 per case for his premium beers such as World Wide Stout. The high-priced microbrews bring in \$800,000 in operating income on revenue of \$7 million. Calagione reports that his raw ingredients and labor costs for one case of World Wide Stout are \$30 and \$16, respectively. Bottling and packaging costs are \$6 per case. Gas and electric costs are about \$10 per case.

If we assume that World Wide Stout is representative of all Dogfish microbrews, then we can compute the company's margin of safety in five steps. First, variable cost as a percentage of sales is 62% [$(\$30 + \$16 + \$6 + \$10)/\$100$]. Second, the contribution margin ratio is 38% ($1 - 0.62$). Third, Dogfish's total fixed cost is \$1,860,000 [$(\$7,000,000 \times 0.38) - \$800,000$]. Fourth, the break-even point in sales dollars is \$4,894,737 ($\$1,860,000/0.38$). Fifth, the margin of safety is \$2,105,263 ($\$7,000,000 - \$4,894,737$).

Source: Patricia Huang, "Château Dogfish," *Forbes*, February 28, 2005, pp. 57–59.

MANAGERIAL ACCOUNTING IN ACTION

The Wrap-up



Prem Narayan and Bob Luchinni met to discuss the results of Bob's analysis.

Prem: Bob, everything you have shown me is pretty clear. I can see what impact the sales manager's suggestions would have on our profits. Some of those suggestions are quite good and others are not so good. I am concerned that our margin of safety is only 50 speakers. What can we do to increase this number?

Bob: Well, we have to increase total sales or decrease the break-even point or both.

Prem: And to decrease the break-even point, we have to either decrease our fixed expenses or increase our unit contribution margin?

Bob: Exactly.

Prem: And to increase our unit contribution margin, we must either increase our selling price or decrease the variable cost per unit?

Bob: Correct.

Prem: So what do you suggest?

Bob: Well, the analysis doesn't tell us which of these to do, but it does indicate we have a potential problem here.

Prem: If you don't have any immediate suggestions, I would like to call a general meeting next week to discuss ways we can work on increasing the margin of safety. I think everyone will be concerned about how vulnerable we are to even small downturns in sales.

CVP Considerations in Choosing a Cost Structure

Cost structure refers to the relative proportion of fixed and variable costs in an organization. Managers often have some latitude in trading off between these two types of costs. For example, fixed investments in automated equipment can reduce variable labor costs. In this section, we discuss the choice of a cost structure. We also introduce the concept of *operating leverage*.

Cost Structure and Profit Stability

Which cost structure is better—high variable costs and low fixed costs, or the opposite? No single answer to this question is possible; each approach has its advantages. To show what we mean, refer to the contribution format income statements given below for two blueberry farms. Bogside Farm depends on migrant workers to pick its berries by hand, whereas Sterling Farm has invested in expensive berry-picking machines. Consequently, Bogside Farm has higher variable costs, but Sterling Farm has higher fixed costs:

	Bogside Farm		Sterling Farm	
	Amount	Percent	Amount	Percent
Sales	\$100,000	100%	\$100,000	100%
Variable expenses	60,000	60%	30,000	30%
Contribution margin	40,000	40%	70,000	70%
Fixed expenses	30,000		60,000	
Net operating income	\$ 10,000		\$ 10,000	

Which farm has the better cost structure? The answer depends on many factors, including the long-run trend in sales, year-to-year fluctuations in the level of sales, and the attitude of the owners toward risk. If sales are expected to exceed \$100,000 in the future, then Sterling Farm probably has the better cost structure. The reason is that its CM ratio is higher, and its profits will therefore increase more rapidly as sales increase. To illustrate, assume that each farm experiences a 10% increase in sales without any increase in fixed costs. The new income statements would be as follows:

	Bogside Farm		Sterling Farm	
	Amount	Percent	Amount	Percent
Sales	\$110,000	100%	\$110,000	100%
Variable expenses	66,000	60%	33,000	30%
Contribution margin	44,000	40%	77,000	70%
Fixed expenses	30,000		60,000	
Net operating income	\$ 14,000		\$ 17,000	

Sterling Farm has experienced a greater increase in net operating income due to its higher CM ratio even though the increase in sales was the same for both farms.

What if sales drop below \$100,000? What are the farms' break-even points? What are their margins of safety? The computations needed to answer these questions are shown below using the formula method:

	Bogside Farm	Sterling Farm
Fixed expenses	\$ 30,000	\$ 60,000
Contribution margin ratio	÷ 0.40	÷ 0.70
Dollar sales to break even	\$ 75,000	\$ 85,714
Total current sales (a)	\$100,000	\$100,000
Break-even sales	75,000	85,714
Margin of safety in sales dollars (b)	\$ 25,000	\$ 14,286
Margin of safety percentage (b) ÷ (a)	25.0%	14.3%

Bogside Farm's margin of safety is greater and its contribution margin ratio is lower than Sterling Farm. Therefore, Bogside Farm is less vulnerable to downturns than Sterling Farm. Due to its lower contribution margin ratio, Bogside Farm will not lose contribution margin as rapidly as Sterling Farm when sales decline. Thus, Bogside Farm's profit will be less volatile. We saw earlier that this is a drawback when sales increase, but it provides more protection when sales drop. And because its break-even point is lower, Bogside Farm can suffer a larger sales decline before losses emerge.

To summarize, without knowing the future, it is not obvious which cost structure is better. Both have advantages and disadvantages. Sterling Farm, with its higher fixed costs and lower variable costs, will experience wider swings in net operating income as sales fluctuate, with greater profits in good years and greater losses in bad years. Bogside Farm, with its lower fixed costs and higher variable costs, will enjoy greater profit stability and will be more protected from losses during bad years, but at the cost of lower net operating income in good years.

Operating Leverage

LEARNING OBJECTIVE 8

Compute the degree of operating leverage at a particular level of sales and explain how it can be used to predict changes in net operating income.

A lever is a tool for multiplying force. Using a lever, a massive object can be moved with only a modest amount of force. In business, *operating leverage* serves a similar purpose. **Operating leverage** is a measure of how sensitive net operating income is to a given percentage change in dollar sales. Operating leverage acts as a multiplier. If operating leverage is high, a small percentage increase in sales can produce a much larger percentage increase in net operating income.

Operating leverage can be illustrated by returning to the data for the two blueberry farms. We previously showed that a 10% increase in sales (from \$100,000 to \$110,000 in each farm) results in a 70% increase in the net operating income of Sterling Farm (from \$10,000 to \$17,000) and only a 40% increase in the net operating income of Bogside Farm (from \$10,000 to \$14,000). Thus, for a 10% increase in sales, Sterling Farm experiences a much greater percentage increase in profits than does Bogside Farm. Therefore, Sterling Farm has greater operating leverage than Bogside Farm.

The **degree of operating leverage** at a given level of sales is computed by the following formula:

$$\text{Degree of operating leverage} = \frac{\text{Contribution margin}}{\text{Net operating income}}$$

The degree of operating leverage is a measure, at a given level of sales, of how a percentage change in sales volume will affect profits. To illustrate, the degree of operating leverage for the two farms at \$100,000 sales would be computed as follows:

$$\text{Bogside Farm: } \frac{\$40,000}{\$10,000} = 4$$

$$\text{Sterling Farm: } \frac{\$70,000}{\$10,000} = 7$$

Because the degree of operating leverage for Bogside Farm is 4, the farm's net operating income grows four times as fast as its sales. In contrast, Sterling Farm's net operating income grows seven times as fast as its sales. Thus, if sales increase by 10%, then we can expect the net operating income of Bogside Farm to increase by four times this amount, or by 40%, and the net operating income of Sterling Farm to increase by seven times this amount, or by 70%. In general, this relation between the percentage change in sales and the percentage change in net operating income is given by the following formula:

$$\text{Percentage change in net operating income} = \text{Degree of operating leverage} \times \text{Percentage change in sales}$$

$$\text{Bogside Farm: Percentage change in net operating income} = 4 \times 10\% = 40\%$$

$$\text{Sterling Farm: Percentage change in net operating income} = 7 \times 10\% = 70\%$$

What is responsible for the higher operating leverage at Sterling Farm? The only difference between the two farms is their cost structure. If two companies have the same total revenue and same total expense but different cost structures, then the company with the higher proportion of fixed costs in its cost structure will have higher operating leverage. Referring back to the original example on page 201, when both farms have sales of \$100,000 and total expenses of \$90,000, one-third of Bogside Farm's costs are fixed but two-thirds of Sterling Farm's costs are fixed. As a consequence, Sterling's degree of operating leverage is higher than Bogside's.

The degree of operating leverage is not a constant; it is greatest at sales levels near the break-even point and decreases as sales and profits rise. The following table shows the degree of operating leverage for Bogside Farm at various sales levels. (Data used earlier for Bogside Farm are shown in color.)

Sales	\$75,000	\$80,000	\$100,000	\$150,000	\$225,000
Variable expenses	45,000	48,000	60,000	90,000	135,000
Contribution margin (a)	30,000	32,000	40,000	60,000	90,000
Fixed expenses	30,000	30,000	30,000	30,000	30,000
Net operating income (b)	\$ 0	\$ 2,000	\$ 10,000	\$ 30,000	\$ 60,000
Degree of operating leverage, (a) ÷ (b)	∞	16	4	2	1.5

Thus, a 10% increase in sales would increase profits by only 15% ($10\% \times 1.5$) if sales were previously \$225,000, as compared to the 40% increase we computed earlier at the \$100,000 sales level. The degree of operating leverage will continue to decrease the farther the company moves from its break-even point. At the break-even point, the degree of operating leverage is infinitely large (\$30,000 contribution margin ÷ \$0 net operating income = ∞).

THE DANGERS OF A HIGH DEGREE OF OPERATING LEVERAGE

In recent years, computer chip manufacturers have poured more than \$75 billion into constructing new manufacturing facilities to meet the growing demand for digital devices such as iPhones and Blackberrys. Because 70% of the costs of running these facilities are fixed, a sharp drop in customer demand forces these companies to choose between two undesirable options. They can slash production levels and absorb large amounts of unused capacity costs, or they can continue producing large volumes of output in spite of shrinking demand, thereby flooding the market with excess supply and lowering prices. Either choice distresses investors who tend to shy away from computer chip makers in economic downturns.

Source: Bruce Einhorn, "Chipmakers on the Edge," *BusinessWeek*, January 5, 2009, pp. 30–31.

IN BUSINESS



The degree of operating leverage can be used to quickly estimate what impact various percentage changes in sales will have on profits, without the necessity of preparing detailed income statements. As shown by our examples, the effects of operating leverage can be dramatic. If a company is near its break-even point, then even small percentage increases in sales can yield large percentage increases in profits. *This explains why management will often work very hard for only a small increase in sales volume.* If the degree of operating leverage is 5, then a 6% increase in sales would translate into a 30% increase in profits.

Structuring Sales Commissions

Companies usually compensate salespeople by paying them a commission based on sales, a salary, or a combination of the two. Commissions based on sales dollars can lead to lower profits. To illustrate, consider Pipeline Unlimited, a producer of surfing equipment. Salespersons sell the company’s products to retail sporting goods stores throughout North America and the Pacific Basin. Data for two of the company’s surfboards, the XR7 and Turbo models, appear below:

	Model	
	XR7	Turbo
Selling price	\$695	\$749
Variable expenses	344	410
Contribution margin	<u>\$351</u>	<u>\$339</u>

Which model will salespeople push hardest if they are paid a commission of 10% of sales revenue? The answer is the Turbo because it has the higher selling price and hence the larger commission. On the other hand, from the standpoint of the company, profits will be greater if salespeople steer customers toward the XR7 model because it has the higher contribution margin.

To eliminate such conflicts, commissions can be based on contribution margin rather than on selling price. If this is done, the salespersons will want to sell the mix of products that maximizes contribution margin. Providing that fixed costs are not affected by the sales mix, maximizing the contribution margin will also maximize the company’s profit.⁴ In effect, by maximizing their own compensation, salespersons will also maximize the company’s profit.

IN BUSINESS

AN ALTERNATIVE APPROACH TO SALES COMMISSIONS

Thrive Networks, located in Concord, Massachusetts, used to pay its three salesmen based on individually earned commissions. This system seemed to be working fine as indicated by the company’s sales growth from \$2.7 million in 2002 to \$3.6 million in 2003. However, the company felt there was a better way to motivate and compensate its salesmen. It pooled commissions across the three salesmen and compensated them collectively. The new approach was designed to build teamwork and leverage each salesman’s individual strengths. Jim Lippie, the director of business development, was highly skilled at networking and generating sales leads. John Barrows, the sales director, excelled at meeting with prospective clients and producing compelling proposals. Nate Wolfson, the CEO and final member of the sales team, was the master at closing the deal. The new approach has worked so well that Wolfson plans to use three-person sales teams in his offices nationwide.

Source: Cara Cannella, “Kill the Commissions,” *Inc. Magazine*, August 2004, p. 38.

Sales Mix

LEARNING OBJECTIVE 9
Compute the break-even point for a multiproduct company and explain the effects of shifts in the sales mix on contribution margin and the break-even point.

Before concluding our discussion of CVP concepts, we need to consider the impact of changes in *sales mix* on a company’s profit.

⁴ This also assumes the company has no production constraint. If it does, the sales commissions should be modified. See the Profitability Appendix at the end of the book.

The Definition of Sales Mix

The term **sales mix** refers to the relative proportions in which a company's products are sold. The idea is to achieve the combination, or mix, that will yield the greatest profits. Most companies have many products, and often these products are not equally profitable. Hence, profits will depend to some extent on the company's sales mix. Profits will be greater if high-margin rather than low-margin items make up a relatively large proportion of total sales.

Changes in the sales mix can cause perplexing variations in a company's profits. A shift in the sales mix from high-margin items to low-margin items can cause total profits to decrease even though total sales may increase. Conversely, a shift in the sales mix from low-margin items to high-margin items can cause the reverse effect—total profits may increase even though total sales decrease. It is one thing to achieve a particular sales volume; it is quite another to sell the most profitable mix of products.

IN BUSINESS

WALMART ATTEMPTS TO SHIFT ITS SALES MIX

Almost 130 million customers shop at **Walmart's** 3,200 U.S. stores each week. However, less than half of them shop the whole store—choosing to buy only low-margin basics while skipping higher-margin departments such as apparel. In an effort to shift its sales mix toward higher-margin merchandise, Walmart has reduced spending on advertising and plowed the money into remodeling the clothing departments within its stores. The company hopes this remodeling effort will entice its customers to add clothing to their shopping lists while bypassing the apparel offerings of competitors such as **Kohl's** and **Target**.

Source: Robert Berner, "Fashion Emergency at Walmart," *BusinessWeek*, July 31, 2006, p. 67.

Sales Mix and Break-Even Analysis

If a company sells more than one product, break-even analysis is more complex than discussed to this point. The reason is that different products will have different selling prices, different costs, and different contribution margins. Consequently, the break-even point depends on the mix in which the various products are sold. To illustrate, consider Virtual Journeys Unlimited, a small company that imports DVDs from France. At present, the company sells two DVDs: the Le Louvre DVD, a tour of the famous art museum in Paris; and the Le Vin DVD, which features the wines and wine-growing regions of France. The company's September sales, expenses, and break-even point are shown in Exhibit 5-4.

As shown in the exhibit, the break-even point is \$60,000 in sales, which was computed by dividing the company's fixed expenses of \$27,000 by its overall CM ratio of 45%. However, this is the break-even only if the company's sales mix does not change. Currently, the Le Louvre DVD is responsible for 20% and the Le Vin DVD for 80% of the company's dollar sales. Assuming this sales mix does not change, if total sales are \$60,000, the sales of the Le Louvre DVD would be \$12,000 (20% of \$60,000) and the sales of the Le Vin DVD would be \$48,000 (80% of \$60,000). As shown in Exhibit 5-4, at these levels of sales, the company would indeed break even. But \$60,000 in sales represents the break-even point for the company only if the sales mix does not change. *If the sales mix changes, then the break-even point will also usually change.* This is illustrated by the results for October in which the sales mix shifted away from the more profitable Le Vin DVD (which has a 50% CM ratio) toward the less profitable Le Louvre CD (which has a 25% CM ratio). These results appear in Exhibit 5-5.

EXHIBIT 5-4

Multiproduct Break-Even Analysis

Virtual Journeys Unlimited Contribution Income Statement For the Month of September						
	Le Louvre DVD		Le Vin DVD		Total	
	Amount	Percent	Amount	Percent	Amount	Percent
Sales	\$20,000	100%	\$80,000	100%	\$100,000	100%
Variable expenses	15,000	75%	40,000	50%	55,000	55%
Contribution margin	<u>\$ 5,000</u>	<u>25%</u>	<u>\$40,000</u>	<u>50%</u>	<u>45,000</u>	<u>45%</u>
Fixed expenses					27,000	
Net operating income					<u>\$ 18,000</u>	

Computation of the break-even point:

$$\frac{\text{Fixed expenses}}{\text{Overall CM ratio}} = \frac{\$27,000}{0.45} = \$60,000$$

Verification of the break-even point:

	Le Louvre DVD	Le Vin DVD	Total
Current dollar sales	\$20,000	\$80,000	\$100,000
Percentage of total dollar sales	20%	80%	100%

Sales at the break-even point

	Le Louvre DVD	Le Vin DVD	Total
Sales	\$12,000	\$48,000	\$ 60,000
Variable expenses	9,000	24,000	33,000
Contribution margin	<u>\$ 3,000</u>	<u>\$24,000</u>	<u>27,000</u>
Fixed expenses			27,000
Net operating income			<u>\$ 0</u>

EXHIBIT 5-5

Multiproduct Break-Even Analysis: A Shift in Sales Mix (see Exhibit 5-4)

Virtual Journeys Unlimited Contribution Income Statement For the Month of October						
	Le Louvre DVD		Le Vin DVD		Total	
	Amount	Percent	Amount	Percent	Amount	Percent
Sales	\$80,000	100%	\$20,000	100%	\$100,000	100%
Variable expenses	60,000	75%	10,000	50%	70,000	70%
Contribution margin	<u>\$20,000</u>	<u>25%</u>	<u>\$10,000</u>	<u>50%</u>	<u>30,000</u>	<u>30%</u>
Fixed expenses					27,000	
Net operating income					<u>\$ 3,000</u>	

Computation of the break-even point:

$$\frac{\text{Fixed expenses}}{\text{Overall CM ratio}} = \frac{\$27,000}{0.30} = \$90,000$$

Although sales have remained unchanged at \$100,000, the sales mix is exactly the reverse of what it was in Exhibit 5–4, with the bulk of the sales now coming from the less profitable Le Louvre DVD. Notice that this shift in the sales mix has caused both the overall CM ratio and total profits to drop sharply from the prior month even though total sales are the same. The overall CM ratio has dropped from 45% in September to only 30% in October, and net operating income has dropped from \$18,000 to only \$3,000. In addition, with the drop in the overall CM ratio, the company's break-even point is no longer \$60,000 in sales. Because the company is now realizing less average contribution margin per dollar of sales, it takes more sales to cover the same amount of fixed costs. Thus, the break-even point has increased from \$60,000 to \$90,000 in sales per year.

In preparing a break-even analysis, an assumption must be made concerning the sales mix. Usually the assumption is that it will not change. However, if the sales mix is expected to change, then this must be explicitly considered in any CVP computations.

Assumptions of CVP Analysis

A number of assumptions commonly underlie CVP analysis:

1. Selling price is constant. The price of a product or service will not change as volume changes.
2. Costs are linear and can be accurately divided into variable and fixed elements. The variable element is constant per unit, and the fixed element is constant in total over the entire relevant range.
3. In multiproduct companies, the sales mix is constant.
4. In manufacturing companies, inventories do not change. The number of units produced equals the number of units sold.

While these assumptions may be violated in practice, the results of CVP analysis are often “good enough” to be quite useful. Perhaps the greatest danger lies in relying on simple CVP analysis when a manager is contemplating a large change in volume that lies outside of the relevant range. For example, a manager might contemplate increasing the level of sales far beyond what the company has ever experienced before. However, even in these situations the model can be adjusted as we have done in this chapter to take into account anticipated changes in selling prices, fixed costs, and the sales mix that would otherwise violate the assumptions mentioned above. For example, in a decision that would affect fixed costs, the change in fixed costs can be explicitly taken into account as illustrated earlier in the chapter in the Acoustic Concepts example on pages 192–195.

Summary

CVP analysis is based on a simple model of how profits respond to prices, costs, and volume. This model can be used to answer a variety of critical questions such as what is the company's break-even volume, what is its margin of safety, and what is likely to happen if specific changes are made in prices, costs, and volume.

A CVP graph depicts the relationships between unit sales on the one hand and fixed expenses, variable expenses, total expenses, total sales, and profits on the other hand. The profit graph is simpler than the CVP graph and shows how profits depend on sales. The CVP and profit graphs are useful for developing intuition about how costs and profits respond to changes in sales.

The contribution margin ratio is the ratio of the total contribution margin to total sales. This ratio can be used to quickly estimate what impact a change in total sales would have on net operating income. The ratio is also useful in break-even analysis.

Target profit analysis is used to estimate how much sales would have to be to attain a specified target profit. The unit sales required to attain the target profit can be estimated by dividing the sum of the target profit and fixed expense by the unit contribution margin. Break-even analysis is

This chapter describes two applications of the contribution format income statements that were introduced in Chapters 2 and 5. First, it explains how manufacturing companies can prepare *variable costing* income statements, which rely on the contribution format, for internal decision making purposes. The variable costing approach will be contrasted with *absorption costing* income statements, which were discussed in Chapter 3 and are generally used for external reports. Ordinarily, variable costing and absorption costing produce different net operating income figures, and the difference can be quite large. In addition to showing how these two methods differ, we will describe the advantages of variable costing for internal reporting purposes and we will show how management decisions can be affected by the costing method chosen.

Second, the chapter explains how the contribution format can be used to prepare segmented income statements. In addition to companywide income statements, managers need to measure the profitability of individual *segments* of their organizations. A **segment** is a part or activity of an organization about which managers would like cost, revenue, or profit data. This chapter explains how to create contribution format income statements that report profit data for business segments, such as divisions, individual stores, geographic regions, customers, and product lines.

Overview of Variable and Absorption Costing

LEARNING OBJECTIVE 1

Explain how variable costing differs from absorption costing and compute unit product costs under each method.

As you begin to read about variable and absorption costing income statements in the coming pages, focus your attention on three key concepts. First, both income statement formats include product costs and period costs, although they define these cost classifications differently. Second, variable costing income statements are grounded in the contribution format. They categorize expenses based on cost behavior—variable costs are reported separately from fixed costs. Absorption costing income statements ignore variable and fixed cost distinctions. Third, as mentioned in the paragraph above, variable and absorption costing net operating income figures often differ from one another. The reason for these differences always relates to the fact the variable costing and absorption costing income statements account for fixed manufacturing overhead differently. *Pay very close attention to the two different ways that variable costing and absorption costing account for fixed manufacturing overhead.*

Variable Costing

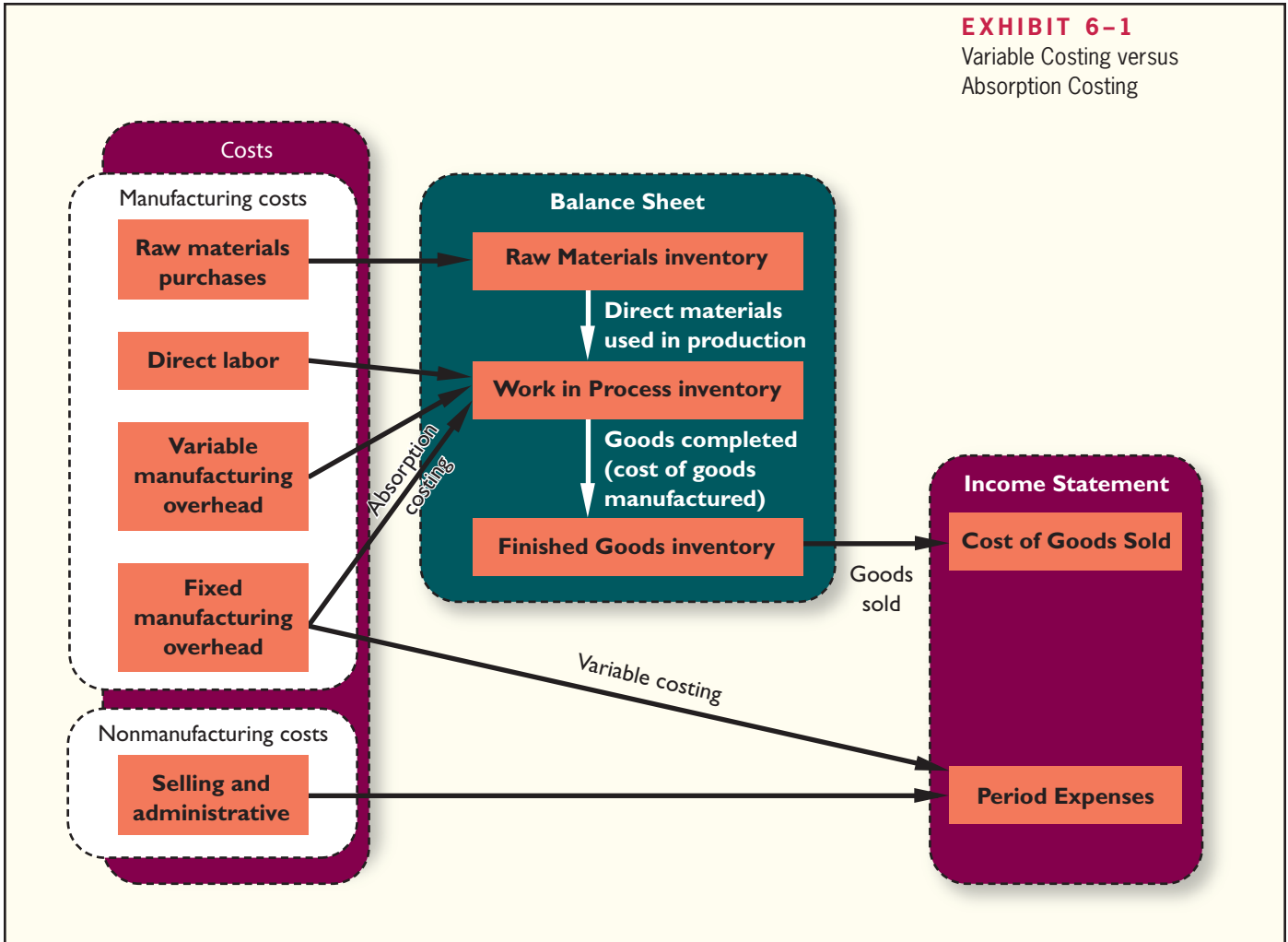
Under **variable costing**, only those manufacturing costs that vary with output are treated as product costs. This would usually include direct materials, direct labor, and the variable portion of manufacturing overhead. Fixed manufacturing overhead is not treated as a product cost under this method. Rather, fixed manufacturing overhead is treated as a period cost and, like selling and administrative expenses, it is expensed in its entirety each period. Consequently, the cost of a unit of product in inventory or in cost of goods sold under the variable costing method does not contain any fixed manufacturing overhead cost. Variable costing is sometimes referred to as *direct costing* or *marginal costing*.

Absorption Costing

As discussed in Chapter 3, **absorption costing** treats *all* manufacturing costs as product costs, regardless of whether they are variable or fixed. The cost of a unit of product under the absorption costing method consists of direct materials, direct labor, and *both* variable and fixed manufacturing overhead. Thus, absorption costing allocates a portion of fixed

EXHIBIT 6-1

Variable Costing versus Absorption Costing



manufacturing overhead cost to each unit of product, along with the variable manufacturing costs. Because absorption costing includes all manufacturing costs in product costs, it is frequently referred to as the *full cost* method.

Selling and Administrative Expenses

Selling and administrative expenses are never treated as product costs, regardless of the costing method. Thus, under absorption and variable costing, variable and fixed selling and administrative expenses are always treated as period costs and are expensed as incurred.

Summary of Differences The essential difference between variable costing and absorption costing, as illustrated in Exhibit 6-1, is how each method accounts for fixed manufacturing overhead costs—all other costs are treated the same under the two methods. In absorption costing, fixed manufacturing overhead costs are included as part of the costs of work in process inventories. When units are completed, these costs are transferred to finished goods and only when the units are sold do these costs flow through to the income statement as part of cost of goods sold. In variable costing, fixed manufacturing overhead costs are considered to be period costs—just like selling and administrative costs—and are taken immediately to the income statement as period expenses.

Variable and Absorption Costing—An Example

To illustrate the difference between variable costing and absorption costing, consider Weber Light Aircraft, a company that produces light recreational aircraft. Data concerning the company's operations appear below:

	Per Aircraft	Per Month
Selling price	\$100,000	
Direct materials	\$19,000	
Direct labor	\$5,000	
Variable manufacturing overhead	\$1,000	
Fixed manufacturing overhead		\$70,000
Variable selling and administrative expenses	\$10,000	
Fixed selling and administrative expenses		\$20,000

	January	February	March
Beginning inventory	0	0	1
Units produced	1	2	4
Units sold	1	1	5
Ending inventory	0	1	0

As you review the data above, it is important to realize that for the months of January, February, and March, the selling price per aircraft, variable cost per aircraft, and total monthly fixed expenses never change. The only variables that change in this example are the number of units produced (January = 1 unit produced; February = 2 units produced; March = 4 units produced) and the number of units sold (January = 1 unit sold; February = 1 unit sold; March = 5 units sold).

We will first construct the company's variable costing income statements for January, February, and March. Then we will show how the company's net operating income would be determined for the same months using absorption costing.

Variable Costing Contribution Format Income Statement

LEARNING OBJECTIVE 2
Prepare income statements using both variable and absorption costing.

To prepare the company's variable costing income statements for January, February, and March we begin by computing the unit product cost. Under variable costing, product costs consist solely of variable production costs. At Weber Light Aircraft, the variable production cost per unit is \$25,000, determined as follows:

Variable Costing Unit Product Cost	
Direct materials	\$19,000
Direct labor	5,000
Variable manufacturing overhead	1,000
Variable costing unit product cost	<u>\$25,000</u>

Since each month's variable production cost is \$25,000 per aircraft, the variable costing cost of goods sold for all three months can be easily computed as follows:

Variable Costing Cost of Goods Sold			
	January	February	March
Variable production cost (a)	\$25,000	\$25,000	\$25,000
Units sold (b)	1	1	5
Variable cost of goods sold (a) × (b)	\$25,000	\$25,000	\$125,000

And the company's total selling and administrative expense would be derived as follows:

Selling and Administrative Expenses			
	January	February	March
Variable selling and administrative expense (@ \$10,000 per unit sold)	\$10,000	\$10,000	\$50,000
Fixed selling and administrative expense	20,000	20,000	20,000
Total selling and administrative expense	<u>\$30,000</u>	<u>\$30,000</u>	<u>\$70,000</u>

Putting it all together, the variable costing income statements would appear as shown in Exhibit 6–2. Notice, the contribution format has been used in these income statements. Also, the monthly fixed manufacturing overhead costs (\$70,000) have been recorded as a period expense in the month incurred.

Variable Costing Contribution Format Income Statements			
	January	February	March
Sales	\$100,000	\$100,000	\$500,000
Variable expenses:			
Variable cost of goods sold	25,000	25,000	125,000
Variable selling and administrative expense	10,000	10,000	50,000
Total variable expenses	35,000	35,000	175,000
Contribution margin	65,000	65,000	325,000
Fixed expenses:			
Fixed manufacturing overhead	70,000	70,000	70,000
Fixed selling and administrative expense . . .	20,000	20,000	20,000
Total fixed expenses	90,000	90,000	90,000
Net operating income (loss)	<u>\$ (25,000)</u>	<u>\$ (25,000)</u>	<u>\$235,000</u>

EXHIBIT 6–2
Variable Costing Income
Statements

A simple method for understanding how Weber Light Aircraft computed its variable costing net operating income figures is to focus on the contribution margin per aircraft sold, which is computed as follows:

Contribution Margin per Aircraft Sold		
Selling price per aircraft		\$100,000
Variable production cost per aircraft	\$25,000	
Variable selling and administrative expense per aircraft	10,000	35,000
Contribution margin per aircraft		<u>\$ 65,000</u>

The variable costing net operating income for each period can always be computed by multiplying the number of units sold by the contribution margin per unit and then subtracting total fixed costs. For Weber Light Aircraft these computations would appear as follows:

	January	February	March
Number of aircraft sold	1	1	5
Contribution margin per aircraft	× \$65,000	× \$65,000	× \$65,000
Total contribution margin	\$65,000	\$65,000	\$325,000
Total fixed expenses	90,000	90,000	90,000
Net operating income (loss)	<u>\$(25,000)</u>	<u>\$(25,000)</u>	<u>\$235,000</u>

Notice, January and February have the same net operating loss. This occurs because one aircraft was sold in each month and, as previously mentioned, the selling price per aircraft, variable cost per aircraft, and total monthly fixed expenses remain constant.

Absorption Costing Income Statement

As we begin the absorption costing portion of the example, remember that the only reason absorption costing income differs from variable costing is that the methods account for fixed manufacturing overhead differently. Under absorption costing, fixed manufacturing overhead is included in product costs. In variable costing, fixed manufacturing overhead is not included in product costs and instead is treated as a period expense just like selling and administrative expenses.

The first step in preparing Weber’s absorption costing income statements for January, February, and March, is to determine the company’s unit product costs for each month as follows¹:

Absorption Costing Unit Product Cost			
	January	February	March
Direct materials	\$19,000	\$19,000	\$19,000
Direct labor	5,000	5,000	5,000
Variable manufacturing overhead	1,000	1,000	1,000
Fixed manufacturing overhead (\$70,000 ÷ 1 unit produced in January; \$70,000 ÷ 2 units produced in February; \$70,000 ÷ 4 units produced in March) . . .	70,000	35,000	17,500
Absorption costing unit product cost	<u>\$95,000</u>	<u>\$60,000</u>	<u>\$42,500</u>

Notice that in each month, Weber’s fixed manufacturing overhead cost of \$70,000 is divided by the number of units produced to determine the fixed manufacturing overhead cost per unit.

Given these unit product costs, the company’s absorption costing net operating income in each month would be determined as shown in Exhibit 6–3.

The sales for all three months in Exhibit 6–3 are the same as the sales shown in the variable costing income statements. The January cost of goods sold consists of one unit produced during January at a cost of \$95,000 according to the absorption costing system. The February cost of goods sold consists of one unit produced during February at a cost of \$60,000 according to the absorption costing system. The March cost of goods sold (\$230,000) consists of one unit produced during February at an absorption cost of \$60,000 plus four units produced in March with a total absorption cost of \$170,000 (= 4 units produced × \$42,500 per unit). The selling and administrative expenses equal the amounts reported in the variable costing income statements; however they are reported as one amount rather than being separated into variable and fixed components.

EXHIBIT 6–3 Absorption Costing Income Statements	Absorption Costing Income Statements			
		January	February	March
	Sales	\$100,000	\$100,000	\$500,000
	Cost of goods sold (\$95,000 × 1 unit; \$60,000 × 1 unit; \$60,000 × 1 unit + \$42,500 × 4 units)	95,000	60,000	230,000
	Gross margin	5,000	40,000	270,000
	Selling and administrative expenses	30,000	30,000	70,000
	Net operating income (loss)	<u>\$(25,000)</u>	<u>\$ 10,000</u>	<u>\$200,000</u>

¹ For simplicity, we assume in this section that an actual costing system is used in which actual costs are spread over the units produced during the period. If a predetermined overhead rate were used, the analysis would be similar, but more complex.

Note that even though sales were exactly the same in January and February and the cost structure did not change, net operating income was \$35,000 higher in February than in January under absorption costing. This occurs because one aircraft produced in February is not sold until March. This aircraft has \$35,000 of fixed manufacturing overhead attached to it that was incurred in February, but will not be recorded as part of cost of goods sold until March.

Contrasting the variable costing and absorption costing income statements in Exhibits 6–2 and 6–3, note that net operating income is the same in January under variable costing and absorption costing, but differs in the other two months. We will discuss this in some depth shortly. Also note that the format of the variable costing income statement differs from the absorption costing income statement. An absorption costing income statement categorizes costs by function—manufacturing versus selling and administrative. All of the manufacturing costs flow through the absorption costing cost of goods sold and all of the selling and administrative costs are listed separately as period expenses. In contrast, in the contribution approach, costs are categorized according to how they behave. All of the variable expenses are listed together and all of the fixed expenses are listed together. The variable expenses category includes manufacturing costs (i.e., variable cost of goods sold) as well as selling and administrative expenses. The fixed expenses category also includes both manufacturing costs and selling and administrative expenses.

IN BUSINESS

THE BEHAVIORAL SIDE OF CALCULATING UNIT PRODUCT COSTS

In 2004, **Andreas STIHL**, a manufacturer of chain saws and other landscaping products, asked its U.S. subsidiary, STIHL Inc., to replace its absorption costing income statements with the variable costing approach. From a computer systems standpoint, the change was not disruptive because STIHL used an enterprise system called SAP that accommodates both absorption and variable costing. However, from a behavioral standpoint, STIHL felt the change could be very disruptive. For example, STIHL's senior managers were keenly aware that the variable costing approach reported lower unit product costs than the absorption costing approach. Given this reality, the sales force might be inclined to erroneously conclude that each product had magically become more profitable, thereby justifying ill-advised price reductions. Because of behavioral concerns such as this, STIHL worked hard to teach its employees how to interpret a variable costing income statement.

Source: Carl S. Smith, "Going for GPK: STIHL Moves Toward This Costing System in the United States," *Strategic Finance*, April 2005, pp. 36–39.

Reconciliation of Variable Costing with Absorption Costing Income

As noted earlier, variable costing and absorption costing net operating incomes may not be the same. In the case of Weber Light Aircraft, the net operating incomes are the same in January, but differ in the other two months. These differences occur because under absorption costing some fixed manufacturing overhead is capitalized in inventories (i.e., included in product costs) rather than currently expensed on the income statement. If inventories increase during a period, under absorption costing some of the fixed manufacturing overhead of the current period will be *deferred* in ending inventories. For example, in February two aircraft were produced and each carried with it \$35,000 ($= \$70,000 \div 2$ aircraft produced) in fixed manufacturing overhead. Since only one aircraft was sold, \$35,000 of this fixed manufacturing overhead was on February's absorption costing income statement as part of cost of goods sold, but \$35,000 would have been on the balance sheet as part of finished goods inventories. In contrast, under variable costing *all* of the \$70,000 of fixed manufacturing overhead appeared

LEARNING OBJECTIVE 3

Reconcile variable costing and absorption costing net operating incomes and explain why the two amounts differ.

on the February income statement as a period expense. Consequently, net operating income was higher under absorption costing than under variable costing by \$35,000 in February. This was reversed in March when four units were produced, but five were sold. In March, under absorption costing \$105,000 of fixed manufacturing overhead was included in cost of goods sold (\$35,000 for the unit produced in February and sold in March plus \$17,500 for each of the four units produced and sold in March), but only \$70,000 was recognized as a period expense under variable costing. Hence, the net operating income in March was \$35,000 lower under absorption costing than under variable costing.

In general, when the units produced exceed unit sales and hence inventories increase, net operating income is higher under absorption costing than under variable costing. This occurs because some of the fixed manufacturing overhead of the period is *deferred* in inventories under absorption costing. In contrast, when unit sales exceed the units produced and hence inventories decrease, net operating income is lower under absorption costing than under variable costing. This occurs because some of the fixed manufacturing overhead of previous periods is *released* from inventories under absorption costing. When the units produced and unit sales are equal, no change in inventories occurs and absorption costing and variable costing net operating incomes are the same.²

Variable costing and absorption costing net operating incomes can be reconciled by determining how much fixed manufacturing overhead was deferred in, or released from, inventories during the period:

Fixed Manufacturing Overhead Deferred in, or Released from, Inventories under Absorption Costing			
	January	February	March
Fixed manufacturing overhead in beginning inventories	\$0	\$ 0	\$ 35,000
Fixed manufacturing overhead in ending inventories	<u>0</u>	<u>35,000</u>	<u>0</u>
Fixed manufacturing overhead deferred in (released from) inventories	<u>\$0</u>	<u>\$35,000</u>	<u>\$(35,000)</u>

The reconciliation would then be reported as shown in Exhibit 6-4:

EXHIBIT 6-4 Reconciliation of Variable Costing and Absorption Costing Net Operating Incomes	Reconciliation of Variable Costing and Absorption Costing Net Operating Incomes			
	January	February	March	
Variable costing net operating income (loss)	\$(25,000)	\$(25,000)	\$235,000	
Add (deduct) fixed manufacturing overhead deferred in (released from) inventory under absorption costing	<u>0</u>	<u>35,000</u>	<u>(35,000)</u>	
Absorption costing net operating income (loss) . . .	<u>\$(25,000)</u>	<u>\$ 10,000</u>	<u>\$200,000</u>	

Again note that the difference between variable costing net operating income and absorption costing net operating income is entirely due to the amount of fixed manufacturing overhead that is deferred in, or released from, inventories during the period

² These general statements about the relation between variable costing and absorption costing net operating income assume LIFO is used to value inventories. Even when LIFO is not used, the general statements tend to be correct. Although U.S. GAAP allows LIFO and FIFO inventory flow assumptions, International Financial Reporting Standards do not allow a LIFO inventory flow assumption.

Relation between Production and Sales for the Period	Effect on Inventories 	Relation between Absorption and Variable Costing Net Operating Incomes
Units produced = Units sold	No change in inventories 	Absorption costing net operating income = Variable costing net operating income
Units produced > Units sold	Inventories increase 	Absorption costing net operating income > Variable costing net operating income*
Units produced < Units sold	Inventories decrease 	Absorption costing net operating income < Variable costing net operating income†
*Net operating income is higher under absorption costing because fixed manufacturing overhead cost is <i>deferred</i> in inventory under absorption costing as inventories increase. †Net operating income is lower under absorption costing because fixed manufacturing overhead cost is <i>released</i> from inventory under absorption costing as inventories decrease.		

EXHIBIT 6-5

Comparative Income Effects—
Absorption and Variable
Costing

under absorption costing. Changes in inventories affect absorption costing net operating income—they do not affect variable costing net operating income, providing that variable manufacturing costs per unit are stable.

The reasons for differences between variable and absorption costing net operating incomes are summarized in Exhibit 6-5. When the units produced equal the units sold, as in January for Weber Light Aircraft, absorption costing net operating income will equal variable costing net operating income. This occurs because when production equals sales, all of the fixed manufacturing overhead incurred in the current period flows through to the income statement under both methods. For companies that use Lean Production, the number of units produced tends to equal the number of units sold. This occurs because goods are produced in response to customer orders, thereby eliminating finished goods inventories and reducing work in process inventory to almost nothing. So, when a company uses Lean Production differences in variable costing and absorption costing net operating income will largely disappear.

When the units produced exceed the units sold, absorption costing net operating income will exceed variable costing net operating income. This occurs because inventories have increased; therefore, under absorption costing some of the fixed manufacturing overhead incurred in the current period is deferred in ending inventories on the balance sheet, whereas under variable costing all of the fixed manufacturing overhead incurred in the current period flows through to the income statement. In contrast, when the units produced are less than the units sold, absorption costing net operating income will be less than variable costing net operating income. This occurs because inventories have decreased; therefore, under absorption costing fixed manufacturing overhead that had been deferred in inventories during a prior period flows through to the current period's income statement together with all of the fixed manufacturing overhead incurred during the current period. Under variable costing, just the fixed manufacturing overhead of the current period flows through to the income statement.

Managers often consider decisions that involve an investment today in the hope of realizing future profits. For example, **Tri-Con Global Restaurants, Inc.**, makes an investment when it opens a new Pizza Hut restaurant. **L. L. Bean** makes an investment when it installs a new computer to handle customer billing. **Chrysler** makes an investment when it redesigns a product such as the Jeep Eagle. **Merck & Co.** invests in medical research. **Amazon.com** makes an investment when it redesigns its website. All of these investments require spending now with the expectation of additional future net cash flows.

The term **capital budgeting** is used to describe how managers plan significant investments in projects that have long-term implications such as the purchase of new equipment or the introduction of new products. Most companies have many more potential projects than can actually be funded. Hence, managers must carefully select those projects that promise the greatest future return. How well managers make these capital budgeting decisions is a critical factor in the long-run financial health of the organization.

Capital Budgeting—Planning Investments

Typical Capital Budgeting Decisions

Any decision that involves an outlay now in order to obtain a future return is a capital budgeting decision. Typical capital budgeting decisions include:

1. Cost reduction decisions. Should new equipment be purchased to reduce costs?
2. Expansion decisions. Should a new plant, warehouse, or other facility be acquired to increase capacity and sales?
3. Equipment selection decisions. Which of several available machines should be purchased?
4. Lease or buy decisions. Should new equipment be leased or purchased?
5. Equipment replacement decisions. Should old equipment be replaced now or later?

Capital budgeting decisions fall into two broad categories—*screening decisions* and *preference decisions*. **Screening decisions** relate to whether a proposed project is acceptable—whether it passes a preset hurdle. For example, a company may have a policy of accepting projects only if they provide a return of at least 20% on the investment. The required rate of return is the minimum rate of return a project must yield to be acceptable. **Preference decisions**, by contrast, relate to selecting from among several acceptable alternatives. To illustrate, a company may be considering several different machines to replace an existing machine on the assembly line. The choice of which machine to purchase is a preference decision. In this chapter, we first discuss screening decisions and then move on to preference decisions toward the end of the chapter.

The Time Value of Money

Capital investments usually earn returns that extend over fairly long periods of time. Consequently, it is important to recognize *the time value of money* when evaluating investment proposals. A dollar today is worth more than a dollar a year from now if for no other reason than you could put a dollar in a bank today and have more than a dollar a year from now. Therefore, projects that promise earlier returns are preferable to those that promise later returns.

Capital budgeting techniques that recognize the time value of money involve *discounting cash flows*. We will spend most of this chapter showing how to use discounted cash flow methods in making capital budgeting decisions. If you are not already familiar with discounting and the use of present value tables, you should read Appendix 13A: The Concept of Present Value, at the end of this chapter, before proceeding any further.

IN BUSINESS

CHOOSING A CAT

Sometimes a long-term decision does not have to involve present value calculations or any other sophisticated analytical technique. **White Grizzly Adventures** of Meadow Creek, British Columbia, needs two snowcats for its powder skiing operations—one for shuttling guests to the top of the mountain and one to be held in reserve in case of mechanical problems with the first. **Bombardier** of Canada sells new snowcats for \$250,000 and used, reconditioned snowcats for \$150,000. In either case, the snowcats are good for about 5,000 hours of operation before they need to be reconditioned. From White Grizzly's perspective, the choice is clear. Because both new and reconditioned snowcats last about 5,000 hours, but the reconditioned snowcats cost \$100,000 less, the reconditioned snowcats are the obvious choice. They may not have all of the latest bells and whistles, but they get the job done at a price a small company can afford.

Bombardier snowcats do not have passenger cabs as standard equipment. To save money, White Grizzly builds its own custom-designed passenger cab for about \$15,000, using recycled Ford Escort seats and industrial-strength aluminum for the frame and siding. If purchased at retail, a passenger cab would cost about twice as much and would not be as well-suited for snowcat skiing.

Source: Brad & Carole Karafil, owners and operators of White Grizzly Adventures, www.whitegrizzly.com.

Discounted Cash Flows—The Net Present Value Method

Two approaches to making capital budgeting decisions use discounted cash flows. One is the *net present value method*, and the other is the *internal rate of return method*. The net present value method is discussed in this section followed by a discussion of the internal rate of return method.

LEARNING OBJECTIVE 1

Evaluate the acceptability of an investment project using the net present value method.

The Net Present Value Method Illustrated

Under the net present value method, the present value of a project's cash inflows is compared to the present value of the project's cash outflows. The difference between the present value of these cash flows, called the **net present value**, determines whether or not the project is an acceptable investment. To illustrate, consider the following data:

Example A: Harper Company is contemplating the purchase of a machine capable of performing some operations that are now performed manually. The machine will cost \$50,000, and it will last for five years. At the end of the five-year period, the machine will have a zero scrap value. Use of the machine will reduce labor costs by \$18,000 per year. Harper Company requires a minimum pretax return of 20% on all investment projects.¹

Should the machine be purchased? Harper Company must determine whether a cash investment now of \$50,000 can be justified if it will result in an \$18,000 reduction in cost in each of the next five years. It may appear that the answer is obvious because the total cost savings is \$90,000 (\$18,000 per year $\times$ 5 years). However, the company can earn a 20% return by investing its money elsewhere. It is not enough that the cost reductions cover just the original cost of the machine; they must also yield a return of at least 20% or the company would be better off investing the money elsewhere.

¹ For simplicity, we ignore inflation and taxes. The impact of income taxes on capital budgeting decisions is discussed in Appendix 13C.

EXHIBIT 13-1Net Present Value Analysis
of a Proposed Project

		Initial cost	\$50,000	
		Life of the project	5 years	
		Annual cost savings	\$18,000	
		Salvage value	\$0	
		Required rate of return	20%	
Item	Year(s)	Amount of Cash Flow	20% Factor	Present Value of Cash Flows
Annual cost savings	1–5	\$18,000	2.991*	\$53,838
Initial investment	Now	\$(50,000)	1.000	(50,000)
Net present value				<u>\$ 3,838</u>

*From Exhibit 13B–2 in Appendix 13B at the end of this chapter.

To determine whether the investment is desirable, the stream of annual \$18,000 cost savings should be discounted to its present value and then compared to the cost of the new machine. Harper Company's minimum required return of 20% is used as the *discount rate* in the discounting process. Exhibit 13–1 illustrates the computation of the net present value of this proposed project. The annual cost savings of \$18,000 is multiplied by 2.991, the present value factor of a five-year annuity at the discount rate of 20%, to obtain \$53,838.² This is the present value of the annual cost savings. The present value of the initial investment is computed by multiplying the investment amount of \$50,000 by 1.000, the present value factor for any cash flow that occurs immediately.

According to the analysis, Harper Company should purchase the new machine. The present value of the cost savings is \$53,838, whereas the present value of the required investment (cost of the machine) is only \$50,000. Deducting the present value of the required investment from the present value of the cost savings yields the *net present value* of \$3,838. Whenever the net present value is zero or greater, as in our example, an investment project is acceptable. Whenever the net present value is negative (the present value of the cash outflows exceeds the present value of the cash inflows), an investment project is not acceptable. In sum:

If the Net Present Value Is . . .	Then the Project Is . . .
Positive	Acceptable because its return is greater than the required rate of return.
Zero	Acceptable because its return is equal to the required rate of return.
Negative	Not acceptable because its return is less than the required rate of return.

There is another way to interpret the net present value. Harper Company could spend up to \$53,838 for the new machine and still obtain the minimum required 20% rate of return. The net present value of \$3,838, therefore, shows the amount of “cushion” or “margin of error.” One way to look at this is that the company could underestimate the cost of the new machine by up to \$3,838, or overestimate the net present value of the future cash savings by up to \$3,838, and the project would still be financially attractive.

² Unless otherwise stated, for the sake of simplicity we will assume in this chapter that all cash flows other than the initial investment occur at the ends of years.

Emphasis on Cash Flows

Accounting net income is based on accruals that ignore when cash flows occur. However, in capital budgeting, the timing of cash flows is critical. The present value of a cash flow depends on when it occurs. For that reason, cash flow rather than accounting net income is the focus in capital budgeting.³ Examples of cash outflows and cash inflows that are often relevant to capital investment decisions are described below.

Typical Cash Outflows Most projects have at least three types of cash outflows. First, they often require an immediate cash outflow in the form of an initial investment in equipment, other assets, and installation costs. Any salvage value realized from the sale of old equipment can be recognized as a reduction in the initial investment or as a cash inflow. Second, some projects require a company to expand its working capital. **Working capital** is current assets (e.g., cash, accounts receivable, and inventory) less current liabilities. When a company takes on a new project, the balances in the current asset accounts often increase. For example, opening a new Nordstrom's department store requires additional cash in sales registers and more inventory. These additional working capital needs are treated as part of the initial investment in a project. Third, many projects require periodic outlays for repairs and maintenance and additional operating costs.

Typical Cash Inflows Most projects also have at least three types of cash inflows. First, a project will normally increase revenues or reduce costs. Either way, the amount involved should be treated as a cash inflow for capital budgeting purposes. Notice that from a cash flow standpoint, a reduction in costs is equivalent to an increase in revenues. Second, cash inflows are also frequently realized from selling equipment for its salvage value when a project ends, although the company may actually have to pay to dispose of some low-value or hazardous items. Third, any working capital that was tied up in the project can be released for use elsewhere at the end of the project and should be treated as a cash inflow at that time. Working capital is released, for example, when a company sells off its inventory or collects its accounts receivable.

BEST BUY'S BIG GAMBLE

Best Buy is overhauling hundreds of its stores in an effort to tailor merchandise offerings and employee skills to meet the needs of each store's target customers. The cost to revamp one department of one store can easily exceed \$600,000 for lighting and fixtures, plus additional costs for employee training. While these initial cash outflows are readily quantifiable, the future cash inflows that they will generate are highly uncertain.

The first few dozen stores overhauled by Best Buy recorded sales growth that was three times greater than nonrenovated stores. Best Buy reacted to these initial results by hastily renovating 154 more stores over the next three months. Shortly after completing these expensive renovations, the company had the misfortune of informing Wall Street that the newly revamped stores' growth rates were only slightly higher than nonrenovated stores. This disappointing news apparently caused the market value of Best Buy's common stock to plummet by almost \$3 billion in one day. Clearly, Wall Street analysts had serious concerns about the future cash flow generating ability of this capital investment project. Despite the "bump in the road," Best Buy remains committed to its course of action; however, the company has decided to slow down the pace of its implementation.

Source: Matthew Boyle, "Best Buy's Giant Gamble," *Fortune*, April, 3, 2006, pp. 69–75.

IN BUSINESS



³ Under certain conditions, capital budgeting decisions can be correctly made by discounting appropriately defined accounting net income. However, this approach requires advanced techniques that are beyond the scope of this book.

In summary, the following types of cash flows are common in business investment projects:

- Cash outflows:
Initial investment (including installation costs).
Increased working capital needs.
Repairs and maintenance.
Incremental operating costs.
- Cash inflows:
Incremental revenues.
Reduction in costs.
Salvage value.
Release of working capital.

Recovery of the Original Investment

The net present value method automatically provides for return of the original investment. Whenever the net present value of a project is positive, the project will recover the original cost of the investment plus sufficient excess cash inflows to compensate the organization for tying up funds in the project. To demonstrate this point, consider the following situation:



Example B: Carver Hospital is considering the purchase of an attachment for its X-ray machine that will cost \$3,170. The attachment will be usable for four years, after which time it will have no salvage value. It will increase net cash inflows by \$1,000 per year in the X-ray department. The hospital’s board of directors requires a rate of return of at least 10% on such investments.

A net present value analysis of the desirability of purchasing the X-ray attachment is presented in Exhibit 13–2. Notice that the attachment has exactly a 10% return on the original investment because the net present value is zero at a 10% discount rate.

Each annual \$1,000 cash inflow arising from use of the attachment is made up of two parts. One part represents a recovery of a portion of the original \$3,170 paid for the attachment, and the other part represents a return on this investment. The breakdown of each year’s \$1,000 cash inflow between recovery of investment and return on investment is shown in Exhibit 13–3.

The first year’s \$1,000 cash inflow consists of a return on investment of \$317 (a 10% return on the \$3,170 original investment), plus a \$683 return of that investment. Because

EXHIBIT 13–2
Carver Hospital—Net Present Value Analysis of X-Ray Attachment

		Initial cost	\$3,170	
		Life of the project	4 years	
		Annual net cash inflow	\$1,000	
		Salvage value	\$0	
		Required rate of return	10%	
Item	Year(s)	Amount of Cash Flow	10% Factor	Present Value of Cash Flows
Annual net cash inflow . . .	1–4	\$1,000	3.170*	\$ 3,170
Initial investment	Now	\$(3,170)	1.000	(3,170)
Net present value				<u>\$ 0</u>
*From Exhibit 13B–2 in Appendix 13B.				

EXHIBIT 13-3Carver Hospital—Breakdown
of Annual Cash Inflows

Year	(1) Investment Outstanding during the Year	(2) Cash Inflow	(3) Return on Investment (1) × 10%	(4) Recovery of Investment during the Year (2) – (3)	(5) Unrecovered Investment at the End of the Year (1) – (4)
1	\$3,170	\$1,000	\$317	\$ 683	\$2,487
2	\$2,487	\$1,000	\$249	751	\$1,736
3	\$1,736	\$1,000	\$173	827	\$909
4	\$909	\$1,000	\$91	909	\$0
Total investment recovered				<u>\$3,170</u>	

the amount of the unrecovered investment decreases each year, the dollar amount of the return on investment also decreases each year. By the end of the fourth year, all \$3,170 of the original investment has been recovered.

Simplifying Assumptions

Two simplifying assumptions are usually made in net present value analysis.

The first assumption is that all cash flows other than the initial investment occur at the end of periods. This is somewhat unrealistic in that cash flows typically occur *throughout* a period rather than just at its end. The purpose of this assumption is to simplify computations.

The second assumption is that all cash flows generated by an investment project are immediately reinvested at a rate of return equal to the discount rate. Unless these conditions are met, the net present value computed for the project will not be accurate. We used a discount rate of 10% for Carver Hospital in Exhibit 13-2. Unless the cash flows in each period are immediately reinvested at a 10% return, the net present value computed for the X-ray attachment will be misstated.

Choosing a Discount Rate

A positive net present value indicates that the project's return exceeds the discount rate. A negative net present value indicates that the project's return is less than the discount rate. Therefore, if the company's minimum required rate of return is used as the discount rate, a project with a positive net present value has a return that exceeds the minimum required rate of return and is acceptable. Contrarily, a project with a negative net present value has a return that is less than the minimum required rate of return and is unacceptable.

What is a company's minimum required rate of return? The company's *cost of capital* is usually regarded as the minimum required rate of return. The **cost of capital** is the average rate of return the company must pay to its long-term creditors and its shareholders for the use of their funds. If a project's rate of return is less than the cost of capital, the company does not earn enough to compensate its creditors and shareholders. Therefore, any project with a rate of return less than the cost of capital should be rejected.

The cost of capital serves as a *screening device*. When the cost of capital is used as the discount rate in net present value analysis, any project with a negative net present value does not cover the company's cost of capital and should be discarded as unacceptable.

IN BUSINESS

BUCK KNIVES PACKS ITS BAGS

Buck Knives was losing money at its plant in San Diego, California. The company responded by loading its entire factory into a caravan of tractor-trailers and moving to Post Falls, Idaho. The relocation cost \$6.5 million, but Buck Knives justified the move based on the annual cost savings summarized below.

	San Diego	Post Falls
Electricity (cost per kilowatt hour)	\$0.118	\$0.031
Workers' compensation (cost per employee per year)	\$2,095	\$210
Median hourly wage	\$15.15	\$12.40
State business tax	8.84%	7.6%
Health insurance (per family per year)	\$9,091	\$8,563
Office space (cost per square foot)	\$34.20	\$1.50
Sales tax	7.75%	5%

How would you analyze the financial viability of this decision? The first step would be to convert the data in the above table into annual lump sum savings. Then, net present value analysis could be used to compare the discounted value of the annual cost savings to the initial cash outlay associated with the relocation.

Source: Chris Lydgate, "The Buck Stopped Here," *Inc.* magazine, May 2006, pp. 86–95.

An Extended Example of the Net Present Value Method

Example C provides an extended example of how the net present value method is used to analyze a proposed project. This example helps tie together and reinforce many of the ideas discussed thus far.



Example C: Under a special licensing arrangement, Swinyard Corporation has an opportunity to market a new product for a five-year period. The product would be purchased from the manufacturer, with Swinyard responsible for promotion and distribution costs. The licensing arrangement could be renewed at the end of the five-year period. After careful study, Swinyard estimated the following costs and revenues for the new product:

Cost of equipment needed	\$60,000
Working capital needed	\$100,000
Overhaul of the equipment in four years	\$5,000
Salvage value of the equipment in five years	\$10,000
Annual revenues and costs:	
Sales revenues	\$200,000
Cost of goods sold	\$125,000
Out-of-pocket operating costs (for salaries, advertising, and other direct costs)	\$35,000

At the end of the five-year period, if Swinyard decides not to renew the licensing arrangement the working capital would be released for investment elsewhere. Swinyard uses a 14% discount rate. Would you recommend that the new product be introduced?

This example involves a variety of cash inflows and cash outflows. The solution is given in Exhibit 13–4.

Notice how the working capital is handled in this exhibit. It is counted as a cash outflow at the beginning of the project and as a cash inflow when it is released at the end of the project. Also notice how the sales revenues, cost of goods sold, and out-of-pocket costs are handled. **Out-of-pocket costs** are actual cash outlays for salaries, advertising, and other operating expenses.

Because the net present value of the proposal is positive, the new product is acceptable.

EXHIBIT 13-4The Net Present Value Method—
An Extended Example

		Sales revenues	\$200,000		
		Less cost of goods sold	125,000		
		Less out-of-pocket costs for salaries, advertising, etc.	35,000		
		Annual net cash inflows	<u>\$ 40,000</u>		
Item	Year(s)	Amount of Cash Flow	14% Factor	Present Value of Cash Flows	
Purchase of equipment	Now	\$(60,000)	1.000	\$ (60,000)	
Working capital needed	Now	\$(100,000)	1.000	(100,000)	
Overhaul of equipment	4	\$(5,000)	0.592*	(2,960)	
Annual net cash inflows from sales of the product line	1–5	\$40,000	3.433 [†]	137,320	
Salvage value of the equipment	5	\$10,000	0.519*	5,190	
Working capital released	5	\$100,000	0.519*	51,900	
Net present value				<u>\$ 31,450</u>	

*From Exhibit 13B–1 in Appendix 13B.

[†]From Exhibit 13B–2 in Appendix 13B.**IN BUSINESS****ECONOMIC WOES SHRINK CAPITAL BUDGETS**

When the health of the economy is uncertain, capital spending declines. **Rite Aid** CEO Mary Sammons cut her company's capital budget by \$50 million due to uncertain economic conditions. **PetroHawk Energy** responded to a weak economy by slashing its \$1.5 billion capital budget by one-third. **Estee Lauder** tightened its belt by challenging managers to defend what they must have and define what they can give up. **YUM! Brands** (owner of **Pizza Hut**, **KFC**, and **Taco Bell**) navigated the difficult economy by abandoning projects that "might come true" in favor of a "must have" capital budgeting mentality.

Source: Matthew Boyle, "The Budget Knives Come Out," *BusinessWeek*, October 13, 2008, p. 30.

Discounted Cash Flows—The Internal Rate of Return Method

The **internal rate of return** is the rate of return of an investment project over its useful life. The internal rate of return is computed by finding the discount rate that equates the present value of a project's cash outflows with the present value of its cash inflows. In other words, the internal rate of return is the discount rate that results in a net present value of zero.

LEARNING OBJECTIVE 2

Evaluate the acceptability of an investment project using the internal rate of return method.

The Internal Rate of Return Method Illustrated

To illustrate the internal rate of return method, consider the following data:

Example D: Glendale School District is considering the purchase of a large tractor-pulled lawn mower. At present, the lawn is mowed using a small hand-pushed gas mower. The large, tractor-pulled mower will cost \$16,950 and will have a useful life of 10 years. It will have a negligible scrap value, which can be ignored. The tractor-pulled mower would do the job faster than the old mower, resulting in labor savings of \$3,000 per year.



To compute the internal rate of return of the new mower, we must find the discount rate that will result in a zero net present value. How do we do this? The simplest and most direct approach *when the net cash inflow is the same every year* is to divide the investment in the project by the expected annual net cash inflow. This computation yields a factor from which the internal rate of return can be determined. The formula is as follows:

$$\text{Factor of the internal rate of return} = \frac{\text{Investment required}}{\text{Annual net cash inflow}} \tag{1}$$

The factor derived from formula (1) is then located in the present value tables to see what rate of return it represents. Using formula (1) and the data for the Glendale School District’s proposed project, we get:

$$\frac{\text{Investment required}}{\text{Annual net cash inflow}} = \frac{\$16,950}{\$3,000} = 5.650$$

Thus, the discount factor that will equate a series of \$3,000 cash inflows with a present investment of \$16,950 is 5.650. Now we need to find this factor in Exhibit 13B–2 in Appendix 13B to see what rate of return it represents. We should use the 10-period line in Exhibit 13B–2 because the cash flows for the project continue for 10 years. If we scan along the 10-period line, we find that a factor of 5.650 represents a 12% rate of return. Therefore, the internal rate of return of the mower project is 12%. We can verify this by computing the project’s net present value using a 12% discount rate as shown in Exhibit 13–5.

Notice from Exhibit 13–5 that using a 12% discount rate equates the present value of the annual net cash inflows with the present value of the investment required for the project, leaving a zero net present value. The 12% rate therefore represents the internal rate of return of the project.

Salvage Value and Other Cash Flows

The technique just demonstrated works if a project’s cash flows are identical every year. But what if they are not? For example, what if a project will have some salvage value at the end of its life in addition to the annual cash inflows? Under these circumstances, a trial-and-error process may be used to find the rate of return that will result in a zero net present value. The trial-and-error process can be carried out by hand; however, computer software programs such as spreadsheets can perform the necessary computations in seconds. Erratic or uneven cash flows should not prevent an analyst from determining a project’s internal rate of return.

EXHIBIT 13–5
Evaluation of the Mower Using a
12% Discount Rate

		Initial cost	\$16,950		
		Life of the project	10 years		
		Annual cost savings	\$3,000		
		Salvage value	\$0		
Item	Year(s)	Amount of Cash Flow	12% Factor	Present Value of Cash Flows	
Annual cost savings	1–10	\$3,000	5.650*	\$16,950	
Initial investment	Now	\$(16,950)	1.000	(16,950)	
Net present value				<u>\$ 0</u>	
*From Exhibit 13B–2 in Appendix 13B.					

Using the Internal Rate of Return

To evaluate a project, the internal rate of return is compared to the company's minimum required rate of return, which is usually the company's cost of capital. If the internal rate of return is equal to or greater than the required rate of return, then the project is acceptable. If the internal rate of return is less than the required rate of return, then the project is rejected.

In the case of the Glendale School District example, let us assume that the district has set a minimum required rate of return of 15% on all projects. Because the large mower's internal rate of return is only 12%, the project does not clear the 15% hurdle and should be rejected.

The Cost of Capital as a Screening Tool

As we have seen in preceding examples, the cost of capital is often used to screen out undesirable investment projects. This screening is accomplished in different ways, depending on whether the company is using the internal rate of return method or the net present value method.

When the internal rate of return method is used, the cost of capital is used as the *hurdle rate* that a project must clear for acceptance. If the internal rate of return of a project is not high enough to clear the cost of capital hurdle, then the project is ordinarily rejected. We saw the application of this idea in the Glendale School District example, where the hurdle rate was set at 15%.

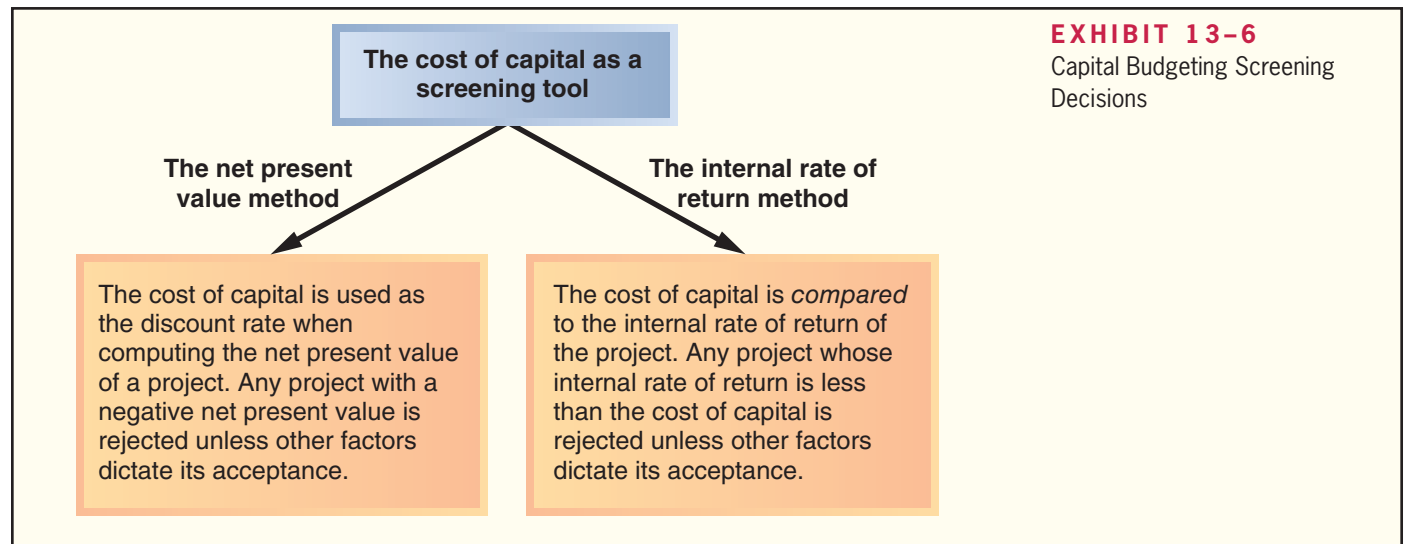
When the net present value method is used, the cost of capital is the *discount rate* used to compute the net present value of a proposed project. Any project yielding a negative net present value is rejected unless other factors are significant enough to warrant its acceptance.

The use of the cost of capital as a screening tool is summarized in Exhibit 13–6.

Comparison of the Net Present Value and Internal Rate of Return Methods

The net present value method has several important advantages over the internal rate of return method.

First, the net present value method is often simpler to use than the internal rate of return method. As mentioned earlier, the internal rate of return method may require hunting for the discount rate that results in a net present value of zero. This can be a very laborious trial-and-error process, although it can be automated using a computer.



Second, the internal rate of return method makes a questionable assumption. Both methods assume that cash flows generated by a project during its useful life are immediately reinvested elsewhere. However, the two methods make different assumptions concerning the rate of return that is earned on those cash flows. The net present value method assumes the rate of return is the discount rate, whereas the internal rate of return method assumes the rate of return earned on cash flows is the internal rate of return on the project. Specifically, if the internal rate of return of the project is high, this assumption may not be realistic. It is generally more realistic to assume that cash inflows can be reinvested at a rate of return equal to the discount rate—particularly if the discount rate is the company's cost of capital or an opportunity rate of return. For example, if the discount rate is the company's cost of capital, this rate of return can be actually realized by paying off the company's creditors and buying back the company's stock with cash flows from the project. In short, when the net present value method and the internal rate of return method do not agree concerning the attractiveness of a project, it is best to go with the net present value method. Of the two methods, it makes the more realistic assumption about the rate of return that can be earned on cash flows from the project.

Expanding the Net Present Value Method

So far, all of our examples have involved an evaluation of a single investment project. In the following section we expand the discussion of the net present value method to include evaluation of two alternative projects. In addition, we integrate relevant cost concepts into the discounted cash flow analysis. We use two approaches to compare competing investment projects—the *total-cost approach* and the *incremental-cost approach*. Each approach is illustrated in the next few pages.

The Total-Cost Approach

The total-cost approach is the most flexible method for comparing competing projects. To illustrate the mechanics of the approach, consider the following data:



Example E: Harper Ferry Company operates a high-speed passenger ferry service across the Mississippi River. One of its ferryboats is in poor condition. This ferry can be renovated at an immediate cost of \$200,000. Further repairs and an overhaul of the motor will be needed five years from now at a cost of \$80,000. In all, the ferry will be usable for 10 years if this work is done. At the end of 10 years, the ferry will have to be scrapped at a salvage value of \$60,000. The scrap value of the ferry right now is \$70,000. It will cost \$300,000 each year to operate the ferry, and revenues will total \$400,000 annually.

As an alternative, Harper Ferry Company can purchase a new ferryboat at a cost of \$360,000. The new ferry will have a life of 10 years, but it will require some repairs costing \$30,000 at the end of 5 years. At the end of 10 years, the ferry will have a scrap value of \$60,000. It will cost \$210,000 each year to operate the ferry, and revenues will total \$400,000 annually.

Harper Ferry Company requires a return of at least 14% on all investment projects.

Should the company purchase the new ferry or renovate the old ferry? Exhibit 13–7 shows the solution using the total-cost approach.

Two points should be noted from the exhibit. First, *all* cash inflows and *all* cash outflows are included in the solution under each alternative. No effort has been made to isolate those cash flows that are relevant to the decision and those that are not relevant. The inclusion of all cash flows associated with each alternative gives the approach its name—the *total-cost approach*.

Second, notice that a net present value is computed for each alternative. This is a distinct advantage of the total-cost approach because an unlimited number of alternatives can be compared side by side to determine the best option. For example, another

EXHIBIT 13-7The Total-Cost Approach
to Project Selection

		New Ferry	Old Ferry		
Annual revenues		\$400,000	\$400,000		
Annual cash operating costs		210,000	300,000		
Annual net cash inflows		<u>\$190,000</u>	<u>\$100,000</u>		
Item	Year(s)	Amount of Cash Flows	14% Factor*	Present Value of Cash Flows	
Buy the new ferry:					
Initial investment	Now	\$(360,000)	1.000	\$(360,000)	
Salvage value of the old ferry	Now	\$70,000	1.000	70,000	
Repairs in five years	5	\$(30,000)	0.519	(15,570)	
Annual net cash inflows	1–10	\$190,000	5.216	991,040	
Salvage value of the new ferry	10	\$60,000	0.270	16,200	
Net present value				<u>701,670</u>	
Keep the old ferry:					
Renovation	Now	\$(200,000)	1.000	(200,000)	
Repairs in five years	5	\$(80,000)	0.519	(41,520)	
Annual net cash inflows	1–10	\$100,000	5.216	521,600	
Salvage value of the old ferry	10	\$60,000	0.270	16,200	
Net present value				<u>296,280</u>	
Net present value in favor of buying the new ferry				<u>\$ 405,390</u>	

*All present value factors are from Exhibits 13B–1 and 13B–2 in Appendix 13B.

alternative for Harper Ferry Company would be to get out of the ferry business entirely. If management desired, the net present value of this alternative could be computed to compare with the alternatives shown in Exhibit 13–7. Still other alternatives might be open to the company. In the case at hand, given only two alternatives, the data indicate that the most profitable choice is to purchase the new ferry.⁴

The Incremental-Cost Approach

When only two alternatives are being considered, the incremental-cost approach offers a simpler and more direct route to a decision. In the incremental-cost approach, only those costs and revenues that *differ* between the two alternatives are included in the analysis. To illustrate, refer again to the data in Example E relating to Harper Ferry Company. The solution using only differential costs is presented in Exhibit 13–8.⁵

Two things should be noted from the data in this exhibit. First, the net present value in favor of buying the new ferry of \$405,390 shown in Exhibit 13–8 agrees with the net present value shown under the total-cost approach in Exhibit 13–7. The two approaches are equivalent.

⁴ The alternative with the highest net present value is not always the best choice, although it is the best choice in this case. For further discussion, see the section Preference Decisions—The Ranking of Investment Projects.

⁵ Technically, the incremental-cost approach is misnamed because it focuses on differential costs (that is, on both cost increases and decreases) rather than just on incremental costs. As used here, the term *incremental costs* should be interpreted broadly to include both cost increases and cost decreases.

EXHIBIT 13-8

The Incremental-Cost Approach to Project Selection

Item	Year(s)	Amount of Cash Flows	14% Factor*	Present Value of Cash Flows
Incremental investment to buy the new ferry	Now	\$(160,000)	1.000	\$(160,000)
Salvage value of the old ferry now	Now	\$70,000	1.000	70,000
Difference in repair costs in five years	5	\$50,000	0.519	25,950
Increase in annual net cash inflows	1-10	\$90,000	5.216	469,440
Difference in salvage value in 10 years	10	\$0	0.270	0
Net present value in favor of buying the new ferry				<u>\$ 405,390</u>

*All present value factors are from Exhibits 13B-1 and 13B-2 in Appendix 13B.

Second, the costs used in Exhibit 13-8 are just the differences between the costs shown for the two alternatives in the prior exhibit. For example, the \$160,000 incremental investment required to purchase the new ferry in Exhibit 13-8 is the difference between the \$360,000 cost of the new ferry and the \$200,000 cost required to renovate the old ferry from Exhibit 13-7. The other numbers in Exhibit 13-8 have been computed in the same way.

Least-Cost Decisions



Some decisions do not involve any revenues. For example, a company may be trying to decide whether to buy or lease an executive jet. The choice would be made on the basis of which alternative—buying or leasing—would be least costly. In situations such as these, where no revenues are involved, the most desirable alternative is the one with the *least total cost* from a present value perspective. Hence, these are known as least-cost decisions. To illustrate a least-cost decision, consider the following data:

Example F: Val-Tek Company is considering replacing an old threading machine with a new threading machine that would substantially reduce annual operating costs. Selected data relating to the old and new machines are presented below:

	Old Machine	New Machine
Purchase cost when new	\$200,000	\$250,000
Salvage value now	\$30,000	—
Annual cash operating costs	\$150,000	\$90,000
Overhaul needed immediately	\$40,000	—
Salvage value in six years	\$0	\$50,000
Remaining life	6 years	6 years

Val-Tek Company uses a 10% discount rate.

Exhibit 13-9 analyzes the alternatives using the total-cost approach. Because this is a least-cost decision, the present values are negative for both alternatives. However, the present value of the alternative of buying the new machine is \$109,500 higher than the other alternative. Therefore, buying the new machine is the less costly alternative.

Exhibit 13-10 presents an analysis of the same alternatives using the incremental-cost approach. Once again, when done correctly, the total-cost and incremental-cost approaches arrive at the same answer.

EXHIBIT 13-9The Total-Cost Approach
(Least-Cost Decision)

Item	Year(s)	Amount of Cash Flows	10% Factor*	Present Value of Cash Flows
Buy the new machine:				
Initial investment	Now	\$(250,000)	1.000	\$(250,000) [†]
Salvage value of the old machine	Now	\$30,000	1.000	30,000 [†]
Annual cash operating costs	1–6	\$(90,000)	4.355	(391,950)
Salvage value of the new machine	6	\$50,000	0.564	28,200
Present value of net cash outflows				<u>(583,750)</u>
Keep the old machine:				
Overhaul needed now	Now	\$(40,000)	1.000	\$ (40,000)
Annual cash operating costs	1–6	\$(150,000)	4.355	(653,250)
Present value of net cash outflows				<u>(693,250)</u>
Net present value in favor of buying the new machine				<u>\$ 109,500</u>

*All factors are from Exhibits 13B-1 and 13B-2 in Appendix 13B.

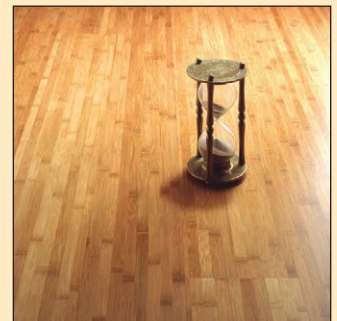
[†]These two items could be netted into a single \$220,000 incremental-cost figure (\$250,000 – \$30,000 = \$220,000).**EXHIBIT 13-10**The Incremental-Cost Approach
(Least-Cost Decision)

Item	Year(s)	Amount of Cash Flows	10% Factor*	Present Value of Cash Flows
Incremental investment required to purchase the new machine				
Incremental investment required to purchase the new machine	Now	\$(210,000)	1.000	\$(210,000) [†]
Salvage value of the old machine	Now	\$30,000	1.000	30,000 [†]
Savings in annual cash operating costs	1–6	\$60,000	4.355	261,300
Difference in salvage value in six years	6	\$50,000	0.564	28,200
Net present value in favor of buying the new machine				<u>\$ 109,500</u>

*All factors are from Exhibits 13B-1 and 13B-2 in Appendix 13B.

[†]These two items could be netted into a single \$180,000 incremental-cost figure (\$210,000 – \$30,000 = \$180,000).**HOME CONSTRUCTION GOES GREEN—OR DOES IT?**

Many homebuyers like the idea of building environmentally friendly homes until they get the bill. **Specpan**, an Indianapolis research firm, estimates a “green” home costs 10%–19% more than a comparable conventional home. For example, installing solar-electric glass-faced tiles on a roof costs \$15,000 per 100 square feet compared to \$1,200 per 100 square feet for standard fiber-cement tiles. Environmentally friendly interior paint costs \$35–\$42 per gallon compared to \$20–\$32 per gallon for standard latex paint. To further complicate this least-cost decision, the average homeowner lives in a house only seven years before moving. Within this time frame, many green investments appear to be financially unattractive. Nonetheless, the American Institute of Architects reports that 63% of their clients expressed an interest in renewable flooring materials such as cork and bamboo, up from 53% a year earlier.

Source: June Fletcher, “The Price of Going Green,” *The Wall Street Journal*, February 29, 2008, p. W8.**IN BUSINESS**

Uncertain Cash Flows

LEARNING OBJECTIVE 3
Evaluate an investment project that has uncertain cash flows.

Thus far, we have assumed that all future cash flows are known with certainty. However, future cash flows are often uncertain or difficult to estimate. A number of techniques are available for handling this complication. Some of these techniques are quite technical—involving computer simulations or advanced mathematical skills—and are beyond the scope of this book. However, we can provide some very useful information to managers without getting too technical.

An Example

As an example of difficult-to-estimate future cash flows, consider the case of investments in automated equipment. The up-front costs of automated equipment and the tangible benefits, such as reductions in operating costs and waste, tend to be relatively easy to estimate. However, the intangible benefits, such as greater reliability, greater speed, and higher quality, are more difficult to quantify in terms of future cash flows. These intangible benefits certainly impact future cash flows—particularly in terms of increased sales and perhaps higher selling prices—but the cash flow effects are difficult to estimate. What can be done?

A fairly simple procedure can be followed when the intangible benefits are likely to be significant. Suppose, for example, that a company with a 12% discount rate is considering purchasing automated equipment that would have a 10-year useful life. Also suppose that a discounted cash flow analysis of just the tangible costs and benefits shows a negative net present value of \$226,000. Clearly, if the intangible benefits are large enough, they could turn this negative net present value into a positive net present value. In this case, the amount of additional cash flow per year from the intangible benefits that would be needed to make the project financially attractive can be computed as follows:

Net present value excluding the intangible benefits (negative)	\$(226,000)
Present value factor for an annuity at 12% for 10 periods (from Exhibit 13B–2 in Appendix 13B)	5.650

$$\frac{\text{Negative net present value to be offset, \$226,000}}{\text{Present value factor, 5.650}} = \$40,000$$

Thus, if the intangible benefits of the automated equipment are worth at least \$40,000 a year to the company, then the automated equipment should be purchased. If, in the judgment of management, these intangible benefits are not worth \$40,000 a year, then the automated equipment should not be purchased.

This technique can be used in other situations in which future cash flows are difficult to estimate. For example, this technique can be used when the salvage value is difficult to estimate. To illustrate, suppose that all of the cash flows from an investment in a supertanker have been estimated—other than its salvage value in 20 years. Using a discount rate of 12%, management has determined that the net present value of all of these cash flows is a negative \$1.04 million. This negative net present value would be offset by the salvage value of the supertanker. How large would the salvage value have to be to make this investment attractive?

Net present value excluding salvage value (negative)	\$(1,040,000)
Present value factor at 12% for 20 periods (from Exhibit 13B–1 in Appendix 13B)	0.104

$$\frac{\text{Negative net present value to be offset, \$1,040,000}}{\text{Present value factor, 0.104}} = \$10,000,000$$

Thus, if the salvage value of the tanker in 20 years is at least \$10 million, its net present value would be positive and the investment would be made. However, if management believes the salvage value is unlikely to be as large as \$10 million, the investment should not be made.

Real Options

The analysis in this chapter has assumed that an investment cannot be postponed and that, once started, nothing can be done to alter the course of the project. In reality, investments can often be postponed. Postponement is a particularly attractive option when the net present value of a project is modest using current estimates of future cash flows and the future cash flows involve a great deal of uncertainty that may be resolved over time. Similarly, once an investment is made, management can often exploit changes in the business environment and take actions that enhance future cash flows. For example, buying a supertanker provides management with a number of options, some of which may become more attractive as time unfolds. Instead of operating the supertanker itself, the company may decide to lease it to another operator if the rental rates become high enough. Or, if a supertanker shortage develops, management may decide to sell the supertanker and take a gain. In the case of an investment in automated equipment, management may initially buy only the basic model without costly add-ons, but keep the option open to add more capacity and capability later. The ability to delay the start of a project, to expand it if conditions are favorable, to cut losses if they are unfavorable, and to otherwise modify plans as business conditions change adds value to many investments. These advantages can be quantified using what is called *real options* analysis, but the techniques are beyond the scope of this book.

Preference Decisions—The Ranking of Investment Projects

Recall that when considering investment opportunities, managers must make two types of decisions—screening decisions and preference decisions. Screening decisions, which come first, pertain to whether or not a proposed investment is acceptable. Preference decisions come *after* screening decisions and attempt to answer the following question: “How do the remaining investment proposals, all of which have been screened and provide an acceptable rate of return, rank in terms of preference? That is, which one(s) would be *best* for the company to accept?”

Sometimes preference decisions are called rationing decisions, or ranking decisions. Limited investment funds must be rationed among many competing alternatives. Hence, the alternatives must be ranked. Either the internal rate of return method or the net present value method can be used in making preference decisions. However, as discussed earlier, if the two methods are in conflict, it is best to use the net present value method, which is more reliable.

LEARNING OBJECTIVE 4

Rank investment projects in order of preference.

Internal Rate of Return Method

When using the internal rate of return method to rank competing investment projects, the preference rule is: *The higher the internal rate of return, the more desirable the project.* An investment project with an internal rate of return of 18% is usually considered preferable to another project that has a return of only 15%. Internal rate of return is widely used to rank projects.

Net Present Value Method

The net present value of one project cannot be directly compared to the net present value of another project unless the initial investments are equal. For example, assume that a company is considering two competing investments, as shown below:

	Investment	
	A	B
Investment required	\$(10,000)	\$(5,000)
Present value of cash inflows	11,000	6,000
Net present value	<u>\$ 1,000</u>	<u>\$ 1,000</u>

Although each project has a net present value of \$1,000, the projects are not equally desirable if the funds available for investment are limited. The project requiring an investment of only \$5,000 is much more desirable than the project requiring an investment of \$10,000. This fact can be highlighted by dividing the net present value of the project by the investment required. The result, shown below in equation form, is called the **project profitability index**.

$$\text{Project profitability index} = \frac{\text{Net present value of the project}}{\text{Investment required}} \quad (2)$$

The project profitability indexes for the two investments above would be computed as follows:

	Investment	
	A	B
Net present value (a)	\$1,000	\$1,000
Investment required (b)	\$10,000	\$5,000
Project profitability index, (a) ÷ (b)	0.10	0.20

When using the project profitability index to rank competing investments projects, the preference rule is: *The higher the project profitability index, the more desirable the project.*⁶ Applying this rule to the two investments above, investment B should be chosen over investment A.

The project profitability index is an application of the techniques for utilizing constrained resources discussed in an earlier chapter. In this case, the constrained resource is the limited funds available for investment, and the project profitability index is similar to the contribution margin per unit of the constrained resource.

A few details should be clarified with respect to the computation of the project profitability index. The “Investment required” refers to any cash outflows that occur at the beginning of the project, reduced by any salvage value recovered from the sale of old equipment. The “Investment required” also includes any investment in working capital that the project may need.

⁶ Because of the “lumpiness” of projects, the project profitability index ranking may not be perfect. Nevertheless, it is a good starting point. For further details, see the Profitability Analysis Appendix at the end of the book.

IN BUSINESS

FEDEX GOES GREEN—WELL, NOT EXACTLY!

In 2003, FedEx announced a 10-year plan to replace 3,000 delivery trucks per year with environmentally friendly hybrid vehicles, thereby eliminating 250,000 tons of greenhouse gases per year. The hybrid vehicles cost 75% more than conventional trucks, but over 10 years they generate fuel savings that offset the higher cost. By 2007, FedEx had purchased less than 100 hybrid vehicles because management decided that the eco-friendly investment was not the most profitable use of the company's resources. FedEx's environmental director justified the company's actions by saying "we do have a fiduciary responsibility to our shareholders . . . we can't subsidize the development of this technology for our competitors."

This example illustrates the challenges that companies face when attempting to satisfy the expectations of various stakeholders. Perhaps FedEx shareholders would applaud the company's decision to retreat from its 10-year plan, whereas environmentally conscious customers may criticize the company's actions. What do you think?

Source: Ben Elgin, "Little Green Lies," *BusinessWeek*, October 29, 2007, pp. 45–52.



Other Approaches to Capital Budgeting Decisions

The net present value and internal rate of return methods are widely used as decision-making tools. However, some managers also use the payback method and simple rate of return method to make capital budgeting decisions. Each of these methods will be discussed in turn.

The Payback Method

The payback method focuses on the *payback period*. The **payback period** is the length of time that it takes for a project to recover its initial cost from the net cash inflows that it generates. This period is sometimes referred to as "the time that it takes for an investment to pay for itself." The basic premise of the payback method is that the more quickly the cost of an investment can be recovered, the more desirable is the investment.

The payback period is expressed in years. *When the annual net cash inflow is the same every year*, the following formula can be used to compute the payback period:

$$\text{Payback period} = \frac{\text{Investment required}}{\text{Annual net cash inflow}} \quad (3)$$

To illustrate the payback method, consider the following data:

Example G: York Company needs a new milling machine. The company is considering two machines: machine A and machine B. Machine A costs \$15,000, has a useful life of ten years, and will reduce operating costs by \$5,000 per year. Machine B costs only \$12,000, will also reduce operating costs by \$5,000 per year, but has a useful life of only five years.

Required:

Which machine should be purchased according to the payback method?

$$\text{Machine A payback period} = \frac{\$15,000}{\$5,000} = 3.0 \text{ years}$$

$$\text{Machine B payback period} = \frac{\$12,000}{\$5,000} = 2.4 \text{ years}$$

According to the payback calculations, York Company should purchase machine B because it has a shorter payback period than machine A.

LEARNING OBJECTIVE 5

Determine the payback period for an investment.

IN BUSINESS

ENTREPRENEURIAL INGENUITY AT ITS BEST

Jonathan Pratt owns two Ümani Cafés in Westchester County, New York. He used to pay \$200 a month to dispose of the vegetable oil that is used to fry foods in his restaurants. Plus, he bought \$700 of gas every month to operate his company's pick-up truck. Then Pratt got an idea. He purchased a diesel-powered Ford F250 on eBay for \$11,000 and paid \$1,500 to haul the truck from Arizona to New York. Next, he installed an \$850 conversion kit on his new truck to enable it to run on vegetable oil. Because he no longer has to pay to dispose of vegetable oil or buy gasoline, Pratt figures that his investment will pay for itself in about 15 months ($\$13,350 \div \$900 = 14.83$ months). Furthermore, he now has the best smelling car in town—it smells like french fries when he drives down the road.

Source: Jean Chatzky, "Out of the Frying Pan, Into the Ford," *Money*, October, 2004, p. 28.

Evaluation of the Payback Method

The payback method is not a true measure of the profitability of an investment. Rather, it simply tells a manager how many years are required to recover the original investment. Unfortunately, a shorter payback period does not always mean that one investment is more desirable than another.

To illustrate, refer back to Example G on the previous page. Machine B has a shorter payback period than machine A, but it has a useful life of only 5 years rather than 10 years for machine A. Machine B would have to be purchased twice—once immediately and then again after the fifth year—to provide the same service as just one machine A. Under these circumstances, machine A would probably be a better investment than machine B, even though machine B has a shorter payback period. Unfortunately, the payback method ignores all cash flows that occur after the payback period.

A further criticism of the payback method is that it does not consider the time value of money. A cash inflow to be received several years in the future is weighed the same as a cash inflow received right now. To illustrate, assume that for an investment of \$8,000 you can purchase either of the two following streams of cash inflows:

Year	0	1	2	3	4	5	6	7	8
Stream 1 . . .					\$8,000	\$2,000	\$2,000	\$2,000	\$2,000
Stream 2 . . .		\$2,000	\$2,000	\$2,000	\$2,000	\$8,000			

Which stream of cash inflows would you prefer to receive in return for your \$8,000 investment? Each stream has a payback period of 4.0 years. Therefore, if payback alone is used to make the decision, the streams would be considered equally desirable. However, from a time value of money perspective, stream 2 is much more desirable than stream 1.

On the other hand, under certain conditions the payback method can be very useful. For one thing, it can help identify which investment proposals are in the "ballpark." That is, it can be used as a screening tool to help answer the question, "Should I consider this proposal further?" If a proposal doesn't provide a payback within some specified period, then there may be no need to consider it further. In addition, the payback period is often important to new companies that are "cash poor." When a company is cash poor, a project with a short payback period but a low rate of return might be preferred over another project with a high rate of return but a long payback period. The reason is that the company may simply need a faster return of its cash investment. And finally, the payback method is sometimes used in industries where products become obsolete very rapidly—such as consumer electronics. Because products may last only a year or two, the payback period on investments must be very short.

THE ECONOMICS OF HYBRID VEHICLES

The table below shows the price premiums (after tax credits) that customers must pay to buy four types of hybrid vehicles. It also depicts the annual gas savings that customers realize by driving a hybrid version of the vehicle instead of a standard model of the same vehicle (assuming the vehicles are driven 15,000 miles per year and gas costs \$2.79 a gallon). Dividing the price premium by the annual gas savings yields the payback period when purchasing the hybrid version of the vehicle.

Make and Model	Price Premium	Annual Gas Savings	Payback Period
Ford Escape	\$1,364	\$438	3.1
Honda Civic	\$1,482	\$317	4.7
Toyota Camry	\$3,763	\$310	12.1
Toyota Highlander	\$4,372	\$388	11.3

The above payback figures highlight the dilemma faced by customers who want to make environmentally friendly purchases, but are constrained by limited financial resources.

Source: Mike Spector, "The Economics of Hybrids," *The Wall Street Journal*, October 29, 2007, pp. R5–R6.

An Extended Example of Payback

As shown by formula (3) on page 597, the payback period is computed by dividing the investment in a project by the project's annual net cash inflows. If new equipment is replacing old equipment, then any salvage value to be received when disposing of the old equipment should be deducted from the cost of the new equipment, and only the *incremental* investment should be used in the payback computation. In addition, any depreciation deducted in arriving at the project's net operating income must be added back to obtain the project's expected annual net cash inflow. To illustrate, consider the following data:

Example H: Goodtime Fun Centers, Inc., operates amusement parks. Some of the vending machines in one of its parks provide very little revenue, so the company is considering removing the machines and installing equipment to dispense soft ice cream. The equipment would cost \$80,000 and have an eight-year useful life with no salvage value. Incremental annual revenues and costs associated with the sale of ice cream would be as follows:



Sales	\$150,000
Variable expenses	90,000
Contribution margin	60,000
Fixed expenses:	
Salaries	27,000
Maintenance	3,000
Depreciation	10,000
Total fixed expenses	40,000
Net operating income	\$ 20,000

The vending machines can be sold for a \$5,000 scrap value. The company will not purchase equipment unless it has a payback period of three years or less. Does the ice cream dispenser pass this hurdle?

Exhibit 13–11 computes the payback period for the ice cream dispenser. Several things should be noted. First, depreciation is added back to net operating income to obtain

EXHIBIT 13-11

Computation of the Payback Period

Step 1: *Compute the annual net cash inflow.* Because the annual net cash inflow is not given, it must be computed before the payback period can be determined:

Net operating income	\$20,000
Add: Noncash deduction for depreciation	10,000
Annual net cash inflow	<u>\$30,000</u>

Step 2: *Compute the payback period.* Using the annual net cash inflow from above, the payback period can be determined as follows:

Cost of the new equipment	\$80,000
Less salvage value of old equipment	5,000
Investment required	<u>\$75,000</u>

$$\begin{aligned}\text{Payback period} &= \frac{\text{Investment required}}{\text{Annual net cash inflow}} \\ &= \frac{\$75,000}{\$30,000} = 2.5 \text{ years}\end{aligned}$$

the annual net cash inflow from the new equipment. Depreciation is not a cash outlay; thus, it must be added back to adjust net operating income to a cash basis. Second, the payback computation deducts the salvage value of the old machines from the cost of the new equipment so that only the incremental investment is used in computing the payback period.

Because the proposed equipment has a payback period of less than three years, the company's payback requirement has been met.

Payback and Uneven Cash Flows

When the cash flows associated with an investment project change from year to year, the simple payback formula that we outlined earlier cannot be used. Consider the following data:

Year	Investment	Cash Inflow
1	\$4,000	\$1,000
2		\$0
3		\$2,000
4	\$2,000	\$1,000
5		\$500
6		\$3,000
7		\$2,000

What is the payback period on this investment? The answer is 5.5 years, but to obtain this figure it is necessary to track the unrecovered investment year by year. The steps involved in this process are shown in Exhibit 13-12. By the middle of the sixth year, sufficient cash inflows will have been realized to recover the entire investment of \$6,000 (\$4,000 + \$2,000).

Year	Investment	Cash Inflow	Unrecovered Investment*
1	\$4,000	\$1,000	\$3,000
2		\$0	\$3,000
3		\$2,000	\$1,000
4	\$2,000	\$1,000	\$2,000
5		\$500	\$1,500
6		\$3,000	\$0
7		\$2,000	\$0

*Year X unrecovered investment = Year X-1 unrecovered investment + Year X investment – Year X cash inflow

EXHIBIT 13-12

Payback and Uneven Cash Flows

The Simple Rate of Return Method

The **simple rate of return** method is another capital budgeting technique that does not involve discounting cash flows. The simple rate of return is also known as the accounting rate of return or the unadjusted rate of return.

Unlike the other capital budgeting methods that we have discussed, the simple rate of return method focuses on accounting net operating income rather than cash flows. To obtain the simple rate of return, the annual incremental net operating income generated by a project is divided by the initial investment in the project, as shown below.

LEARNING OBJECTIVE 6

Compute the simple rate of return for an investment.

$$\text{Simple rate of return} = \frac{\text{Annual incremental net operating income}}{\text{Initial investment}} \quad (4)$$

Two additional points should be made. First, depreciation charges that result from making the investment should be deducted when determining the annual incremental net operating income. Second, the initial investment should be reduced by any salvage value realized from the sale of old equipment.

Example 1: Brigham Tea, Inc., is a processor of low-acid tea. The company is contemplating purchasing equipment for an additional processing line that would increase revenues by \$90,000 per year. Incremental cash operating expenses would be \$40,000 per year. The equipment would cost \$180,000 and have a nine-year life with no salvage value.

To apply the formula for the simple rate of return, we must first determine the annual incremental net operating income from the project:

Annual incremental revenues		\$90,000
Annual incremental cash operating expenses	\$40,000	
Annual depreciation (\$180,000 – \$0)/9	<u>20,000</u>	
Annual incremental expenses		<u>60,000</u>
Annual incremental net operating income		<u>\$30,000</u>

Given that the annual incremental net operating income from the project is \$30,000 and the initial investment is \$180,000, the simple rate of return is 16.7% as shown below:

$$\begin{aligned}
 \text{Simple rate of return} &= \frac{\text{Annual incremental net operating income}}{\text{Initial investment}} \\
 &= \frac{\$30,000}{\$180,000} \\
 &= 16.7\%
 \end{aligned}$$



Example J: Midwest Farms, Inc., hires people on a part-time basis to sort eggs. The cost of hand sorting is \$30,000 per year. The company is investigating an egg-sorting machine that would cost \$90,000 and have a 15-year useful life. The machine would have negligible salvage value, and it would cost \$10,000 per year to operate and maintain. The egg-sorting equipment currently being used could be sold now for a scrap value of \$2,500.

This project is slightly different from the preceding project because it involves cost reductions with no additional revenues. Nevertheless, the annual incremental net operating income can be computed by treating the annual cost savings as if it were incremental revenues as follows:

Annual incremental cost savings		\$30,000
Annual incremental cash operating expenses	\$10,000	
Annual depreciation (\$90,000 – \$0)/15	<u>6,000</u>	
Annual incremental expenses		<u>16,000</u>
Annual incremental net operating income		<u>\$14,000</u>

Thus, even though the new equipment would not generate any additional revenues, it would reduce costs by \$14,000 a year. This would have the effect of increasing net operating income by \$14,000 a year.

Finally, the salvage value of the old equipment offsets the initial cost of the new equipment as follows:

Cost of the new equipment	\$90,000
Less salvage value of the old equipment	<u>2,500</u>
Initial investment	<u>\$87,500</u>

Given the annual incremental net operating income of \$14,000 and the initial investment of \$87,500, the simple rate of return is 16.0% computed as follows:

$$\begin{aligned}
 \text{Simple rate of return} &= \frac{\text{Annual incremental net operating income}}{\text{Initial investment}} \\
 &= \frac{\$14,000}{\$87,500} \\
 &= 16.0\%
 \end{aligned}$$

Criticisms of the Simple Rate of Return

The simple rate of return method ignores the time value of money. It considers a dollar received 10 years from now to be as valuable as a dollar received today. Thus, the simple rate of return method can be misleading if the alternatives have different cash flow patterns. Additionally, many projects do not have constant incremental revenues and expenses over their useful lives. As a result, the simple rate of return will fluctuate from year to year, with the possibility that a project may appear to be desirable in some years and undesirable in others. In contrast, the net present value method provides a single number that summarizes all of the cash flows over the entire useful life of the project.

Postaudit of Investment Projects

After an investment project has been approved and implemented, a *postaudit* should be conducted. A **postaudit** involves checking whether or not expected results are actually realized. This is a key part of the capital budgeting process because it helps keep managers honest in their investment proposals. Any tendency to inflate the benefits or downplay

the costs in a proposal should become evident after a few postaudits have been conducted. The postaudit also provides an opportunity to reinforce and possibly expand successful projects and to cut losses on floundering projects.

The same capital budgeting method should be used in the postaudit as was used in the original approval process. That is, if a project was approved on the basis of a net present value analysis, then the same procedure should be used in performing the postaudit. However, the data used in the postaudit analysis should be *actual observed data* rather than estimated data. This gives management an opportunity to make a side-by-side comparison to see how well the project has succeeded. It also helps assure that estimated data received on future proposals will be carefully prepared because the persons submitting the data knows that their estimates will be compared to actual results in the postaudit process. Actual results that are far out of line with original estimates should be carefully reviewed.

ROYAL CARIBBEAN CRUISES LAUNCHES OASIS OF THE SEAS

Royal Caribbean Cruises invested \$1.4 billion to build the *Oasis of the Seas*, a cruise ship that carries 5,400 passengers and stands 20 stories above the sea. The vessel is a third larger than any other cruise ship and contains 21 pools, 24 restaurants, 13 retail shops, and 300-foot water slides. The company hopes the ship's extraordinary amenities will attract large numbers of customers willing to pay premium prices. However, the economic downturn has caused many customers to refrain from spending on lavish vacations.

Source: Mike Esterl, "Huge Cruise Ships Prepare for Launch but Face Uncertain Waters," *The Wall Street Journal*, December 4, 2009, pp. B1–B2.

IN BUSINESS



Summary

Investment decisions should take into account the time value of money because a dollar today is more valuable than a dollar received in the future. The net present value and internal rate of return methods both reflect this fact. In the net present value method, future cash flows are discounted to their present value. The difference between the present value of the cash inflows and the present value of the cash outflows is called a project's net present value. If the net present value of a project is negative, the project is rejected. The discount rate in the net present value method is usually based on a minimum required rate of return such as a company's cost of capital.

The internal rate of return is the rate of return that equates the present value of the cash inflows and the present value of the cash outflows, resulting in a zero net present value. If the internal rate of return is less than a company's minimum required rate of return, the project is rejected.

After rejecting projects whose net present values are negative or whose internal rates of return are less than the minimum required rate of return, more projects may remain than can be supported with available funds. The remaining projects can be ranked using either the project profitability index or internal rate of return. The project profitability index is computed by dividing the net present value of the project by the required initial investment.

Some companies prefer to use either the payback method or the simple rate of return to evaluate investment proposals. The payback period is the number of periods that are required to fully recover the initial investment in a project. The simple rate of return is determined by dividing a project's accounting net operating income by the initial investment in the project.

Review Problem: Comparison of Capital Budgeting Methods

Lamar Company is considering a project that would have an eight-year life and require a \$2,400,000 investment in equipment. At the end of eight years, the project would terminate and

the equipment would have no salvage value. The project would provide net operating income each year as follows:

Sales		\$3,000,000
Variable expenses		<u>1,800,000</u>
Contribution margin		1,200,000
Fixed expenses:		
Advertising, salaries, and other fixed out-of-pocket costs	\$700,000	
Depreciation	<u>300,000</u>	
Total fixed expenses		1,000,000
Net operating income		<u>\$ 200,000</u>

The company's discount rate is 12%.

Required:

1. Compute the annual net cash inflow from the project.
2. Compute the project's net present value. Is the project acceptable?
3. Find the project's internal rate of return to the nearest whole percent.
4. Compute the project's payback period.
5. Compute the project's simple rate of return.

Solution to Review Problem

1. The annual net cash inflow can be computed by deducting the cash expenses from sales:

Sales	\$3,000,000
Variable expenses	<u>1,800,000</u>
Contribution margin	1,200,000
Advertising, salaries, and other fixed out-of-pocket costs	<u>700,000</u>
Annual net cash inflow	<u>\$ 500,000</u>

Or the annual net cash inflow can be computed by adding depreciation back to net operating income:

Net operating income	\$200,000
Add: Noncash deduction for depreciation . . .	<u>300,000</u>
Annual net cash inflow	<u>\$500,000</u>

2. The net present value is computed as follows:

Item	Year(s)	Amount of Cash Flows	12% Factor	Present Value of Cash Flows
Cost of new equipment	Now	\$(2,400,000)	1.000	\$(2,400,000)
Annual net cash inflow	1–8	\$500,000	4.968	<u>2,484,000</u>
Net present value				<u>\$ 84,000</u>

Yes, the project is acceptable because it has a positive net present value.

3. The formula for computing the factor of the internal rate of return is:

$$\begin{aligned}
 \text{Factor of the internal rate of return} &= \frac{\text{Investment required}}{\text{Annual net cash inflow}} \\
 &= \frac{\$2,400,000}{\$500,000} = 4.800
 \end{aligned}$$

Looking in Exhibit 13B–2 in Appendix 13B at the end of the chapter and scanning along the 8-period line, we find that a factor of 4.800 represents a rate of return of about 13%.

4. The formula for the payback period is:

$$\begin{aligned}\text{Payback period} &= \frac{\text{Investment required}}{\text{Annual net cash flow}} \\ &= \frac{\$2,400,000}{\$500,000} \\ &= 4.8 \text{ years}\end{aligned}$$

5. The formula for the simple rate of return is:

$$\begin{aligned}\text{Simple rate of return} &= \frac{\text{Annual incremental net operating income}}{\text{Initial investment}} \\ &= \frac{\$200,000}{\$2,400,000} \\ &= 8.3\%\end{aligned}$$

Glossary

Capital budgeting The process of planning significant investments in projects that have long-term implications such as the purchase of new equipment or the introduction of a new product. (p. 580)

Cost of capital The average rate of return a company must pay to its long-term creditors and shareholders for the use of their funds. (p. 585)

Internal rate of return The discount rate at which the net present value of an investment project is zero; the rate of return of a project over its useful life. (p. 587)

Net present value The difference between the present value of an investment project's cash inflows and the present value of its cash outflows. (p. 581)

Out-of-pocket costs Actual cash outlays for salaries, advertising, repairs, and similar costs. (p. 586)

Payback period The length of time that it takes for a project to fully recover its initial cost out of the net cash inflows that it generates. (p. 597)

Postaudit The follow-up after a project has been approved and implemented to determine whether expected results were actually realized. (p. 602)

Preference decision A decision in which the alternatives must be ranked. (p. 580)

Project profitability index The ratio of the net present value of a project's cash flows to the investment required. (p. 596)

Screening decision A decision as to whether a proposed investment project is acceptable. (p. 580)

Simple rate of return The rate of return computed by dividing a project's annual incremental accounting net operating income by the initial investment required. (p. 601)

Working capital Current assets less current liabilities. (p. 583)

Questions

- 13–1 What is the difference between capital budgeting screening decisions and capital budgeting preference decisions?
- 13–2 What is meant by the term *time value of money*?
- 13–3 What is meant by the term *discounting*?
- 13–4 Why isn't accounting net income used in the net present value and internal rate of return methods of making capital budgeting decisions?
- 13–5 Why are discounted cash flow methods of making capital budgeting decisions superior to other methods?
- 13–6 What is net present value? Can it ever be negative? Explain.
- 13–7 Identify two simplifying assumptions associated with discounted cash flow methods of making capital budgeting decisions.

Appendix 13C: Income Taxes in Capital Budgeting Decisions

We ignored income taxes in this chapter for two reasons. First, many organizations do not pay income taxes. Not-for-profit organizations, such as hospitals and charitable foundations, and governmental agencies are exempt from income taxes. Second, capital budgeting is complex and is best absorbed in small doses. Now that we have a solid foundation in the concepts of present value and discounting, we can explore the effects of income taxes on capital budgeting decisions.

The U.S. income tax code is enormously complex. We only scratch the surface here. To keep the subject within reasonable bounds, we have made many simplifying assumptions about the tax code. Among the most important of these assumptions are: (1) taxable income equals net income as computed for financial reports; and (2) the tax rate is a flat percentage of taxable income. The actual tax code is far more complex than this; indeed, experts acknowledge that no one person knows or can know it all. However, the simplifications that we make throughout this appendix allow us to cover the most important implications of income taxes for capital budgeting without getting bogged down in details.

LEARNING OBJECTIVE 8

Include income taxes in a capital budgeting analysis.

The Concept of After-Tax Cost

Businesses, like individuals, must pay income taxes. In the case of businesses, the amount of income tax that must be paid is determined by the company's net taxable income. Tax deductible expenses (tax deductions) decrease the company's net taxable income and, hence, reduce the taxes the company must pay. For this reason, expenses are often stated on an *after-tax* basis. For example, if a company pays rent of \$10 million a year but this expense results in a reduction in income taxes of \$3 million, the after-tax cost of the rent is \$7 million. An expenditure net of its tax effect is known as **after-tax cost**.

To illustrate, assume that a company with a tax rate of 30% is contemplating a training program that costs \$60,000. What impact will this have on the company's taxes? To keep matters simple, let's suppose the training program has no immediate effect on sales. How much does the company actually pay for the training program after taking into account the impact of this expense on taxes? The answer is **\$42,000** as shown in Exhibit 13C-1. While the training program costs **\$60,000** before taxes, it would reduce the company's taxes by **\$18,000**, so its *after-tax* cost would be only \$42,000.

	Without Training Program	With Training Program
Sales	\$850,000	\$850,000
Less tax deductible expenses:		
Salaries, insurance, and other	700,000	700,000
New training program		60,000
Total expenses	<u>700,000</u>	<u>760,000</u>
Taxable income	<u>\$150,000</u>	<u>\$ 90,000</u>
Income taxes (30%)	<u>\$ 45,000</u>	<u>\$ 27,000</u>
Cost of new training program		\$60,000
Less: Reduction in income taxes (\$45,000 – \$27,000)		<u>18,000</u>
After-tax cost of the new training program . . .		<u><u>\$42,000</u></u>

EXHIBIT 13C-1

The Computation of After-Tax Cost

The after-tax cost of any tax-deductible cash expense can be determined using the following formula:¹

$$\text{After-tax cost (net cash outflow)} = (1 - \text{Tax rate}) \times \text{Tax-deductible cash expense} \quad (1)$$

We can verify the accuracy of this formula by applying it to the \$60,000 training program expenditure:

$$(1 - 0.30) \times \$60,000 = \$42,000 \text{ after-tax cost of the training program}$$

This formula is very useful because it provides the actual amount of cash a company must pay after considering tax effects. It is this actual, after-tax, cash outflow that should be used in capital budgeting decisions.

Similar reasoning applies to revenues and other *taxable* cash inflows. Because these cash receipts are taxable, the company must pay out a portion of them in taxes. The **after-tax benefit**, or net cash inflow, realized from a particular cash receipt can be obtained by applying a simple variation of the cash expenditure formula used above:

$$\text{After-tax benefit (net cash outflow)} = (1 - \text{Tax rate}) \times \text{Taxable cash receipt} \quad (2)$$

We emphasize the term *taxable cash receipts* because not all cash inflows are taxable. For example, the release of working capital at the end of an investment project would not be a taxable cash inflow. It is not counted as income for either financial accounting or income tax reporting purposes because it is simply a recovery of the initial investment.

Depreciation Tax Shield

Depreciation is not a cash flow. For this reason, depreciation was ignored in Chapter 13 in all discounted cash flow computations. However, depreciation does affect the taxes that must be paid and, therefore, has an effect on a company's cash flows.

To illustrate the effect of depreciation deductions on tax payments, consider a company with annual cash sales of \$500,000 and cash operating expenses of \$310,000. In addition, the company has a depreciable asset on which the depreciation deduction is \$90,000 per year. The tax rate is 30%. As shown in Exhibit 13C–2, the depreciation deduction reduces the company's taxes by \$27,000. In effect, the depreciation deduction of \$90,000 *shields* \$90,000 in revenues from taxation and thereby *reduces* the amount of taxes that the company must pay. Because depreciation deductions shield revenues from taxation, they are generally referred to as a **depreciation tax shield**.² The reduction in tax payments made possible by the depreciation tax shield is equal to the amount of the depreciation deduction, multiplied by the tax rate as follows:

$$\text{Tax savings from the depreciation tax shield} = \text{Tax rate} \times \text{Depreciation deduction} \quad (3)$$

We can verify this formula by applying it to the \$90,000 depreciation deduction in our example:

$$0.30 \times \$90,000 = \$27,000 \text{ reduction in tax payments}$$

¹ This formula assumes that a company is operating at a profit; if it is operating at a loss, the tax situation can be very complex. For simplicity, we assume in all examples, exercises, and problems that the company is operating at a profit.

² The term *depreciation tax shield* may convey the impression that there is something underhanded about depreciation deductions—that companies are getting some sort of a special tax break. However, to use the depreciation deduction, a company must have already acquired a depreciable asset—which typically requires a cash outflow. Essentially, the tax code requires companies to delay recognizing the cash outflow as an expense until depreciation charges are recorded.

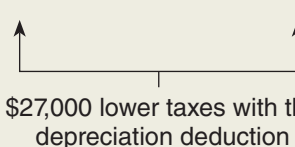
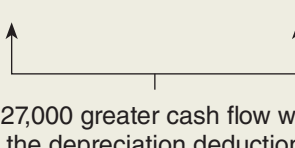
	Without Depreciation Deduction	With Depreciation Deduction
Sales	\$500,000	\$500,000
Cash operating expenses	310,000	310,000
Cash flow from operations	190,000	190,000
Depreciation expense	—	90,000
Taxable income	<u>\$190,000</u>	<u>\$100,000</u>
Income taxes (30%)	<u>\$ 57,000</u>	<u>\$ 30,000</u>
 \$27,000 lower taxes with the depreciation deduction		
Cash flow comparison:		
Cash flow from operations (above)	\$190,000	\$190,000
Income taxes (above)	57,000	30,000
Net cash flow	<u>\$133,000</u>	<u>\$160,000</u>
 \$27,000 greater cash flow with the depreciation deduction		

EXHIBIT 13C-2

The Impact of Depreciation Deductions on Tax Payments

In this appendix, when we estimate after-tax cash flows for capital budgeting decisions, we will include the tax savings provided by the depreciation tax shield.

To keep matters simple, we will assume in all of our examples and problem materials that depreciation reported for tax purposes is straight-line depreciation, with no deduction for salvage value. In other words, we will assume that the entire original cost of the asset is written off evenly over its useful life. Because the net book value of the asset at the end of its useful life will be zero under this depreciation method, we will assume that any proceeds received on disposal of the asset at the end of its useful life will be taxed as ordinary income.

In actuality, the rules for depreciation are more complex than this and most companies take advantage of accelerated depreciation methods allowed under the tax code. These accelerated methods usually result in a reduction in current taxes and an offsetting increase in future taxes. This shifting of part of the tax burden from the current year to future years is advantageous from a present value point of view because a dollar today is worth more than a dollar in the future. A summary of the concepts we have introduced so far is given in Exhibit 13C-3.

Item	Treatment
Tax-deductible cash expense*	Multiply by $(1 - \text{Tax rate})$ to get after-tax cost.
Taxable cash receipt*	Multiply by $(1 - \text{Tax rate})$ to get after-tax cash inflow.
Depreciation deduction	Multiply by the tax rate to get the tax savings from the depreciation tax shield.
*Cash expenses can be deducted from the cash receipts and the difference multiplied by $(1 - \text{Tax rate})$. See the example at the top of Exhibit 13C-4.	

EXHIBIT 13C-3

Tax Adjustments Required in a Capital Budgeting Analysis

Example of Income Taxes and Capital Budgeting

Armed with an understanding of after-tax cost, after-tax revenue, and the depreciation tax shield, we are now prepared to examine a comprehensive example of income taxes and capital budgeting.

Holland Company owns the mineral rights to land that has a deposit of ore. The company is uncertain if it should purchase equipment and open a mine on the property. After careful study, the company assembled the following data.

Cost of equipment needed	\$300,000
Working capital needed	\$75,000
Estimated annual cash receipts from sales of ore	\$250,000
Estimated annual cash expenses for salaries, insurance, utilities, and other cash expenses of mining the ore	\$170,000
Cost of road repairs needed in 6 years	\$40,000
Salvage value of the equipment in 10 years.	\$100,000

The ore in the mine would be exhausted after 10 years of mining activity, at which time the mine would be closed. The equipment would then be sold for its salvage value. Holland Company uses the straight-line method, assuming no salvage value, to compute depreciation deductions for tax purposes. The company's after-tax cost of capital is 12% and its tax rate is 30%. To be consistent, when we take the net present value of after-tax cash flows, we use the *after-tax* cost of capital as the discount rate.

Should Holland Company purchase the equipment and open a mine on the property? The solution to the problem is given in Exhibit 13C-4. We suggest that you go through this solution item by item and note the following points:

Cost of new equipment. The initial investment of \$300,000 in the new equipment is included in full with no reductions for taxes. This represents an *investment*,

EXHIBIT 13C-4

Example of Income Taxes and Capital Budgeting

				Per Year		
				Cash receipts from sales of ore	\$250,000	
				Less payments for salaries, insurance, utilities, and other cash expenses	170,000	
				Net cash receipts	<u>\$ 80,000</u>	
Items and Computations	Year(s)	(1) Amount	(2) Tax Effect*	After-Tax Cash Flows (1) × (2)	12% Factor	Present Value of Cash Flows
Cost of new equipment	Now	\$(300,000)	—	\$(300,000)	1.000	\$(300,000)
Working capital needed	Now	\$(75,000)	—	\$(75,000)	1.000	(75,000)
Annual net cash receipts (above)	1–10	\$80,000	1 – 0.30	\$56,000	5.650	316,400
Road repairs	6	\$(40,000)	1 – 0.30	\$(28,000)	0.507	(14,196)
Annual depreciation deductions	1–10	\$30,000	0.30	\$9,000	5.650	50,850
Salvage value of equipment	10	\$100,000	1 – 0.30	\$70,000	0.322	22,540
Release of working capital	10	\$75,000	—	\$75,000	0.322	24,150
Net present value						<u>\$ 24,744</u>

*Taxable cash receipts and tax-deductible cash expenses are multiplied by (1 – Tax rate) to determine the after-tax cash flow. Depreciation deductions are multiplied by the tax rate itself to determine the after-tax cash flow (i.e., tax savings from the depreciation tax shield).

not an expense, so no tax adjustment is made. (Only revenues and expenses are adjusted for the effects of taxes.) However, this investment does affect taxes through the depreciation deductions that are considered below.

Working capital. Observe that the working capital needed for the project is included in full with no reductions for taxes. Like the cost of new equipment, working capital is an investment and not an expense, so no tax adjustment is made. Also observe that no tax adjustment is made when the working capital is released at the end of the project's life. The release of working capital is not a taxable cash flow because it is a return of investment funds back to the company.

Annual net cash receipts. The annual net cash receipts from sales of ore are adjusted for the effects of income taxes, as discussed earlier in the chapter. Note at the top of Exhibit 13C–4 that the annual cash expenses are deducted from the annual cash receipts to obtain the net cash receipts. This simplifies computations.

Road repairs. Because the road repairs occur just once (in the sixth year), they are treated separately from other expenses. Road repairs would be a tax-deductible cash expense, and, therefore, they are adjusted for the effects of income taxes, as discussed earlier in the chapter.

Depreciation deductions. The tax savings provided by depreciation deductions is essentially an annuity that is included in the present value computations in the same way as other cash flows.

Salvage value of equipment. Because the company does not consider salvage value when computing depreciation deductions, book value will be zero at the end of the life of an asset. Thus, any salvage value received is taxable as income to the company. The after-tax benefit is determined by multiplying the salvage value by $(1 - \text{Tax rate})$.

Because the net present value of the proposed mining project is positive, the equipment should be purchased and the mine opened. Study Exhibit 13C–4 thoroughly. *Exhibit 13C–4 is the key exhibit!*

Summary (Appendix 13C)

Unless a company is a tax-exempt organization, such as a not-for-profit school or a governmental unit, income taxes should be considered in making capital budgeting decisions. Tax-deductible cash expenditures and taxable cash receipts are placed on an after-tax basis by multiplying them by $(1 - \text{Tax rate})$. Only the after-tax amount should be used in determining the desirability of an investment proposal.

Although depreciation is not a cash outflow, it is a valid deduction for tax purposes and as such affects income tax payments. The depreciation tax shield—computed by multiplying the depreciation deduction by the tax rate itself—also results in savings in income taxes.

Glossary (Appendix 13C)

After-tax benefit The amount of net cash inflow realized from a taxable cash receipt after income tax effects have been considered. The amount is determined by multiplying the taxable cash receipt by $(1 - \text{Tax rate})$. (p. 632)

After-tax cost The amount of net cash outflow resulting from a tax-deductible cash expense after income tax effects have been considered. The amount is determined by multiplying the tax-deductible cash expense by $(1 - \text{Tax rate})$. (p. 631)

Depreciation tax shield A reduction in tax that results from depreciation deductions. The reduction in tax is computed by multiplying the depreciation deduction by the tax rate. (p. 632)

approach the pricing problem from a completely different perspective.⁶ They prefer to mark up some version of full, not variable, costs, and the markup is based on desired profits rather than on factors related to demand. This approach is called the *absorption costing approach to cost-plus pricing*.

IN BUSINESS

COMPETITION INFLUENCES PRICES

The Department of Transportation says Cincinnati has the most expensive airport in the United States. Cincinnati passengers pay an average fare of 20.63¢ per mile while more than 20 airports, such as Buffalo, Oakland, Reno, and Tampa, offer flights at less than 12¢ per mile. Why the higher prices in Cincinnati? Delta Airlines controls more than 80% of the Cincinnati market so the lack of competition enables it to charge higher fares. For example, Delta charges \$529 to fly direct from Cincinnati to Las Vegas because it is the only airline to offer nonstop service on this route. However, Delta can only charge \$258 to fly direct from Atlanta to Las Vegas because AirTran Airways also flies nonstop on this route. Of course, Delta must balance the desire to raise prices at the Cincinnati airport with the realization that its customers can defect to competing airline departures from nearby airports in Dayton, Columbus, Louisville, Lexington, and Indianapolis.

Source: Scott McCartney, "The Most Expensive City to Leave: Cincinnati," *The Wall Street Journal*, December 11, 2007, pp. B9–B10.

The Absorption Costing Approach to Cost-Plus Pricing

LEARNING OBJECTIVE 2
Compute the selling price of a product using the absorption costing approach.

The absorption costing approach to cost-plus pricing differs from the economists' approach both in what costs are marked up and in how the markup is determined. Under the absorption approach to cost-plus pricing, the cost base is the absorption costing unit product cost, as defined in Chapter 3, rather than variable cost.

Setting a Target Selling Price Using the Absorption Costing Approach

To illustrate, assume that the management of Ritter Company wants to set the selling price on a product that has just undergone some design modifications. The accounting department has provided cost estimates for the redesigned product as shown below:

	Per Unit	Total
Direct materials	\$6	
Direct labor	\$4	
Variable manufacturing overhead	\$3	
Fixed manufacturing overhead		\$70,000
Variable selling and administrative expenses	\$2	
Fixed selling and administrative expenses		\$60,000

⁶ One study found that 83% of the 504 large manufacturing companies surveyed used some form of full cost (either absorption cost or absorption cost plus selling and administrative expenses) as the basis for pricing. The remaining 17% used only variable costs as a basis for pricing decisions. See V. Govindarajan and Robert N. Anthony, "How Firms Use Cost Data in Pricing Decisions," *Management Accounting*, July 1983, pp. 30–36. A less extensive survey by Eunsup Shim and Ephraim F. Sudit, "How Manufacturers Price Products," *Management Accounting*, February 1995, pp. 37–39, found similar results.

On the other hand, a survey of small-company executives summarized in *Inc.* magazine, November 1996, p. 84, revealed that only 41% set prices based on cost. The others charge what they think customers are willing to pay or what the market demands.

The first step in the absorption costing approach to cost-plus pricing is to compute the unit product cost. For Ritter Company, this amounts to \$20 per unit at a volume of 10,000 units, as computed below:

Direct materials	\$ 6
Direct labor	4
Variable manufacturing overhead	3
Fixed manufacturing overhead (\$70,000 ÷ 10,000 units)	7
Unit product cost	<u>\$20</u>

Ritter Company has a general policy of marking up unit product costs by 50%. A price quotation sheet for the company prepared using the absorption approach is presented in Exhibit A-2. Note that selling and administrative expenses are not included in the cost base. Instead, the markup is supposed to cover these expenses.

Direct materials	\$ 6	EXHIBIT A-2 Price Quotation Sheet— Absorption Basis (10,000 Units)
Direct labor	4	
Variable manufacturing overhead	3	
Fixed manufacturing overhead (based on 10,000 units)	7	
Unit product cost	20	
Markup to cover selling and administrative expenses and desired profit—50% of unit manufacturing cost	10	
Target selling price	<u>\$30</u>	

Determining the Markup Percentage

Ritter Company's markup percentage of 50% could be a widely used rule of thumb in the industry or just a company tradition that seems to work. The markup percentage may also be the result of an explicit computation. As we have discussed, the markup over cost should be largely determined by market conditions. However, many companies base their markup on cost and desired profit. The reasoning goes like this. The markup must be large enough to cover selling and administrative expenses and provide an adequate return on investment (ROI). Given the forecasted unit sales, the markup can be computed as follows:

$$\text{Markup percentage on absorption cost} = \frac{(\text{Required ROI} \times \text{Investment}) + \text{Selling and administrative expenses}}{\text{Unit product cost} \times \text{Unit sales}}$$

To show how this formula is applied, assume Ritter Company invests \$100,000 in operating assets such as equipment to produce and market 10,000 units of the product each year. If Ritter Company requires a 20% ROI, then the markup for the product would be determined as follows:

$$\begin{aligned} \text{Markup percentage on absorption cost} &= \frac{(20\% \times \$100,000) + (\$2 \text{ per unit} \times 10,000 \text{ units} + \$60,000)}{\$20 \text{ per unit} \times 10,000 \text{ units}} \\ &= \frac{(\$20,000) + (\$80,000)}{\$200,000} = 50\% \end{aligned}$$

As shown earlier, this markup of 50% leads to a target selling price of \$30 for Ritter Company. *If the company actually sells 10,000 units* of the product at this price, the company's ROI on this product will indeed be 20%. This is verified in Exhibit A-3. However,

EXHIBIT A-3

Income Statement and ROI
Analysis—Ritter Company
Actual Unit Sales = 10,000 Units;
Selling Price = \$30

Direct materials	\$ 6
Direct labor	4
Variable manufacturing overhead	3
Fixed manufacturing overhead (\$70,000 ÷ 10,000 units)	7
Unit product cost	<u>\$20</u>

Ritter Company
Absorption Costing Income Statement

Sales (\$30 per unit × 10,000 units)	\$300,000
Cost of goods sold (\$20 per unit × 10,000 units)	<u>200,000</u>
Gross margin	100,000
Selling and administrative expenses (\$2 per unit × 10,000 units + \$60,000)	<u>80,000</u>
Net operating income	<u>\$ 20,000</u>

ROI

$$\begin{aligned}
 \text{ROI} &= \frac{\text{Net operating income}}{\text{Average operating assets}} \\
 &= \frac{\$20,000}{\$100,000} \\
 &= 20\%
 \end{aligned}$$

if it turns out that more than 10,000 units are sold at this price, the ROI will be greater than 20%. If less than 10,000 units are sold, the ROI will be less than 20%. *The required ROI will be attained only if the forecasted unit sales volume is attained.*

IN BUSINESS

THE LIMITATIONS OF COST-PLUS PRICING

The invention of flat-panel televisions has destroyed customer demand for the entertainment armoires that used to house the bulky televisions of yesteryear. Hotel chains have dumped more than 40,000 armoires helping create a huge supply of obsolete units that practically nobody wants to buy. From a cost-plus pricing perspective, the cost to manufacture an armoire has probably increased in recent years as lumber prices have climbed. However, this cost-focused perspective is irrelevant in a marketplace where demand for armoires has all but disappeared.

Source: Juliet Chung, "Au Revoir, Armoire," *The Wall Street Journal*, November 10–11, 2007, pp. W1 and W3.

Problems with the Absorption Costing Approach

The absorption costing approach makes pricing decisions look deceptively simple. All a company needs to do is compute its unit product cost, decide how much profit it wants, and then set its price. It appears that a company can ignore demand and arrive at a price that will safely yield whatever profit it wants. However, as noted above, the absorption costing approach relies on a forecast of unit sales. Neither the markup nor the unit product cost can be computed without such a forecast.

The absorption costing approach essentially assumes that customers *need* the forecasted unit sales and will pay whatever price the company decides to charge. However, customers have a choice. If the price is too high, they can buy from a competitor or they may choose not to buy at all. Suppose, for example, that when Ritter Company sets its

Direct materials	\$ 6
Direct labor	4
Variable manufacturing overhead	3
Fixed manufacturing overhead (\$70,000 ÷ 7,000 units)	10
Unit product cost	<u>\$23</u>

Ritter Company
Absorption Costing Income Statement

Sales (\$30 per unit × 7,000 units)	\$210,000
Cost of goods sold (\$23 per unit × 7,000 units)	<u>161,000</u>
Gross margin	49,000
Selling and administrative expenses (\$2 per unit × 7,000 units + \$60,000)	<u>74,000</u>
Net operating loss	<u>\$ (25,000)</u>

ROI

$$\begin{aligned}
 \text{ROI} &= \frac{\text{Net operating income}}{\text{Average operating assets}} \\
 &= \frac{-\$25,000}{\$100,000} \\
 &= -25\%
 \end{aligned}$$

EXHIBIT A-4

Income Statement and ROI
Analysis—Ritter Company
Actual Unit Sales = 7,000 Units;
Selling Price = \$30

price at \$30, it sells only 7,000 units rather than the 10,000 units forecasted. As shown in Exhibit A-4, the company would then have a loss of \$25,000 on the product instead of a profit of \$20,000.⁷ Some managers believe that the absorption costing approach to pricing is safe. This is an illusion. The absorption costing approach is safe only if customers choose to buy at least as many units as managers forecasted they would buy.

Target Costing

Our discussion thus far has presumed that a product has already been developed, has been costed, and is ready to be marketed as soon as a price is set. In many cases, the sequence of events is just the reverse. That is, the company already *knows* what price should be charged, and the problem is to *develop* a product that can be marketed profitably at the desired price. Even in this situation, where the normal sequence of events is reversed, cost is still a crucial factor. The company can use an approach called *target costing*. **Target costing** is the process of determining the maximum allowable cost for a new product and then developing a prototype that can be profitably made for that maximum target cost figure. A number of companies have used target costing, including **Compaq, Culp, Cummins Engine, Daihatsu Motors, Chrysler, Ford, Isuzu Motors, ITT Automotive, Komatsu, Matsushita Electric, Mitsubishi Kasei, NEC, Nipponenso, Nissan, Olympus, Sharp, Texas Instruments, and Toyota.**

LEARNING OBJECTIVE 3

Compute the target cost for a new product or service.

⁷ It may be *impossible* to break even using an absorption costing approach when the company has more than one product—even when it would be possible to make substantial profits using the economists' approach to pricing. For details, see Eric Noreen and David Burgstahler, "Full Cost Pricing and the Illusion of Satisficing," *Journal of Management Accounting Research* 9 (1997).

The target cost for a product is computed by starting with the product's anticipated selling price and then deducting the desired profit, as follows:

$$\text{Target cost} = \text{Anticipated selling price} - \text{Desired profit}$$

The product development team is then given the responsibility of designing the product so that it can be made for no more than the target cost.

Reasons for Using Target Costing

The target costing approach was developed in recognition of two important characteristics of markets and costs. The first is that many companies have less control over price than they would like to think. The market (i.e., supply and demand) really determines price, and a company that attempts to ignore this does so at its peril. Therefore, the anticipated market price is taken as a given in target costing. The second observation is that most of a product's cost is determined in the design stage. Once a product has been designed and has gone into production, not much can be done to significantly reduce its cost. Most of the opportunities to reduce cost come from designing the product so that it is simple to make, uses inexpensive parts, and is robust and reliable. If the company has little control over market price and little control over cost once the product has gone into production, then it follows that the major opportunities for affecting profit come in the design stage where valuable features that customers are willing to pay for can be added and where most of the costs are really determined. So that is where the effort is concentrated—in designing and developing the product. The difference between target costing and other approaches to product development is profound. Instead of designing the product and then finding out how much it costs, the target cost is set first and then the product is designed so that the target cost is attained.

IN BUSINESS

MANAGING COSTS IN THE PRODUCT DESIGN STAGE

The Boeing Company is building the airframe of its 787 Dreamliner jet using carbon fiber-reinforced plastic. While this type of plastic has been used in golf club shafts and tennis rackets, it has never been used to construct the exterior of an airplane. Boeing is excited about this innovative raw material because it allows enormous cost savings. For example, Boeing's Dreamliner should be 20% more fuel efficient than the Boeing 767 or Airbus A330, its maintenance costs should be 30% less than aluminum planes, and the number of fasteners needed to assemble its fuselage should be 80% less than conventional airplanes. In addition, aluminum airplanes require costly corrosion inspections after 6 years of service, while the Dreamliner can fly 12 years before it would need a comparable inspection. To Boeing's delight, the Dreamliner's sales have "taken off" because "customers get tremendous bang for their bucks. For \$120 million—about what they paid for the comparable Boeing 767-300 back in the 1980s—airlines get an all-new aircraft that flies faster than the competition and costs substantially less to operate."

Source: Stanley Holmes, "A Plastic Dream Machine," *BusinessWeek*, June 20, 2005, pp. 32–36.

An Example of Target Costing

To provide a simple example of target costing, assume the following situation: Handy Company wishes to invest \$2,000,000 to design, develop, and produce a new hand mixer. The company's Marketing Department surveyed the features and prices of competing products and determined that a price of \$30 would enable Handy to sell an estimated

40,000 hand mixers per year. Because the company desires a 15% ROI, the target cost to manufacture, sell, distribute, and service one mixer is \$22.50 as computed below:

Projected sales (40,000 mixers $\times$ \$30 per mixer)	\$1,200,000
Less desired profit (15% $\times$ \$2,000,000)	300,000
Target cost for 40,000 mixers	<u>\$ 900,000</u>
Target cost per mixer (\$900,000 $\div$ 40,000 mixers)	<u>\$22.50</u>

This \$22.50 target cost would be broken down into target costs for the various functions: manufacturing, marketing, distribution, after-sales service, and so on. Each functional area would be responsible for keeping its actual costs within target.

Summary

Pricing involves a delicate balancing act. Higher prices result in more revenue per unit but drive down unit sales. Exactly where to set prices to maximize profit is a difficult problem, but, in general, the markup over cost should be highest for those products where customers are least sensitive to price. The demand for such products is said to be price inelastic.

Managers often rely on cost-plus formulas to set target prices. From the economists' point of view, the cost base for the markup should be variable cost. In contrast, in the absorption costing approach the cost base is the absorption costing unit product cost and the markup is computed to cover both nonmanufacturing costs and to provide an adequate return on investment. With the absorption approach, costs will not be covered and return on investment will not be adequate unless the unit sales forecast used in the cost-plus formula is accurate. If applying the cost-plus formula results in a price that is too high, the unit sales forecast will not be attained.

Some companies take a different approach to pricing. Instead of starting with costs and then determining prices, they start with prices and then determine allowable costs. Companies that use target costing estimate what a new product's market price is likely to be based on its anticipated features and prices of products already on the market. They subtract desired profit from the estimated market price to arrive at the product's target cost. The design and development team is then given the responsibility of ensuring that the actual cost of the new product does not exceed the target cost.

Glossary

Cost-plus pricing A pricing method in which a predetermined markup is applied to a cost base to determine the target selling price. (p. 716)

Markup The difference between the selling price of a product or service and its cost. The markup is usually expressed as a percentage of cost. (p. 716)

Price elasticity of demand A measure of the degree to which a change in price affects the unit sales of a product or service. (p. 717)

Target costing The process of determining the maximum allowable cost for a new product and then developing a prototype that can be profitably made for that maximum target cost figure. (p. 723)

Questions

- A-1 What is cost-plus pricing?
- A-2 What does the price elasticity of demand measure? What is inelastic demand? What is elastic demand?
- A-3 According to the economists' approach to setting prices, the profit-maximizing price should depend on what two factors?
- A-4 Which product should have a larger markup over variable cost, a product whose demand is elastic or a product whose demand is inelastic?